

Exoplanet Habitable Zone Analysis

58 messages · exported Sep 13, 2026 10:18 PM

As an industry expert data scientist and expert in CRISP-DM methodology for data science, and also an expert astrophysicist familiar with exoplanet habitability research, given the attached Kaggle "All Exoplanets Dataset" (all_exoplanets_2021.csv, sourced from the NASA Exoplanet Archive, ~4,575 confirmed exoplanets with columns including Planet Name, Planet Host, Discovery Method, Discovery Year, Discovery Facility, Orbital Period, Orbit Semi-Major Axis, Eccentricity, Planet Mass, Planet Radius, Equilibrium Temperature, Insolation Flux, Stellar Effective Temperature, Stellar Radius, Stellar Mass, Stellar Metallicity, and Distance), can you do a step by step data science analysis playing the role of an expert data scientist professor teaching a masters program in data science.

The goal of this analysis is to answer: what stellar and orbital features predict whether an exoplanet falls within its host star's habitable zone, and specifically whether planets with moderate orbital periods around cooler stars are more likely to be habitable-zone candidates. Also investigate whether this relationship is affected by detection bias across different discovery methods (transit vs radial velocity vs imaging).

Your response should be textbook quality. Make sure you prepare all the steps and reports of the methodology in principled fashion on:

- data understanding (describe each column, units, missingness)
- data analysis and visualization (including a plot of Orbital Period vs Stellar Effective Temperature colored by habitable-zone status, faceted by Discovery Method)
- data cleaning and preprocessing
- feature engineering (compute a habitable-zone flag using Insolation Flux between 0.35 and 1.75 Earth flux, and/or an Earth Similarity Index from planet radius, mass, and equilibrium temperature)
- feature selection
- data clustering (cluster planets into types such as hot Jupiters, super-Earths, sub-Neptunes, and see which cluster overlaps most with the habitable zone)
- outlier analysis and processing
- classification/regression to predict habitable-zone membership (or Earth Similarity Index) using a variety of methods, comparing to a naive baseline, with proper metrics
- feature importance analysis (e.g. SHAP or coefficient/importance ranking) to quantify which features matter most, specifically testing the stellar-temperature/orbital-period interaction
- a clear discussion separating the real astrophysical trend from potential detection/selection bias introduced by the discovery method
- final recommendation and conclusion stated in plain language suitable for a general

audience

Use your most popular ML packages like scikit-learn, etc. Please respond in many reasonable small sized chunks, starting with the initial steps of CRISP-DM methodology and continuing through all the steps on this dataset. I will ask you to continue after each chunk. It is important to chunk this carefully and repeat the requirement above at the end of each chunk, and explain where you are (mindmap) in the process and what you will do next. Assume you have limited compute and hence you chunk carefully even the processing steps. I will ask you to continue each chunk.

CHATGPT · SEP 01, 2026 12:10 AM

I'll begin with CRISP-DM business understanding and a reproducible data-understanding audit, establishing the target, schema, units, missingness, data quality, and modeling risks before preprocessing or modeling.

CHATGPT · SEP 01, 2026 12:13 AM

The upload is recognized, but its runtime path is not yet accessible; I'm resolving the mount without substituting another dataset so all reported statistics remain file-derived.

Exoplanet Habitability Drivers

Chunk 1 of 7 — CRISP-DM Business Understanding and Initial Data Understanding

I audited the uploaded `all_exoplanets_2021.csv` file directly. [filecite turn2file0 L1-L4](#)

This first chunk establishes the scientific question, defines valid targets, documents every field and its missingness, and identifies the principal risks that must be handled before visualization or modeling.

1. CRISP-DM Phase 1: Business understanding

1.1 The scientific question

The primary question is:

> **Which stellar and orbital features predict whether a detected exoplanet receives an amount of stellar radiation consistent with the adopted habitable-zone interval?**

The specific hypothesis is:

> **Planets with moderate orbital periods around cooler stars are more likely to be habitable-zone candidates.**

A secondary question is:

> **Does the apparent relationship change across discovery methods—particularly transit, radial velocity, and direct imaging—because those methods detect different parts of the exoplanet population?**

This is an important distinction. We are studying two related but different processes:

1. **The astrophysical process:** how stellar luminosity and planetary orbit determine incident flux.
2. **The observation process:** which planets astronomers are likely to detect and characterize.

The discovery method is not itself a physical cause of habitability. It is primarily a **selection mechanism**.

1.2 What “habitable-zone candidate” means here

Following the requested operational definition:

$$\begin{aligned} & \backslash \\ \text{HZ} = & \\ & \backslash \text{begin}{cases} \\ & 1, \& 0.35 \leq S/S_{\oplus} \leq 1.75 \\ & 0, \& S/S_{\oplus} < 0.35 \text{ or } S/S_{\oplus} > 1.75 \\ & \backslash \text{end}{cases} \\ & \backslash \end{aligned}$$

where $S/S_{\oplus}$ is the planet’s incident stellar flux relative to Earth.

This is a **radiative habitable-zone screening criterion**, not a determination that the planet is actually habitable.

A planet in this interval could still be unsuitable for surface liquid water because of atmospheric composition, pressure, greenhouse effects, albedo, orbital variability, stellar flares, tidal locking, water inventory, or its being a gas giant rather than a terrestrial planet.

Therefore, throughout the analysis I will use the terms:

- **HZ candidate**
- **habitable-zone membership**
- **radiatively temperate candidate**

I will not equate the label with “habitable” or “life-bearing.”

1.3 The expected astrophysical relationship

The incident flux approximately satisfies:

$$\begin{aligned} & \left[\frac{S}{S_{\odot}} \right] \\ &= \\ & \left[\frac{L_{\star}/L_{\odot}}{(a/\mathrm{AU})^2} \right] \end{aligned}$$

Stellar luminosity can be estimated from stellar radius and temperature:

$$\begin{aligned} & \left[\frac{L_{\star}}{L_{\odot}} \right] \\ & \approx \\ & \left(\frac{R_{\star}}{R_{\odot}} \right)^2 \\ & \left(\frac{T_{\mathrm{eff}}}{5772\,\mathrm{K}} \right)^4 \end{aligned}$$

For a planet whose mass is negligible compared with its star, Kepler’s third law gives:

$$\begin{aligned} & \left[a_{\mathrm{AU}} \right] \\ & \approx \\ & \left[\left(\frac{M_{\star}}{M_{\odot}} \right) \right. \\ & \left. \left(\frac{P}{1\,\mathrm{year}} \right)^2 \right]^{1/3} \end{aligned}$$

Combining these relationships gives the approximate scaling:

$$\sqrt{\frac{R_{\star}^2 T_{\mathrm{eff}}^4}{M_{\star}^{-2/3} P^{-4/3}}}$$

This has two major implications.

Implication 1: the period–temperature relationship should be nonlinear

The habitable zone should appear as a **curved or sloping band** in orbital-period versus stellar-temperature space.

A single global definition of “moderate orbital period,” such as 20–100 days, would be scientifically inadequate:

- For a cool, low-luminosity M dwarf, an HZ orbit may have a period of only several days or tens of days.
- For a Sun-like star, an HZ orbit may have a period closer to hundreds of days.
- For a luminous giant star, even a long-period planet can receive very high flux.

Thus, “moderate period” must be interpreted **relative to the host star**, not as one universal period interval.

Implication 2: there is a genuine interaction to test

The effect of orbital period on HZ probability should depend on stellar temperature, radius, and mass. Our models must therefore include at least one of:

- a $\log(\text{period}) \times \text{stellar temperature}$ interaction;
- nonlinear polynomial or spline terms;
- a tree-based response surface;
- a period normalized to the star’s Earth-equivalent-insolation period.

A simple linear model with only additive period and temperature terms would be misspecified.

1.4 Statistical estimand

Because this is a catalog of already detected planets, the estimand is:

$$\mathbb{P}(\text{planet was detected and entered into the catalog} \mid X)$$

It is **not** automatically:

$$\mathbb{P}(\text{all planets in the Galaxy} \mid X)$$

Correcting from the detected catalog to the underlying planet population would require information such as:

- stars surveyed by each program;
- survey duration and cadence;
- instrumental sensitivity;
- transit injection-and-recovery completeness;
- radial-velocity sampling and noise;
- imaging contrast curves and inner working angles.

Those denominator and completeness data are not in this CSV. Consequently, this project can diagnose and partially adjust for catalog selection effects, but it should not claim unbiased planet-occurrence rates.

Transit catalogs are geometrically and observationally biased toward shorter-period systems; radial-velocity signal amplitude depends on quantities including planet mass, period, stellar mass, inclination, and eccentricity; direct imaging predominantly samples luminous young giant planets at comparatively wide separations.

1.5 Planned analytical targets

We will create several related targets rather than forcing every scientific question into one binary label.

Target A: reported-flux HZ membership

```
HZ_reported = 1      when 0.35 <= Insolation Flux <= 1.75
HZ_reported = 0      when flux is observed but outside that interval
HZ_reported = missing when Insolation Flux is missing
```

Missing flux must **not** be encoded as non-HZ. That would incorrectly assign 4,205 planets to the negative class.

Target B: three-zone flux class

To distinguish the two physically different ways of being outside the HZ:

```
too_hot : Insolation Flux > 1.75
HZ       : 0.35 <= Insolation Flux <= 1.75
too_cold: Insolation Flux < 0.35
```

This is useful because the binary negative class combines:

- close, highly irradiated planets; and
- distant, weakly irradiated planets.

The relationship between period and a binary HZ outcome is consequently expected to be approximately **inverted-U-shaped**, not monotonic.

Target C: logarithm of insolation flux

For regression:

$$\begin{aligned} & \backslash \\ y &= \log_{10}(S/S_{\oplus}) \\ & \backslash \end{aligned}$$

This retains more information than the binary threshold. The estimated flux can subsequently be converted to HZ membership.

Target D: physics-derived flux

Where reported flux is missing, we can estimate:

$$\begin{aligned} & \backslash \\ S_{\mathrm{derived}} & \\ = & \\ \frac{R_{\star}^2 (T_{\mathrm{eff}}/5772)^4}{a^2} & \\ & \backslash \end{aligned}$$

A second sensitivity version can infer $\sqrt{a}$ from period and stellar mass.

This broader target will be especially useful for descriptive discovery-method comparisons. It must, however, be labeled as **derived**, and models using it must avoid circular claims. A model predicting a flux label that was itself calculated from temperature, radius, and semi-major axis is largely learning the known physical equation.

1.6 Earth Similarity Index limitation

The prompt describes a `Planet Radius` field, but the uploaded file does **not** contain that column.

The actual dataset has:

- exactly **4,575 planet rows**;
- **23 columns**;
- no planet-radius measurement;
- no parameter-uncertainty columns;
- no mass-provenance column.

Therefore, a conventional radius–mass–temperature Earth Similarity Index cannot be calculated directly from this file.

Later we may construct one or both of these sensitivity measures:

1. a clearly labeled **partial ESI**, using mass and equilibrium temperature only;
2. an estimated-radius ESI using an empirical mass–radius relationship.

The second option would introduce extra modeling assumptions and should not be treated as an observed ESI.

2. CRISP-DM Phase 2: Data understanding

2.1 Dataset-level audit

The uploaded table contains:

Property	Result
---	---
Planet rows	4,575
Columns	23
Unique planet names	4,575
Unique host names	3,392
Duplicate complete rows	0
Duplicate planet names	0
Discovery methods	11
Discovery facilities	63
Discovery years	1989–2021
Rows with orbital period and stellar temperature	4,195
Rows with reported insolation flux	370
Reported-flux HZ candidates	32

The NASA Exoplanet Archive notes that composite parameter tables can select values from different literature references to make each planet row more complete. Such rows are useful for statistical analyses but are not necessarily internally self-consistent at precision-astrophysics level. [\[cite\]turn773934search5](#)

That warning matters here because, for example, a stellar radius and temperature may come from one reference while an orbital parameter comes from another.

2.2 Data dictionary: identifiers and discovery information

Column	Description	Unit/type	Missing
---	---	---	---
No.	Sequential source-row identifier	Integer ID	0 — 0.0%
Planet Name	Catalog name of the planet	Identifier	0 — 0.0%
Planet Host	Name of the stellar host or host system	Identifier/group	0 — 0.0%
Num Stars	Number of known stellar components in the system	Count	0 — 0.0%
Num Planets	Number of known planets in the system	Count	0 — 0.0%
Discovery Method	Method under which the planet was discovered	Categorical	0 — 0.0%

Discovery Year	Calendar year of discovery	Year	0 — 0.0%	
Discovery Facility	Observatory, telescope, mission, or facility credited with discovery			
Categorical	0 — 0.0%			

Modeling roles

- **No.** is administrative and will be dropped.
- **Planet Name** is an identifier, not a predictive feature.
- **Planet Host** will be retained for **grouped train/test splitting**.
- **Discovery Method**, **Discovery Facility**, and **Discovery Year** are primarily observation-process variables.
- **Num Planets** is partly a detection-history variable: a system may contain undiscovered planets, so it is not a complete intrinsic system property.

2.3 Data dictionary: planetary, orbital, and stellar variables

The physical units are consistent with the NASA Exoplanet Archive’s planetary-systems column definitions. [\[cite turn 773934 search 1\]](#)

Column	Description	Unit/type	Missing
---	---	---	---
Orbital Period Days	Time required for one orbit	Days	162 — 3.5%
Orbit Semi-Major Axis	Characteristic orbital size	Astronomical units, AU	1,812 — 39.6%
Mass	Best available planet mass or minimum-mass-like value	Earth masses, $(M_{\oplus})$	2,569 — 56.2%
Eccentricity	Orbital eccentricity	Dimensionless, normally $(0 \leq e < 1)$	2,868 — 62.7%
Insolation Flux	Incident flux relative to Earth	Earth flux, $(S_{\oplus})$	4,205 — 91.9%
Equilibrium Temperature	Estimated radiative equilibrium temperature	Kelvin	3,650 — 79.8%
Spectral Type	Host-star spectral classification	Categorical/string	3,647 — 79.7%
Stellar Effective Temperature	Blackbody-equivalent stellar surface temperature	Kelvin	349 — 7.6%

	Stellar Radius		Host-star radius Solar radii, $(R_{\odot})$		447 — 9.8%
	Stellar Mass		Host-star mass Solar masses, $(M_{\odot})$		731 — 16.0%
	Stellar Metallicity		Logarithmic heavy-element abundance relative to solar dex		1,371 — 30.0%
	Stellar Metallicity Ratio		Abundance convention, usually $[Fe/H]$ or $[M/H]$		Categorical notation 1,383 — 30.2%
	Stellar Surface Gravity		Base-10 logarithm of stellar surface gravity $(\log_{10}(\mathrm{cm\,s^{-2}}))$		603 — 13.2%
	Distance		Distance to the host system Parsecs		104 — 2.3%
	Gaia Magnitude		Apparent brightness in the Gaia passband Magnitudes		175 — 3.8%

Important mass caveat

The generic `Mass` field is not completely homogeneous across methods:

- radial velocity often measures $(M \sin i)$, a minimum mass when inclination is unknown;
- transiting planets can have more direct mass constraints after RV or timing follow-up;
- imaging masses are frequently inferred from luminosity and evolutionary models.

The original archive provides provenance fields that help distinguish these cases, but this Kaggle snapshot does not include them. We will therefore use mass cautiously in cross-method clustering and feature-importance interpretation.

2.4 Discovery-method composition

	Discovery method		Planets		Share of catalog
	---		---		---
	Transit		3,444		75.3%
	Radial Velocity		899		19.7%
	Microlensing		120		2.6%
	Imaging		54		1.2%
	All other methods		58		1.3%

The catalog is consequently dominated by transit discoveries, particularly Kepler and K2:

Leading discovery facility	Planets
Kepler	2,402
K2	477
La Silla Observatory	260
Multiple Observatories	184
W. M. Keck Observatory	181
TESS	172

This means an unweighted “all planets” model will be driven primarily by the Kepler-like transiting population unless the validation and interpretation explicitly account for discovery method.

2.5 Missingness is strongly method-dependent

Selected non-missing coverage rates are:

	Method	Period	Semi-major axis	Mass	Eccentricity	Insolation	Equilibrium temperature	Stellar temperature	Stellar radius
	---	---	---	---	---	---	---	---	---
Transit	99.9%	49.5%	25.7%	23.1%	9.0%	23.6%	97.5%	99.3%	
Radial Velocity	100.0%	94.8%	100.0%	97.2%	6.7%	8.2%	90.1%	74.0%	
Imaging	16.7%	94.4%	92.6%	9.3%	0.0%	57.4%	57.4%	20.4%	
Microlensing	6.7%	99.2%	100.0%	1.7%	0.0%	2.5%	1.7%	0.0%	

This is a textbook example of **missing not at random**:

- transit observations frequently provide radius-like information but not mass or eccentricity without follow-up;
- radial velocity strongly characterizes mass and eccentricity;
- imaging gives projected separation or orbital-scale estimates and model-derived masses, but periods are often poorly constrained;
- microlensing usually provides mass and separation estimates but sparse stellar

characterization in this snapshot.

A simple complete-case analysis would select a highly unrepresentative subset. Similarly, global median imputation without missingness indicators could erase method-specific information or produce physically implausible synthetic planets.

3. Feasibility of the requested HZ target

3.1 Reported insolation coverage

Only **370 of 4,575 planets**, or **8.1%**, have a reported insolation flux.

Among those 370:

- **32** lie between 0.35 and 1.75 Earth flux;
- HZ prevalence is therefore **8.65%**;
- **338** are outside the adopted interval.

The direct-label target is highly imbalanced.

An always-non-HZ classifier would obtain:

$$\begin{aligned} \backslash[\\ \text{Accuracy} &= \frac{338}{370} = 91.35\% \\ \backslash] \end{aligned}$$

but it would identify no HZ planets. This shows why raw accuracy will not be an acceptable primary metric.

The naive prevalence baseline for precision–recall area is approximately:

$$\begin{aligned} \backslash[\\ \text{PR-AUC}_{\text{baseline}} &= 32/370 = 0.0865 \end{aligned}$$

\]

3.2 Target availability by method

| Discovery method | Total planets | Reported flux available | HZ candidates | HZ rate among flux-known planets |

|---|---:|---:|---:|---:|

| Transit | 3,444 | 310 | 19 | 6.1% |

| Radial Velocity | 899 | 60 | 13 | 21.7% |

| Imaging | 54 | 0 | 0 observed | Not estimable |

| Microlensing | 120 | 0 | 0 observed | Not estimable |

| Other methods | 58 | 0 | 0 observed | Not estimable |

The radial-velocity percentage must **not** be interpreted as evidence that RV planets are intrinsically more likely to lie in the HZ. It is the rate among the small and selectively characterized subset with non-missing flux.

Most importantly:

> The direct reported-flux target provides no labeled imaging planets.

Therefore, the reported-target classifier cannot estimate an imaging-specific HZ relationship. Imaging will instead be investigated through:

- method-specific covariate distributions;
- derived stellar-flux estimates where sufficient stellar data exist;
- clustering and outlier analysis;
- sensitivity comparisons rather than a directly labeled imaging classifier.

4. Preliminary signal—not yet a statistical result

There are **366 planets** with all three of:

- reported insolation flux;
- orbital period;
- stellar effective temperature.

Of these, 32 meet the HZ definition.

Pooled, unadjusted comparison

Group	Median orbital period	Median stellar temperature
--- ---	---	---
HZ candidates	32.37 days	3,466 K
Outside HZ	6.73 days	5,055 K

This is directionally consistent with the hypothesis: the HZ candidates in the labeled subset tend to have longer periods and cooler host stars.

Within transit discoveries

Group	Median period	Median stellar temperature
--- ---	---	---
Transit HZ candidates	37.42 days	3,740 K
Other labeled transit planets	6.00 days	5,215 K

Within radial-velocity discoveries

Group	Median period	Median stellar temperature
--- ---	---	---
RV HZ candidates	17.87 days	3,342 K
Other labeled RV planets	14.24 days	3,933 K

These comparisons are only preliminary because:

1. the label is missing for more than 90% of the catalog;
2. the labeled samples are not random;
3. only 32 positive planets are available;
4. multiple planets share the same host;
5. the outside-HZ group combines both extremely hot and extremely cold planets;
6. period spans many orders of magnitude, making medians incomplete summaries;
7. stellar radius and mass differ across groups and must be controlled.

The final answer must come from nonlinear, host-grouped, method-aware analysis—not these medians alone.

5. Initial data-quality findings

5.1 No obvious duplicate records

The audit found:

- no duplicated complete rows;
- no duplicated planet names;
- a complete sequential `No.` field from 1 through 4,575.

The `No.` field therefore contains no scientific information and will be removed.

5.2 A categorical typo requires correction

`Stellar Metallicity Ratio` contains:

- `[Fe/H]`
- `[M/H]`
- `[m/H]`

- [Me/H]
- one malformed value: [Fe/H[

The malformed value occurs for T0I-837 b and should be standardized to [Fe/H]. The capitalization variants must be handled carefully rather than automatically assuming identical scientific meaning.

5.3 Numerical ranges are physically valid but extremely skewed

No non-null values were found with:

- nonpositive orbital period;
- nonpositive semi-major axis;
- nonpositive mass;
- eccentricity below zero or at least one;
- nonpositive stellar temperature, radius, or mass;
- nonpositive insolation or equilibrium temperature;
- nonpositive distance.

However, several distributions have enormous upper tails:

- orbital periods range from approximately 0.09 days to 402 million days;
- semi-major axes extend to thousands of AU;
- stellar temperatures extend to 40,000 K;
- stellar radii extend above 80 solar radii;
- stellar surface gravity includes a value of 7.92, consistent with a compact stellar remnant rather than an ordinary main-sequence star.

These are not automatically data-entry errors. They reflect very different populations, including:

- ultra-short-period planets;
- wide directly imaged companions;
- giant-star systems;
- white-dwarf or hot-subdwarf systems;
- possible high-mass companions near the planet/brown-dwarf boundary.

Outlier processing will therefore be **physics-informed**. We will not blindly delete or winsorize every extreme value.

6. Modeling implications established before preprocessing

6.1 Host-grouped validation is mandatory

There are 4,575 planets but only 3,392 hosts. Planets orbiting the same host share:

- stellar temperature;
- stellar radius;
- stellar mass;
- metallicity;
- distance;
- magnitude;
- survey history.

A random row-level split could place one planet from a system in training and another planet from the same system in testing. That would leak host information.

The primary cross-validation strategy will therefore use:

```
groups = df["Planet Host"]
```

with `StratifiedGroupKFold` or an equivalent host-grouped strategy.

The 32 direct HZ planets belong to only **26 host systems**, making the effective positive sample even smaller than the row count suggests.

6.2 Discovery variables must be separated from astrophysical variables

We will define at least three feature sets.

Astrophysical feature set

Possible variables:

- log orbital period;
- log semi-major axis;
- eccentricity;
- log planet mass;
- stellar effective temperature;
- stellar radius;
- stellar mass;
- metallicity;
- surface gravity;
- system multiplicity.

Observation and selection feature set

- discovery method;
- discovery year;
- discovery facility;
- distance;
- Gaia magnitude;
- missingness indicators.

These variables help explain **what was detected and characterized**, but distance and magnitude are not intrinsic habitability drivers.

Leakage or target-proxy fields

For a classifier whose label is defined from insolation:

- Insolation Flux must not be a predictor;
- HZ_reported must not be a predictor;
- Equilibrium Temperature is a near-direct thermal proxy and will be excluded from the primary leakage-safe model;
- physics-derived flux must not be included in a model predicting a label constructed from that same flux.

6.3 Separate models will answer separate questions

We will eventually compare:

1. Interpretive interaction model

Tests $\log(\text{period}) \times \text{stellar temperature}$, with nonlinear terms and limited covariates.

2. Broader astrophysical classifier

Uses regularized logistic regression, random forest, and gradient boosting.

3. Flux regression model

Predicts $\log_{10}(\text{Insolation Flux})$.

4. Bias-augmented model

Adds discovery method, distance, magnitude, and year.

5. Method-interaction model

Tests whether period and temperature relationships differ between transit and RV discoveries.

6. Derived-label sensitivity model

Broadens coverage using physically estimated flux, while explicitly acknowledging its deterministic construction.

6.4 Primary evaluation metrics

For binary HZ classification:

- precision–recall AUC;
- ROC AUC;
- balanced accuracy;
- HZ recall or sensitivity;
- precision;
- F1 score;
- Matthews correlation coefficient;
- Brier score and calibration curves.

For flux or ESI regression:

- mean absolute error;
- root mean squared error on log flux;
- $\sqrt{R^2}$;
- calibration around the two HZ boundaries;
- rate at which predicted and observed flux fall in the same hot/HZ/cold category.

Confidence intervals will be estimated through host-level bootstrap or repeated grouped validation where computationally feasible.

7. Reproducible initial audit code


```

from pathlib import Path

import numpy as np
import pandas as pd

DATA_PATH = Path("all_exoplanets_2021.csv")

df = pd.read_csv(DATA_PATH)

# Basic integrity checks
assert len(df) == 4575
assert df.shape[1] == 23
assert df["Planet Name"].is_unique
assert df["No."].is_unique

audit = pd.DataFrame(
    {
        "dtype": df.dtypes.astype(str),
        "non_null": df.notna().sum(),
        "missing": df.isna().sum(),
        "missing_pct": 100 * df.isna().mean(),
        "unique_non_null": df.nunique(dropna=True),
    }
).sort_values("missing_pct", ascending=False)

print(audit)

# Correct construction of the reported-flux target:
# missing flux remains an unknown label, not a negative label.
df["HZ_reported"] = pd.Series(pd.NA, index=df.index, dtype="boolean")

flux_known = df["Insolation Flux"].notna()

df.loc[flux_known, "HZ_reported"] = (
    df.loc[flux_known, "Insolation Flux"]
    .between(0.35, 1.75, inclusive="both")
)

# Three-class target
df["Flux_Zone"] = pd.Series(pd.NA, index=df.index, dtype="string")

df.loc[flux_known & (df["Insolation Flux"] < 0.35), "Flux_Zone"] = "too_cold"

```

```

df.loc[
    flux_known & df["Insolation Flux"].between(0.35, 1.75),
    "Flux_Zone"
] = "habitable_zone"
df.loc[flux_known & (df["Insolation Flux"] > 1.75), "Flux_Zone"] = "too_hot"

# Host groups for future leakage-safe validation
groups = df["Planet Host"]

print(df["HZ_reported"].value_counts(dropna=False))
print(df.groupby("Discovery Method")["HZ_reported"].value_counts(dropna=False))

```

Expected reported-target counts are:

False	338
True	32
Missing	4205

8. CRISP-DM process mindmap

Exoplanet habitability drivers

- |
 - | — 1. Business understanding COMPLETE
 - | — Define HZ candidate
 - | — Define moderate-period/cool-star hypothesis
 - | — Separate astrophysics from observation
 - | — Define scientific and predictive success
- |
 - | — 2. Data understanding INITIAL AUDIT COMPLETE
 - | — 4,575 planets × 23 columns
 - | — Column definitions and units
 - | — Missingness and method dependence
 - | — Target availability and class imbalance
 - | — Integrity and initial outlier checks
- |
 - | — 3. Data preparation NEXT
 - | — Standardize categories and types
 - | — Construct nullable HZ targets
 - | — Estimate stellar luminosity and derived flux
 - | — Add log and interaction features
 - | — Create missingness indicators
 - | — Define leakage-safe modeling datasets
- |
 - | — 4. Exploratory analysis and visualization
 - | — Required period-versus-temperature plot
 - | — Facets by discovery method
 - | — HZ / too-hot / too-cold distributions
 - | — Method-specific selection patterns
- |
 - | — 5. Clustering and outlier analysis
 - | — Hot Jupiters
 - | — Super-Earth-like objects
 - | — Sub-Neptunes
 - | — Wide imaged giants
 - | — HZ overlap by cluster
- |
 - | — 6. Modeling and evaluation
 - | — Naive baselines
 - | — Logistic regression
 - | — Random forest
 - | — Gradient boosting
 - | — Flux regression

- | └─ Host-grouped validation
- |
- └─ 7. Interpretation and recommendations
 - └─ Feature importance and SHAP
 - └─ Period × temperature interaction
 - └─ Transit versus RV versus imaging bias
 - └─ Astrophysical conclusions
 - └─ Plain-language recommendations

Next chunk

Chunk 2 will perform data cleaning, preprocessing, and feature engineering. It will construct the reported and physics-derived HZ flags, estimate stellar luminosity and flux, define leakage-safe feature sets, create log and interaction features, and specify how missing values will be handled without destroying discovery-method information.

Standing requirement carried into every chunk: determine which stellar and orbital features predict habitable-zone membership, test the moderate-period × cooler-star relationship, and distinguish the real astrophysical pattern from transit, radial-velocity, and imaging selection bias while using a staged, limited-compute CRISP-DM workflow.

Reply **continue** for Chunk 2.

PROMPT · SEP 01, 2026 12:29 AM

continue.

Before moving fully into data preparation, please add a couple of professional exploratory charts on the raw data now, as if presenting to an audience — specifically the orbital-period (log scale) versus stellar-temperature scatter plot you identified as central to the hypothesis, colored by Flux_Zone where known and faceted by Discovery Method, plus a bar chart of missingness by column. This gives us a visual baseline before any transformation.

As you proceed with the rest of data preparation, please critique your planned imputation strategy before applying it, the same way you'd critique it as a reviewer: go column by column and justify whether you will impute, flag-and-impute with a missingness indicator, or drop the column entirely for certain models — especially given several fields exceed 50-90% missingness (Insolation Flux 91.9%, Equilibrium Temperature 79.8%, Spectral Type 79.7%, Eccentricity 62.7%, Mass 56.2%, Semi-Major Axis 39.6%). Consider whether some of these are better left out of the leakage-safe astrophysical feature set rather than imputed, and how each choice interacts with the host-grouped validation strategy you already defined.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test the moderate-period × cooler-star relationship, and distinguish the real astrophysical pattern from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow. Explain the mindmap of where we are and what's next at the end of the chunk.

CHATGPT · SEP 01, 2026 12:29 AM

I'll establish the raw visual baseline first, then audit each high-missingness field and adopt conservative, model-specific preprocessing while keeping reported and derived flux analyses separate.

CHATGPT · SEP 01, 2026 12:32 AM

The raw audit confirms missingness is structurally method-dependent; discovery method will therefore enter only bias-aware model variants, not the primary astrophysical specification.

Exoplanet Habitability Drivers

Chunk 2 of 7 — Raw Visual Baseline, Data-Preparation Review, and Feature Engineering

All counts and plots in this chunk were computed directly from the uploaded

`all_exoplanets_2021.csv`. [filecite turn2file0L1-L4](#)

The main principle for this phase is:

> Missing values will not be filled merely because an algorithm requires a rectangular matrix. Every treatment must be justified scientifically, statistically, and with respect to discovery-method bias.

1. Raw-data visual baseline

1.1 Orbital period versus stellar temperature

The figure below uses:

- orbital period on a logarithmic scale;
- the original, untransformed stellar effective temperature;
- gray for planets without reported insolation;
- blue, green, and orange for too-cold, HZ-candidate, and too-hot planets;
- separate panels for Transit, Radial Velocity, Imaging, Microlensing, and the remaining sparse methods.

The seven smallest methods were combined into `Other methods` to avoid several nearly empty panels. The three focal methods—transit, radial velocity, and imaging—remain

separate.

![Raw orbital period versus stellar temperature by discovery method]

(sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.png)

[High-resolution PDF of the period–temperature chart]

(sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.pdf)

What the figure shows

The number of planets that can actually enter this plot differs substantially by discovery method:

Discovery-method panel	Total catalog rows	Rows with period and T_{eff}	
Rows also having reported flux			
--- --- --- ---			
Transit	3,444	3,356	306
Radial velocity	899	810	60
Imaging	54	5	0
Microlensing	120	1	0
Other methods	58	23	0

The blankness of the imaging and microlensing panels is itself an important result. It does not imply that those methods find no HZ planets. It means this particular table rarely supplies the combination of period, stellar temperature, and reported flux needed to place those planets in the same empirical comparison.

Preliminary support for the period–temperature interaction

The green HZ points do not occupy one universal “moderate period” interval.

Instead, they approximately form a sloping region:

- around cooler stars, some HZ candidates have periods of days to tens of days;
- around warmer stars, HZ candidates generally require longer periods;
- at a similar stellar temperature, short-period planets tend to fall on the too-hot side;
- sufficiently long-period planets move toward the too-cold side.

That pattern is qualitatively consistent with the expected interaction between period and stellar luminosity.

It also shows why the hypothesis should not be modeled as:

$$\text{HZ} \sim \beta_0 + \beta_1 P + \beta_2 T_{\text{eff}}$$

A more appropriate specification must allow curvature and interaction, for example:

$$\begin{aligned} \text{operatorname{logit}}\{P(\text{HZ}=1)\} \\ = \\ \beta_0 \\ + \beta_1 (\log P) \\ + \beta_2 (T_{\text{eff}}) \\ + \beta_3 (\log P, T_{\text{eff}}) \\ + \dots \end{aligned}$$

The chart alone is not sufficient evidence because the colored subset is highly selected:

- only 370 planets have reported flux;
- 366 have both reported flux and the two plotted variables;
- all flux labels come from transit or radial-velocity discoveries;
- transit discoveries dominate the short-period region.

The displayed temperature window is 1,500–12,000 K for readability. Ten complete rows with exotic host temperatures lie outside the visual window; none have been removed from the data.

1.2 Missingness by raw column

This figure contains only the original 23 source columns. No engineered target or helper field is included.

![Raw-data missingness by column]

(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.png)

[High-resolution PDF of the missingness chart]

(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.pdf)

The main lesson is not merely that some percentages are large. It is that the missingness follows the observation process:

- insolation and equilibrium temperature are available mainly for selected, well-characterized systems;
- planet mass and eccentricity are far more common for radial-velocity planets than for transit-only planets;
- stellar characterization is sparse for many microlensing and imaging discoveries;
- discovery method therefore predicts which fields are missing.

This is incompatible with an uncritical “missing completely at random” assumption.

2. Reviewer’s critique of the imputation plan

2.1 The naive plan I would reject

A common preprocessing recipe would be:

1. median-impute every numeric field;
2. assign an `Unknown` category to missing categorical fields;
3. fit a classifier on the resulting complete matrix.

That plan would be poor for this dataset.

Problem 1: missingness is method-dependent

A median-imputed planet mass for a transit planet is not an innocuous estimate. It substitutes a typical mass from the characterized sample into a population whose masses are often absent because follow-up radial-velocity measurements were unavailable or unsuccessful.

The same issue applies to eccentricity, spectral type, and equilibrium temperature.

Problem 2: some fields are target proxies

`Equilibrium Temperature` is closely related to incident flux. Imputing it and feeding it to an HZ classifier would not create useful new information; it would introduce a noisy, partly reconstructed version of the target.

Problem 3: median values can be physically misleading

Examples include:

- imputing eccentricity as zero or near zero, which silently asserts a circular orbit;
- imputing stellar radius with a main-sequence median for a giant or compact host;
- imputing semi-major axis with a catalog-wide median despite orbital periods spanning roughly ten orders of magnitude;
- imputing planet mass across transit, RV, imaging, and microlensing populations whose measurement processes differ.

Problem 4: global missingness is not the same as modeling-cohort missingness

The reported-flux cohort is a strongly selected subset. Within its 370 rows:

Field	Missing in full catalog	Missing among reported-flux rows
---	---:	---:
Orbital period	3.5%	0.0%
Semi-major axis	39.6%	7.0%
Planet mass	56.2%	34.3%
Eccentricity	62.7%	31.4%
Equilibrium temperature	79.8%	36.2%
Spectral type	79.7%	53.2%
Stellar effective temperature	7.6%	1.1%
Stellar radius	9.8%	0.0%
Stellar mass	16.0%	0.0%
Stellar metallicity	30.0%	21.6%

The core reported-label model therefore does **not** need broad statistical imputation. Its essential features are almost complete.

Problem 5: imputation before grouped validation would contaminate evaluation

Any learned operation—including medians, scaling constants, category frequencies, or iterative imputers—must be fitted on the training hosts only.

The correct order is:

```
Host-grouped split
↓
Fit preprocessing on training hosts
↓
Apply the fitted preprocessing to validation hosts
↓
Fit and evaluate the model
```

It would be incorrect to impute or standardize the full table before cross-validation.

3. Column-by-column treatment decision

The table below distinguishes:

- the **primary leakage-safe astrophysical model**;
- secondary physical sensitivity analyses;
- observation-bias models.

Percentages are shown as `full catalog / reported-flux cohort`.

| Raw column | Missing | Primary treatment and review rationale |

|---|---:|---|

| `No.` | 0.0% / 0.0% | **Drop.** Administrative row number with no scientific information. |

| `Planet Name` | 0.0% / 0.0% | Retain as an audit key, but **drop as a predictor**. One-hot encoding planet names would memorize records rather than generalize. |

| `Planet Host` | 0.0% / 0.0% | Retain exclusively as the **grouping variable** for train/test splitting. Never use as a predictor. |

| `Num Stars` | 0.0% / 0.0% | No imputation required. Exclude from the minimal period-temperature model; consider in an enriched system-architecture sensitivity model. |

| `Num Planets` | 0.0% / 0.0% | No imputation. Treat mainly as an **observation-process/context variable**, since catalog multiplicity depends on survey completeness and confirmation history. |

| `Discovery Method` | 0.0% / 0.0% | Exclude from the core astrophysical model. One-hot encode inside each fold for bias-adjusted and method-interaction models. |

| `Discovery Year` | 0.0% / 0.0% | Exclude from the core model. Include in the bias model because follow-up capabilities and target populations have changed over time. Center and scale within folds. |

| `Discovery Facility` | 0.0% / 0.0% | **Drop from principal models.** Sixty-three levels are too sparse for 370 labeled rows and could cause mission-specific memorization. Use descriptively or in a carefully coarsened sensitivity analysis. |

| `Orbital Period Days` | 3.5% / 0.0% | **Use without imputation** in the reported-label

model. Apply $\log_{10}$ because of the enormous range. Full-catalog rows without period are omitted from period-based analysis rather than statistically filled. |

| Orbit Semi-Major Axis | 39.6% / 7.0% | Exclude from the principal period model to avoid redundancy and preserve the hypothesis test. For physics-derived flux, use the catalog value where available and a Kepler-derived value otherwise. Never median-impute it. |

| Mass | 56.2% / 34.3% | **Exclude from the primary classifier.** Do not globally impute heterogeneous planet masses. Use the 2,006 mass-known planets for an explicitly labeled, mass-characterized clustering analysis. |

| Eccentricity | 62.7% / 31.4% | **Exclude from the primary classifier.** Missing eccentricity must not be replaced with zero. Use observed-only, method-specific sensitivity analyses; a foldwise median-plus-indicator version may be tested only as a predictive robustness check. |

| Insolation Flux | 91.9% / 0.0% by definition | **Target only. Never statistically impute.** Missing flux remains an unknown target. Physics-derived flux is stored as a distinct variable with its own provenance. |

| Equilibrium Temperature | 79.8% / 36.2% | **Drop from leakage-safe HZ classifiers.** It is close to a target proxy and is highly incomplete. Retain only for descriptive analysis or a limited partial-ESI sensitivity exercise. |

| Spectral Type | 79.7% / 53.2% | **Drop from predictive models.** It has inconsistent granularity and is largely redundant with measured effective temperature. Do not infer a supposedly observed spectral class from temperature. |

| Stellar Effective Temperature | 7.6% / 1.1% | **Use without statistical imputation.** The four labeled missing rows are V1298 Tau b–e, all belonging to one host. Dropping that host leaves every reported HZ positive intact. |

| Stellar Radius | 9.8% / 0.0% | **Use without imputation** in the reported-label model. Do not median-fill it for flux construction because luminosity depends on $(R_{\text{star}})^2$. |

| Stellar Mass | 16.0% / 0.0% | **Use without imputation** in the reported-label model. It supports the period-to-semi-major-axis conversion through Kepler's law. |

| Stellar Metallicity | 30.0% / 21.6% | Exclude from the minimal core. Compare two enriched sensitivities: a complete-case model and a foldwise median-plus-missing-indicator model. Any missingness importance must be interpreted as possible observation bias. |

| Stellar Metallicity Ratio | 30.2% / 21.6% | Correct notation errors for auditing, but **drop from models.** It describes the abundance convention rather than an independent physical quantity. |

| Stellar Surface Gravity | 13.2% / 15.9% | Drop the measured field from the primary model because it is strongly redundant with stellar mass and radius. Calculate a consistency-check value from (M_{star}) and (R_{star}) ; do not impute the measured field. |

| Distance | 2.3% / 4.9% | Exclude from the astrophysical model. In the bias model, use $\log_{10}(\text{distance})$, foldwise median imputation, and a missingness indicator. |

| Gaia Magnitude | 3.8% / 2.2% | Exclude from the astrophysical model. Include in the bias model with foldwise median imputation, a missingness indicator, and foldwise scaling. |

The complete machine-readable version of this review is available here:

[Preprocessing decision register]

(sandbox:/mnt/data/exoplanet_habitability/preprocessing_decision_register.csv)

4. A note on host-level filling

One tempting strategy is to copy a stellar value from another planet orbiting the same host.

For example, if one planet row lacks stellar temperature but another row for the same host contains it, the value could be propagated within the system.

I am not adopting that as the default strategy for three reasons.

First, it offers little benefit to the primary labeled model: the four labeled planets missing stellar temperature all orbit V1298 Tau, and no sibling row provides the missing temperature.

Second, the composite catalog contains host-level inconsistencies:

| Stellar field | Hosts having more than one distinct non-null value |

|---|---:|

| Effective temperature | 90 |

| Stellar radius | 98 |

| Stellar mass | 72 |

| Metallicity | 47 |

| Surface gravity | 86 |

These differences can arise because different planet rows use parameter values selected from different literature references.

Third, copying a value from a sibling row creates an appearance of measurement certainty that the row itself does not possess.

Host-level coalescing could be tested as a sensitivity analysis. Because all planets from a host remain in the same fold, it would not cross the train-validation boundary. It is nevertheless not sufficiently clean to be the baseline.

5. Cleaning applied in the prepared master table

5.1 Integrity checks

The following checks found no violations among non-missing values:

- orbital period greater than zero;
- semi-major axis greater than zero;
- planet mass greater than zero;
- $0 \leq e < 1$;
- insolation greater than zero;
- equilibrium temperature greater than zero;
- stellar temperature, radius, and mass greater than zero;
- distance greater than zero.

There are also:

- zero duplicated full rows;
- zero duplicated planet names;
- one malformed metallicity-ratio label, `[Fe/H]`, for which a cleaned parallel value of `[Fe/H]` was created.

The original raw columns remain preserved.

5.2 No statistical imputation has been applied

The prepared master table retains missing values.

This is deliberate. Statistical imputation belongs inside a fitted modeling pipeline, not in the permanent cleaned dataset.

The only gap-filling operation applied is a **physics-based orbital derivation**, stored with explicit provenance.

6. Feature engineering completed

6.1 Reported-flux targets

Two target representations were created.

Binary target

```
\[
HZ_{\mathrm{reported}} =
\begin{cases}
1, & 0.35 \leq S_{\mathrm{reported}} \leq 1.75 \\
0, & S_{\mathrm{reported}} < 0.35 \text{ or } S_{\mathrm{reported}} > 1.75 \\
\mathrm{missing}, & S_{\mathrm{reported}} \text{ unavailable}
\end{cases}
\]
```

Three-zone target

too_cold	$S < 0.35$
habitable_zone	$0.35 \leq S \leq 1.75$
too_hot	$S > 1.75$
unknown	S is missing

The three-zone version will be important for exploratory work. A binary non-HZ label combines two physically opposite populations, so a monotonic relationship with period should not be expected.

6.2 Logarithmic variables

The following deterministic transformations were created:

$$\log_{10}(P \text{ in days})$$

$$\log_{10}(a \text{ in AU})$$

$$\log_{10}(M_p/M_{\oplus})$$

$$\log_{10}(S/S_{\oplus})$$

For orbital period, the log transformation is essential: observed periods range from approximately 0.09 days to (4.02×10^8) days.

No capping or winsorization was applied.

6.3 Stellar luminosity estimate

Where stellar radius and temperature are available:

$$\begin{aligned} & \left[\frac{L_{\text{star}}}{L_{\odot}} \right] \\ &= \\ & \left(\frac{R_{\text{star}}}{R_{\odot}} \right)^2 \\ & \left(\frac{T_{\text{eff}}}{5772 \text{ K}} \right)^4 \end{aligned}$$

This estimate is available for 4,024 rows.

It will be used for:

- physical consistency checks;
- derived-flux sensitivity analysis;
- visualizing the period required for Earth-like irradiation.

It will not be included alongside both stellar radius and temperature in the main coefficient model, because that would create deterministic multicollinearity.

6.4 Kepler-derived semi-major axis

Where period and stellar mass are available:

$$\begin{aligned} & \left[a_{\text{Kep}} \right] \\ &= \end{aligned}$$

$$\left[\frac{M_{\text{star}}}{M_{\text{Jup}}} \left(\frac{P}{365.25 \text{ days}} \right)^2 \right]^{1/3} \text{AU}$$

The prepared table contains:

- catalog semi-major axis;
- Kepler-derived semi-major axis;
- best available semi-major axis;
- a source flag identifying whether the catalog or derived value was used.

This is preferable to generic imputation because it respects the physical scale of the system.

Empirical consistency check

Among 1,910 planets having both a catalog semi-major axis and sufficient information for a Kepler estimate:

- the median ratio $a_{\text{Kep}}/a_{\text{catalog}}$ is approximately 0.9999;
- 97.4% agree within 20%.

The remaining discrepancies may reflect rounding, parameter choices, multiple-star systems, unusual objects, or heterogeneous literature values.

6.5 Physics-derived insolation

The derived estimate is:

$$S_{\text{derived}}$$

$$= \frac{L_{\text{star}}/L_{\odot}}{(a_{\text{best}}/\text{AU})^2}$$

It is available for **4,013 planets**:

- 2,319 use a catalog semi-major axis;
- 1,694 use the Kepler-derived semi-major axis;
- 562 remain unavailable.

A parallel `HZ_derived` flag and three-zone class were created.

Comparison with reported insolation

There are 366 planets with both reported and derived flux:

- correlation between their logarithms: approximately 0.9989;
- median derived-to-reported flux ratio: approximately 1.0035;
- reported versus derived HZ-zone agreement: 363 of 366, or 99.2%.

The three classification disagreements are:

- `GJ 3323 b` ;
- `Kepler-1704 b` ;
- `Kepler-436 b` .

All lie near an HZ boundary or show a notable discrepancy among the component catalog parameters.

This agreement is a validation of the data-preparation equation—not a machine-learning result. The archive’s reported flux may itself have been calculated from similar quantities, so the comparison is partly circular.

6.6 Earth-equivalent period

The orbital period corresponding approximately to one Earth flux was calculated as:

$$\begin{aligned} & P_{\oplus} \\ &= \\ & 365.25 \\ & \frac{L_{\star}^{3/4}}{M_{\star}^{1/2}} \\ & \text{days} \end{aligned}$$

We then define:

$$\begin{aligned} & P_{\text{relative}} \\ &= \\ & \frac{P_{\text{observed}}}{P_{\oplus}} \end{aligned}$$

This produces a physically meaningful interpretation of “moderate period”:

- $P_{\text{relative}} \ll 1$: the planet orbits much faster and closer than the star’s Earth-equivalent orbit;
- $P_{\text{relative}} \approx 1$: the period is near the Earth-equivalent irradiation orbit;
- $P_{\text{relative}} \gg 1$: the planet is farther out and likely more weakly irradiated.

This feature is valuable as a physics benchmark, but it will **not** be used in the principal feature-importance model. It nearly collapses period, temperature, radius, and stellar mass into the target-defining flux equation and would make the feature-ranking exercise tautological.

6.7 Calculated stellar surface gravity

For quality checking:

$$\begin{aligned} & \backslash[\\ & \backslash \log g_{\mathrm{calc}} \\ & = \\ & 4.438 + \\ & \backslash \log_{10}(M_{\mathrm{star}}/M_{\odot}) \\ & - \\ & 2 \backslash \log_{10}(R_{\mathrm{star}}/R_{\odot}) \\ & \backslash] \end{aligned}$$

This field helps identify inconsistencies and unusual stellar hosts.

It is not an additional independent predictor if stellar mass and radius are already in the model.

6.8 Missingness indicators

Indicators were created for fields with missing values, but creating an indicator does not automatically authorize it as a predictor.

They are assigned primarily to the **observation-process model**, where they can answer questions such as:

> Does a planet's probability of receiving an HZ label depend on whether mass, eccentricity, or detailed stellar characterization was available?

An importance score for such an indicator would be evidence of catalog characterization bias, not evidence of an astrophysical habitability driver.

7. Final feature-set architecture

7.1 Core astrophysical interaction model

The principal interpretive model will use:

- $\log_{10}$ orbital period;
- stellar effective temperature;
- stellar radius;
- stellar mass;
- a period $\times$ temperature interaction;
- nonlinear terms or splines for period and temperature.

This set directly tests the stated hypothesis without requiring high-missingness fields.

The minimal form will be compared with variants containing:

- period squared or spline terms;
- temperature squared or spline terms;
- interactions of period with stellar radius or luminosity.

7.2 Enriched astrophysical sensitivity model

Candidate additions are:

- stellar metallicity;
- number of stellar components;
- eccentricity in an observed-only or method-specific analysis;
- planet mass in a mass-characterized subset.

These features will not be allowed to redefine the primary result because their availability is selective.

7.3 Bias-augmented model

The observation-process model will add:

- discovery method;
- discovery year;
- log distance;
- Gaia magnitude;
- detected system multiplicity;
- selected missingness indicators.

Comparing core and bias-augmented models will help distinguish:

- a physically expected period–temperature boundary;
- changes in apparent HZ probability caused by which planets were detectable and sufficiently characterized.

7.4 Explicitly excluded leakage or near-leakage fields

The principal HZ classifier will not use:

- reported insolation flux;
- derived insolation flux;
- equilibrium temperature;
- `HZ_reported` or `HZ_derived` ;
- Earth-equivalent period ratio;
- flux-zone labels.

Those variables are targets, near-target proxies, or deterministic physical summaries of the target.

8. Modeling cohorts after preparation

	Analysis cohort	Rows	Unique hosts	HZ-positive rows	HZ-positive hosts	Transit	RV	Imaging	Microlensing
	---	---	---	---	---	---	---	---	---
All reported-flux labels	370	265	32	26	310	60	0	0	
Core reported model	366	264	32	26	306	60	0	0	
Complete-case enriched model with metallicity	290	209	27	23	231	59	0	0	
Derived-flux sensitivity cohort	4,013	3,018	186	177	3,339	645	10	0	

8.1 Why the 366-row core cohort is preferable

Dropping the four rows missing stellar temperature:

- removes only 1.1% of the labeled sample;
- removes zero HZ positives;
- removes one host group, not four statistically independent hosts;
- avoids inventing the central explanatory variable;
- leaves stellar radius, stellar mass, and period completely observed.

This is substantially more defensible than imputation.

8.2 Why the derived cohort is useful but secondary

The derived cohort expands coverage from 370 to 4,013 planets, but its HZ flag is constructed from the same physical predictors we intend to study.

A high-performing model on that target would largely be recovering the equation used to construct it.

The derived cohort is therefore appropriate for:

- visualizing the physically expected HZ surface;
- testing whether reported-flux findings are geometrically consistent;
- broad descriptive comparisons of transit and RV populations;
- limited examination of imaging systems with sufficient stellar data.

It is not appropriate as the sole evidence for feature importance.

8.3 Imaging remains poorly identified

Only 10 imaging discoveries receive a derived flux estimate, and no imaging planet has a reported-flux label.

Microlensing receives no derived flux estimate because the required stellar quantities are absent.

Consequently, direct statistical claims about the imaging HZ relationship will remain weak. The honest result may be:

> The catalog demonstrates strong method-dependent coverage, but it does not contain enough consistently characterized imaging planets to estimate a comparable imaging-specific HZ model.

That limitation is itself part of the detection-bias conclusion.

9. Implications for host-grouped validation

The modeling pipeline will use host-level groups:

```
groups = df["planet_host"]
```

A suitable split is a `StratifiedGroupKFold`, with folds checked manually for:

- HZ-positive host counts;
- transit and radial-velocity representation;
- no host appearing in more than one fold;

- reasonable class balance.

Because only 26 host systems contain reported HZ positives, five folds would give only about five positive hosts per validation fold. Metrics will therefore have meaningful uncertainty.

All learned preprocessing will be nested inside the model pipeline:

```
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler

numeric_bias_pipeline = Pipeline(
    steps=[
        (
            "imputer",
            SimpleImputer(
                strategy="median",
                add_indicator=True,
            ),
        ),
        ("scaler", StandardScaler()),
    ]
)

categorical_bias_pipeline = Pipeline(
    steps=[
        (
            "imputer",
            SimpleImputer(strategy="most_frequent"),
        ),
        (
            "encoder",
            OneHotEncoder(
                handle_unknown="ignore",
                min_frequency=5,
            ),
        ),
    ]
)
```

This imputation pipeline will be used only for approved bias or sensitivity fields. The core physical predictors require no imputer after the four incomplete rows are removed.

10. Earth Similarity Index decision

A conventional Earth Similarity Index cannot be calculated from this table because planet radius is absent.

Moreover:

- planet mass is missing for 56.2% of the catalog;
- equilibrium temperature is missing for 79.8%;
- both fields are strongly method-selected.

I will therefore not invent a full ESI by globally imputing radius, mass, and temperature.

A later sensitivity analysis may calculate a clearly labeled **partial similarity score** among planets with observed mass and equilibrium temperature. It will not be presented as a conventional ESI and will not replace the insolation-based HZ target.

This also affects clustering: without planet radius, labels such as “super-Earth” and “sub-Neptune” can only be approximate when based on mass, period, and irradiation. The cluster interpretation will explicitly state that limitation.

11. Reproducibility artifacts

- [Prepared dataset with engineered fields and no statistical imputation]
(sandbox:/mnt/data/exoplanet_habitability/exoplanets_prepared_no_statistical_imputation.csv
)

- [Complete preparation script]
(sandbox:/mnt/data/exoplanet_habitability/prepare_exoplanets.py)
- [Modeling cohort sizes]
(sandbox:/mnt/data/exoplanet_habitability/modeling_cohort_sizes.csv)
- [Raw missingness summary]
(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_summary.csv)
- [Raw scatter-panel summary]
(sandbox:/mnt/data/exoplanet_habitability/raw_scatter_panel_summary.csv)

12. CRISP-DM process mindmap

Exoplanet habitability drivers

	└─ 1. Business understanding	COMPLETE
	└─ HZ candidate definition	
	└─ Moderate-period × cooler-star hypothesis	
	└─ Detection-bias question	
	└─ Conditional catalog estimand	
	└─ 2. Data understanding	COMPLETE
	└─ 4,575 planets × 23 raw fields	
	└─ Units and field meanings	
	└─ Overall and method-specific missingness	
	└─ Target availability	
	└─ Class and host imbalance	
	└─ 2A. Raw visual baseline	COMPLETE
	└─ Period-temperature scatter	
	└─ HZ status overlay	
	└─ Discovery-method facets	
	└─ Missingness chart	
	└─ 3. Data preparation	COMPLETE
	└─ Integrity and physical-bound checks	
	└─ Category normalization	
	└─ No global statistical imputation	
	└─ Host-grouped preprocessing policy	
	└─ Leakage-safe feature architecture	
	└─ 3A. Feature engineering	COMPLETE
	└─ Reported HZ and three-zone targets	
	└─ Log orbital variables	
	└─ Stellar luminosity	
	└─ Kepler-derived semi-major axis	
	└─ Physics-derived flux	
	└─ Earth-equivalent period	
	└─ Provenance and missingness indicators	
	└─ 4. Feature selection and outlier analysis	NEXT
	└─ Distributional and correlation diagnostics	
	└─ Collinearity and target-proxy audit	
	└─ Main-sequence, giant, and compact-host outliers	
	└─ Period, mass, and separation extremes	

	└─ Method-aware inclusion rules	
	└─ 5. Clustering	NEXT
	└─ Define an observed mass-characterized cohort	
	└─ Robust scaling and log features	
	└─ Select cluster number	
	└─ Approximate hot-Jupiter/super-Earth/sub-Neptune groups	
	└─ Measure HZ overlap by cluster and method	
	└─ 6. Classification and regression	
	└─ Naive baselines	
	└─ Interaction logistic model	
	└─ Random forest and gradient boosting	
	└─ Flux regression	
	└─ Host-grouped evaluation	
	└─ 7. Interpretation and recommendations	
	└─ Coefficients, permutation importance, and SHAP	
	└─ Period × temperature response surface	
	└─ Transit versus RV versus imaging bias	
	└─ Astrophysical versus catalog conclusions	
	└─ General-audience recommendation	

Next chunk

Chunk 3 will address feature selection, multicollinearity, outlier analysis, and method-aware clustering. It will determine which extreme values are astrophysically meaningful rather than erroneous, define the clusterable mass-characterized sample, compare candidate cluster counts, and examine which approximate planet group overlaps most with the reported and derived HZ.

Standing requirement carried forward: determine which stellar and orbital features predict habitable-zone membership, test the moderate-orbital-period × cooler-star relationship, and distinguish the real astrophysical pattern from transit, radial-velocity, and imaging selection bias using a staged, limited-compute CRISP-DM workflow.

Reply **continue** for Chunk 3.

PROMPT · SEP 01, 2026 08:22 AM

continue.

As you work through feature selection, outlier analysis, and clustering in this chunk, please keep producing professional, presentation-quality charts the way you did for the period-temperature panel — specifically a correlation/collinearity heatmap for the candidate core features, a boxplot or scatter highlighting the physically meaningful outliers you flag (giant hosts, compact remnants, ultra-short-period planets) before deciding whether to keep or separately annotate them, and a chart showing cluster assignments overlaid with HZ_reported and HZ_derived status. These will likely be reused directly in the Medium article, so treat them as final-quality, not draft.

Given the core reported-label cohort has only 26 HZ-positive hosts, please be explicit about statistical power throughout this chunk: use bootstrap or permutation-based confidence intervals rather than naive p-values wherever you test feature relevance or cluster-HZ overlap, and flag any result that should be read as suggestive rather than conclusive because of small-sample uncertainty.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test the moderate-orbital-period $\times$ cooler-star relationship, and distinguish the real astrophysical pattern from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow. Explain the mindmap of where we are and what's next at the end of the chunk.

CHATGPT · SEP 01, 2026 08:22 AM

I'll combine a leakage/collinearity audit, physics-aware outlier mapping, and method-aware clustering, using host-level bootstrap/permutation uncertainty while keeping reported-label inference separate from derived-flux sensitivity.

CHATGPT · SEP 01, 2026 08:25 AM

I'll also produce publication-ready PNG/PDF figures and carry the resulting feature, outlier, and clustering decisions into the CRISP-DM decision log.

CHATGPT · SEP 01, 2026 08:34 AM

The diagnostics show strong stellar collinearity and period–semi-major-axis redundancy; identified extreme populations do not drive the core result. The period–temperature interaction remains specification-sensitive and suggestive.

Exoplanet Habitability Drivers

Chunk 3 of 7 — Feature Selection, Physics-Aware Outlier Analysis, and Clustering

All statistics and figures in this chunk were recomputed from the uploaded

`all_exoplanets_2021.csv`. [filecite turn2file0 L1-L4](#)

Executive findings

This stage produced four main conclusions.

First, **orbital period carries information that is largely independent of the stellar measurements**, but stellar temperature, radius, and mass are extremely collinear. We should retain those physical components in regularized models while avoiding naive interpretation of their individual coefficients.

Second, the reported-label subset continues to support the hypothesis that HZ candidates tend to have **longer periods around cooler, smaller, lower-mass stars**. However, the period separation is strong in the transit subset and statistically unresolved in the much smaller radial-velocity subset.

Third, the extreme rows are overwhelmingly **real astrophysical subpopulations**, not generic data-entry errors. Giant stars, compact remnants, hot subdwarfs, ultra-short-period planets, high-mass companions, and extremely wide imaged objects should be flagged and handled through sensitivity analysis rather than indiscriminately deleted or winsorized.

Fourth, mass–period clustering clearly isolates hot/warm giants from longer-period systems. The hot/warm-giant cluster has essentially no HZ overlap. However, the reported labels are too sparse to decide whether an intermediate-mass, moderate-period cluster or a long-period giant cluster has the highest true HZ rate.

1. Feature selection and collinearity

1.1 Analysis design

The collinearity analysis uses the 366-planet reported-label core cohort:

- 366 planets;
- 264 host systems;
- 32 HZ-positive planets;
- 26 HZ-positive hosts.

The following transformed candidate variables were examined:

$$\begin{aligned} & \log_{10}(P), \quad \\ & \log_{10}(T_{\mathrm{eff}}), \quad \\ & \log_{10}(R_{\star}), \quad \\ & \log_{10}(M_{\star}) \end{aligned}$$

Two derived alternatives were included in the audit:

$$\log_{10}(L_{\star})$$

and

$$\log g_{\mathrm{calculated}}.$$

To avoid multi-planet systems dominating the correlations, each host was given a total statistical weight of one. A host with four planets therefore contributed one-quarter weight per planet.

The heatmap reports a host-balanced Spearman correlation: the variables are ranked first, and the correlations between the ranks are calculated using the host weights.

![[Host-balanced correlation and VIF audit]

(sandbox:/mnt/data/exoplanet_habitability/core_feature_collinearity_heatmap.png)

[Vector PDF of the correlation and VIF figure]

(sandbox:/mnt/data/exoplanet_habitability/core_feature_collinearity_heatmap.pdf)

1.2 What the heatmap shows

Orbital period is almost uncorrelated with the stellar variables in the core cohort:

| Pair | Host-balanced Spearman correlation |

|---|---:|

| Period–temperature | 0.00 |

| Period–radius | 0.00 |

| Period–stellar mass | –0.02 |

| Period–luminosity | 0.01 |

This does **not** mean orbital period is irrelevant to HZ membership.

It means there is no simple marginal relationship between period and stellar properties across the selected catalog. HZ membership depends on their **combination**. For example, a 30-day period may be highly irradiated around one star and approximately temperate around another.

The stellar properties, in contrast, are strongly correlated:

| Pair | Host-balanced Spearman correlation |

|---|---:|

| Temperature–stellar mass | 0.98 |

| Radius–stellar mass | 0.98 |

| Temperature–radius | 0.95 |

| Radius–luminosity | 0.99 |

| Radius-calculated $\log g$ | -0.99 |

This is astrophysically expected for a sample dominated by ordinary main-sequence stars: hotter stars tend to be more massive, larger, and more luminous.

1.3 Variance inflation

The host-weighted variance inflation factors for the four base features are:

Candidate feature	VIF	Interpretation
---	---	---
Log orbital period	1.0	No meaningful collinearity
Log stellar radius	5.4	Material collinearity
Log stellar temperature	7.8	High collinearity
Log stellar mass	11.2	Very high collinearity

VIF thresholds such as five or ten are diagnostic conventions rather than automatic deletion rules. Here, blindly dropping stellar variables would discard physically meaningful information.

A more serious issue arises when adding luminosity and calculated surface gravity. In logarithmic form,

$$\begin{aligned} \log L_{\star} &= \\ &2 \log R_{\star} \\ &+ \\ &4 \log T_{\mathrm{eff}} \\ &+ \\ &\mathrm{constant} \end{aligned}$$

and

$\log$

$$\begin{aligned} & \log g \\ & = \\ & \mathrm{constant} \\ & + \\ & \log M_{\star} \\ & - \\ & 2 \log R_{\star}. \end{aligned}$$

Consequently, the six-column feature matrix has rank four rather than six.

Feature-selection decision

We will **not** place these algebraically redundant representations in the same model:

- temperature, radius, and luminosity;
- mass, radius, and calculated surface gravity.

Instead, the next phase will compare two representations.

Component representation

$$\begin{aligned} & \log P, \quad \\ & \log T_{\mathrm{eff}}, \quad \\ & \log R_{\star}, \quad \\ & \log M_{\star} \end{aligned}$$

This is the primary representation because it permits direct testing of the period–temperature interaction.

Physics-compressed sensitivity representation

$$\sqrt{\log P, \log L_{\star}, \log M_{\star}}$$

This is more compact, but it obscures the specific role of temperature and moves closer to reconstructing the flux equation directly.

1.4 Preliminary feature relevance with host-bootstrap uncertainty

For descriptive feature screening, I calculated Cliff’s delta:

$$\Delta = \frac{P(X_{\mathrm{HZ}} > X_{\mathrm{non-HZ}}) - P(X_{\mathrm{HZ}} < X_{\mathrm{non-HZ}})}{1}$$

A value near:

- $+1$ means HZ planets almost always have larger values;
- 0 means little rank separation;
- -1 means HZ planets almost always have smaller values.

Confidence intervals were calculated by resampling entire host systems 4,000 times. No row-level p-values were used.

Discovery-method scope	Feature	Cliff’s delta	Host-bootstrap 95% interval
All methods	Log period	+0.623	+0.453 to +0.793

All methods	Log stellar temperature	-0.579	-0.735 to -0.395
Transit	Log period	$+0.748$	$+0.496$ to $+0.928$
Transit	Log stellar temperature	-0.577	-0.744 to -0.362
Radial velocity	Log period	$+0.123$	-0.197 to $+0.470$
Radial velocity	Log stellar temperature	-0.502	-0.847 to -0.027

Stellar radius and stellar mass have effects similar to temperature:

- pooled radius delta: -0.588 , with 95% interval -0.744 to -0.402 ;
- pooled stellar-mass delta: -0.589 , with 95% interval -0.750 to -0.401 .

Interpretation

The pooled reported-label sample strongly favors:

- longer periods;
- cooler stars;
- smaller stars;
- lower-mass stars.

The same period separation is especially strong among transit discoveries.

For radial velocity, however, the period interval includes zero. That does not demonstrate that period is irrelevant for RV planets. It means the available RV sample—only 60 planets, 13 HZ planets, and 10 positive hosts—cannot estimate the period effect precisely.

The negative temperature direction remains visible in the RV subset, but its interval is wide and comes from a selectively characterized sample.

These are **marginal, unadjusted effects**, not yet the formal period–temperature interaction test.

1.5 Selected feature strategy for modeling

The following hierarchy will be used in Chunk 4.

Model stage	Features
Minimal orbital model	Centered $\log P$
Additive stellar model	$\log P + \log T_{\text{eff}}$
Interaction model	Add $\log P \times \log T_{\text{eff}}$
Nonlinear interaction model	Add low-order period and temperature curvature
Full core physical model	Add $\log R_{\text{star}}$ and $\log M_{\text{star}}$
Bias-augmented model	Add discovery method, year, distance, and magnitude
Method-interaction model	Add period $\times$ method and temperature $\times$ method terms

Because of stellar collinearity, the full linear model will use regularization. We will interpret:

- joint stellar effects;
- response surfaces;
- predicted-probability contrasts;
- grouped permutation importance;

rather than treating one unstable stellar coefficient as the definitive physical explanation.

Stepwise selection based on ordinary p-values is not appropriate with only 26 positive hosts.

2. Physics-aware outlier analysis

2.1 Outlier-screening definitions

The screening rules were set using astrophysical considerations rather than target values.

Giant or evolved-host flag

A row was flagged when at least one of the following applied:

$$\begin{aligned} &[\\ &R_{\text{star}} \geq 3R_{\odot} \\ &] \end{aligned}$$

or

$$\begin{aligned} &[\\ &\log g < 3.5. \\ &] \end{aligned}$$

Compact-remnant or hot-subdwarf flag

This flag used a combination of:

- compact-object host names such as pulsar or white-dwarf systems;
- very high surface gravity;
- hot, unusually small stellar hosts.

It is an analytical screening class, not a definitive stellar taxonomy.

Ultra-short-period flag

$$\begin{aligned} &[\\ &P < 1 \, \mathrm{day}. \\ &] \end{aligned}$$

Two additional cluster-sensitivity flags were created:

- mass above 13 Jupiter masses;
- orbital period above 100,000 days.

![Physics-aware outlier landscape]

(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.png)

[Vector PDF of the outlier figure]

(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.pdf)

2.2 Outlier counts and coverage

Flag	Planets	Hosts	In reported core	Reported HZ positives	Derived HZ positives
--- ---	--- ---	--- ---	---	---	---
Giant/evolved host	192	172	0	0	12
Compact remnant/hot subdwarf	18	12	1	0	1
Ultra-short period	114	111	10	0	0
Above 13 Jupiter masses	74	73	3	0	2
Period above 100,000 days	4	4	0	0	0

The discovery-method composition is also revealing:

- 176 of the 192 giant/evolved-host planets were found by radial velocity;
- 106 of the 114 ultra-short-period planets were found by transit;
- all four extremely wide-period objects were imaging discoveries;
- 16 of the 18 compact or hot-subdwarf systems fall into the sparse **Other methods** category.

These tails therefore encode both astrophysics and detection method.

2.3 Treatment decisions

Giant and evolved hosts

Decision: keep in the master data; annotate separately; exclude in an ordinary-dwarf sensitivity model.

They are not obvious errors. Radial-velocity surveys have detected many companions to evolved stars.

However, an instantaneous flux label does not capture:

- the star's prior luminosity history;
- how quickly its HZ moved during stellar evolution;
- whether a planet remained temperate for a biologically meaningful period.

The reported-label core happens to contain no giant/evolved hosts, so they cannot drive the primary reported-target model. They matter mainly in the larger derived-flux sensitivity analysis.

Compact remnants and hot subdwarfs

Decision: retain for descriptive work but separate from the ordinary-star interaction model.

The core cohort contains only one such planet, WD 1856+534 b , and it is not an HZ candidate under the adopted reported-flux rule.

These systems have qualitatively different evolutionary and irradiation histories. Combining them with ordinary main-sequence stars would ask a single model to represent physically different regimes with almost no examples from the unusual regime.

Ultra-short-period planets

Decision: retain in the primary model and run a sensitivity analysis excluding them.

All ten ultra-short-period planets with reported labels are non-HZ, as are all 100 with derived labels.

They provide legitimate examples of intensely irradiated planets. Deleting them merely because they are extreme would make the training population less representative of transit discoveries.

However, they could make classification appear easier. We will therefore report performance for:

1. the full core cohort;
2. an otherwise identical ≥ 1 -day sensitivity cohort.

Very massive and extremely wide companions

Decision: retain, flag, and use log transformations with robust scaling.

Some very massive objects may be near or beyond the conventional planet–brown-dwarf boundary. The dataset lacks sufficient mass-provenance information to reclassify them definitively.

The four extreme-period observations are imaging systems. They are valuable evidence of imaging's very different selection function and should not be erased by winsorization.

2.4 Why no automatic winsorization was applied

Winsorizing physical tails would convert real systems into artificial values:

- a $70\text{-}(R_{\odot})$ giant would be changed into a less evolved star;
- a 0.1-day planet would be assigned an orbit it does not have;
- a wide imaging companion would be moved inward;
- a high-mass companion would be made less massive.

Instead, the analysis uses:

- logarithmic transformations;
- robust scaling;
- target-blind physical flags;
- all-data versus ordinary-star sensitivity models;
- cluster-stability checks after removing extreme tails.

This preserves scientific meaning while still testing whether the extremes dominate the result.

3. Mass–period clustering

3.1 Purpose and clustering cohort

Clustering addresses a different question from classification:

> What broad planet populations exist in mass–period space, and which of those populations overlap the adopted HZ interval?

The clustering cohort contains:

- 1,850 planets;
- 1,384 hosts;
- 899 radial-velocity planets;
- 886 transit planets;
- 8 imaging planets;
- 8 microlensing planets;
- 49 planets from other methods.

The apparent balance between transit and RV is misleading. Essentially all RV planets have mass measurements, while only about one-quarter of the full transit catalog does. The clustering cohort is therefore a **mass-characterized subset**, not the full exoplanet population.

Only two variables entered the clustering:

$$x_1 = \log_{10}(M_p/M_{\oplus})$$

$$x_2 = \log_{10}(P/\mathrm{days}).$$

They were robustly scaled before K-means fitting.

Neither reported nor derived HZ status was used to create the clusters.

3.2 Selecting the number of clusters

Number of clusters	Silhouette	Median host-bootstrap ARI	95% bootstrap interval
---: ---: ---: ---:			
3	0.563	0.990	0.979–0.998
4	0.504	0.944	0.697–0.993
5	0.435	0.946	0.701–0.985

The three-cluster solution has the strongest purely geometric separation and the greatest stability.

The four-cluster solution was selected as a **scientifically useful descriptive refinement** because it separates:

- a low-mass, short-period population;
- an intermediate-mass, moderate-period population;
- hot/warm giants;
- long-period giants.

This choice is not a claim that four is the uniquely correct number of planet types.

The wider lower bound for the four-cluster adjusted Rand index shows that some bootstrap samples move the boundary between the low-mass and intermediate-mass groups. Those

two labels should therefore be treated as approximate.

The hot/warm-giant and long-period-giant populations are much more visually distinct.

3.3 Sensitivity to extreme observations

| Sensitivity analysis | Removed | ARI versus full solution |

|---|---:|---:|

| Remove outer 0.5% of both log mass and log period | 37 | 0.984 |

| Remove masses $(>13M_{\mathrm{Jup}})$ or periods $(>100{,}000)$ days | 55 | 0.944 |

An adjusted Rand index near one indicates that the cluster assignments for the remaining planets barely changed.

Therefore, the segmentation is not primarily an artifact of the few widest imaging systems or highest-mass companions.

3.4 Cluster profiles

| Descriptive cluster | Planets | Median mass, $(M_{\oplus})$ [IQR] | Median period, days [IQR] |

Transit share | RV share |

|---|---:|---:|---:|---:|---:|

| Low-mass, short-period | 441 | 6.49 [3.70–10.44] | 8.04 [4.05–14.03] | 59.4% | 39.0% |

| Intermediate-mass, moderate-period | 255 | 22.5 [13.1–53.3] | 57.3 [34.3–130.1] | 39.2% | 56.1% |

| Hot/warm giants | 589 | 305 [178–607] | 3.65 [2.70–5.21] | 86.1% | 12.2% |

| Long-period giants | 565 | 826 [451–2,000] | 871 [416–1,928] | 3.0% | 90.6% |

The low-mass cluster plausibly contains both super-Earths and sub-Neptunes. Those classes cannot be separated reliably here because planet radius is absent.

Similarly, the intermediate cluster includes a mixture of Neptune-like, sub-Saturn-like, and other moderate-mass systems. The labels are intentionally descriptive rather than taxonomic.

4. Cluster assignments and HZ overlap

![Mass-period clusters with reported and derived HZ overlays]

(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.png)

[Vector PDF of the clustering and HZ-overlap figure]

(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.pdf)

In the left panel:

- point color represents cluster assignment;
- an open black ring marks a derived-flux HZ candidate;
- a green star marks a reported-flux HZ candidate.

The right panel reports HZ proportions with uncertainty.

The confidence intervals resample entire hosts rather than individual planets. For zero-event cells, an ordinary nonparametric bootstrap would produce a misleading zero-width interval, so a Jeffreys interval is shown instead.

4.1 Statistical power for cluster overlap

The clustering cohort is large, but the reported-label overlap cohort is not:

HZ target	Eligible clustered planets	HZ planets	HZ-positive hosts
-----------	----------------------------	------------	-------------------

--- ---: ---: ---:

Reported flux	243	18	13
---------------	-----	----	----

Derived flux	1,490	120	113
--------------	-------	-----	-----

The mass requirement reduces the reported HZ sample from 32 to only 18 planets and from 26 to 13 positive hosts.

This means cluster-reported-HZ comparisons have substantially less statistical power than the forthcoming core classifier.

4.2 HZ overlap estimates

Cluster	Reported HZ rate, 95% interval	Derived HZ rate, 95% interval
Low-mass, short-period	11/126 = 8.7% [2.8%–15.3%]	13/348 = 3.7% [1.4%–6.3%]
Intermediate-mass, moderate-period	5/25 = 20.0% [4.2%–37.9%]	23/153 = 15.0% [9.3%–21.3%]
Hot/warm giants	0/81 = 0.0% [0.0%–3.0%]	0/557 = 0.0% [0.0%–0.45%]
Long-period giants	2/11 = 18.2% [0.0%–44.4%]	84/432 = 19.4% [15.9%–23.3%]

Reported-label conclusion

The highest reported point estimate belongs to the intermediate-mass, moderate-period cluster, at 20.0%.

However:

- it is based on only five positive planets;
- the long-period-giant estimate is based on only two;
- the confidence intervals overlap broadly.

Across the host-bootstrap replicates, the probability of having the highest reported HZ rate was:

- 53.3% for the intermediate-mass, moderate-period cluster;
- 43.2% for the long-period-giant cluster;

- 3.5% for the low-mass, short-period cluster.

That is not decisive evidence for a unique winning cluster.

The defensible reported-label conclusion is:

> Hot, short-period giants are strongly depleted in the HZ, while the available data cannot precisely rank the two longer-period clusters.

Derived-label conclusion

The long-period-giant cluster has the highest derived HZ rate in approximately 89.6% of host-bootstrap samples.

This result is much more precise, but it is not independent predictive evidence:

- orbital period enters the clustering;
- orbital period also enters the calculation of derived flux;
- the long-period cluster is overwhelmingly composed of radial-velocity planets.

It is best understood as a **physical consistency check**.

4.3 HZ giant does not mean Earth-like

The largest derived HZ overlap occurs among long-period giant planets.

That does not imply that those planets are the most promising Earth analogues. They are generally hundreds of Earth masses and are likely gas giants.

An HZ flag means only that the orbit receives a selected range of stellar flux. It does not establish:

- a solid surface;

- Earth-like gravity;
- a suitable atmosphere;
- stable liquid surface water;
- an Earth-like climate.

The cluster result therefore separates two questions:

1. Which orbits receive temperate flux?
2. Which planets have terrestrial properties?

This dataset addresses the first question much better than the second.

5. Separating astrophysics from detection bias

5.1 Cluster composition reveals strong method selection

The two most distinct clusters are almost method-specific:

- **86.1% of hot/warm giants are transit discoveries;**
- **90.6% of long-period giants are radial-velocity discoveries.**

Thus, a pooled cluster comparison partly compares discovery methods.

The cluster rate cannot be interpreted as a purely intrinsic planet-type effect unless method is controlled.

5.2 Within-method HZ comparisons

The larger derived-label sample illustrates how the apparent cluster relationship changes

within discovery method.

Transit-derived HZ rates

Cluster	HZ rate
Low-mass, short-period	1.7%
Intermediate-mass, moderate-period	10.7%
Hot/warm giants	0.0%
Long-period giants	11.8%, based on only 17 planets

Radial-velocity-derived HZ rates

Cluster	HZ rate
Low-mass, short-period	8.4%
Intermediate-mass, moderate-period	20.3%
Hot/warm giants	0.0%
Long-period giants	19.7%

Within radial velocity, the intermediate and long-period clusters have almost identical derived HZ rates:

$20.3\% \quad \text{versus} \quad 19.7\%$

Their larger pooled difference is therefore partly caused by discovery-method composition rather than an intrinsic cluster effect.

This is an important preview of the formal bias analysis.

5.3 Imaging and microlensing remain underidentified

There are only eight imaging planets in the mass–period clustering cohort.

Of those:

- three have a derived-flux label;
- none has a reported-flux label;
- none of the three derived-labeled planets is an HZ candidate.

Three observations are insufficient to estimate an imaging-specific HZ rate.

Microlensing contributes eight clusterable planets but lacks the stellar information needed for the derived-flux comparison.

Therefore:

> The imaging and microlensing panels are scientifically valuable demonstrations of missing support, but they cannot currently provide comparable HZ-effect estimates.

A model that reports a precise imaging coefficient from these data would be overclaiming.

6. Current assessment of the hypothesis

6.1 Evidence consistent with a real astrophysical trend

The following observations are consistent with the expected physics:

- HZ candidates in the reported-label sample orbit cooler, smaller, lower-mass stars;
- HZ candidates generally have longer periods than the dominant too-hot population;
- the intermediate-period population has more HZ overlap than short-period giant planets;
- ultra-short-period planets are uniformly outside the adopted HZ;
- hot/warm giants have essentially zero HZ overlap;

- within both transit and RV derived samples, intermediate and longer-period systems have higher HZ rates than short-period systems.

The direction is astrophysically sensible: cooler, less luminous stars permit temperate irradiation at shorter absolute periods.

6.2 Evidence affected by detection and characterization bias

At the same time:

- the strong marginal period effect is concentrated in the transit subset;
- RV period uncertainty is large because there are only ten positive RV hosts;
- reported flux exists only for transit and RV planets;
- planet mass availability selects a small subset of transit discoveries;
- hot-giant clusters are transit-dominated;
- long-period giant clusters are RV-dominated;
- imaging and microlensing have almost no comparable HZ labels.

6.3 Reviewer-style verdict at this stage

Relatively robust

- Short-period giant planets are much less likely to lie within the adopted flux interval.
- Cooler and smaller stellar hosts are associated with the reported HZ candidates.
- The extreme values are real subpopulations and should not be automatically removed.

Suggestive but not yet conclusive

- A moderate-period $\times$ cooler-star combination is more favorable than either feature considered without context.
- The period separation appears stronger for transit discoveries than for RV discoveries.

- Intermediate-mass moderate-period systems may have elevated reported HZ overlap.

Not supported with adequate precision

- A universal period effect that is identical across discovery methods.
- A unique cluster with the highest reported HZ rate.
- Any reliable imaging-specific or microlensing-specific HZ relationship.
- A conclusion that the long-period giant cluster contains the most Earth-like planets.

7. Decisions carried into modeling

The classification phase will implement the following safeguards.

Core feature policy

Use:

$$\begin{aligned} & \backslash[\\ & \log P, \quad \\ & \log T_{\mathrm{eff}}, \quad \\ & \log R_{\star}, \quad \\ & \log M_{\star} \\ & \backslash] \end{aligned}$$

with centered interaction and nonlinear terms.

Do not simultaneously include:

- luminosity with temperature and radius;
- calculated surface gravity with mass and radius;

- derived flux or equilibrium temperature as HZ predictors.

Collinearity policy

Use ridge-type regularization for coefficient models and grouped permutation importance for interpretation.

Report stellar-feature importance both:

- individually;
- jointly as a correlated stellar-structure group.

Outlier policy

Compare:

1. all eligible core planets;
2. an ordinary-dwarf sensitivity set;
3. a $(P \geq 1)$ -day sensitivity set.

Bias policy

Compare:

1. astrophysical-only models;
2. bias-augmented models;
3. transit-only models;
4. RV-only models;
5. discovery-method interaction models.

Statistical-inference policy

Use:

- host-grouped cross-validation;
- host-bootstrap performance intervals;
- host-bootstrap predicted-probability contrasts;
- permutation tests or intervals for incremental interaction value.

Do not rely on ordinary row-level p-values.

8. Reproducibility artifacts

| Artifact | Link |

|---|---|

| Full dataset with outlier flags and cluster assignments | [Chunk 3 analysis dataset]
(sandbox:/mnt/data/exoplanet_habitability/exoplanets_chunk3_feature_outlier_cluster.csv) |

| Planet-level cluster assignments | [Cluster assignment table]
(sandbox:/mnt/data/exoplanet_habitability/mass_period_cluster_assignments.csv) |

| Host-balanced correlation matrix | [Correlation matrix]
(sandbox:/mnt/data/exoplanet_habitability/core_host_balanced_spearman_correlation.csv) |

| Host-weighted VIF results | [VIF table]
(sandbox:/mnt/data/exoplanet_habitability/core_host_weighted_vif.csv) |

| Feature-effect bootstrap intervals | [Feature relevance results]
(sandbox:/mnt/data/exoplanet_habitability/core_feature_relevance_host_bootstrap.csv) |

| Physics-aware outlier counts | [Outlier summary]
(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_summary.csv) |

| Cluster-number and stability diagnostics | [Cluster diagnostics]
(sandbox:/mnt/data/exoplanet_habitability/cluster_solution_diagnostics.csv) |

| Cluster profiles | [Cluster profile table]
(sandbox:/mnt/data/exoplanet_habitability/cluster_profiles.csv) |

| Cluster-HZ bootstrap intervals | [HZ-overlap uncertainty table]
(sandbox:/mnt/data/exoplanet_habitability/cluster_hz_overlap_host_bootstrap.csv) |

| Discovery-method-specific cluster rates | [Method-specific overlap table]
(sandbox:/mnt/data/exoplanet_habitability/cluster_hz_overlap_by_discovery_method.csv) |

| Outlier-trim stability analysis | [Cluster sensitivity table]
(sandbox:/mnt/data/exoplanet_habitability/cluster_outlier_sensitivity.csv) |

| Reproducible feature/outlier script | [Feature and outlier analysis script]
(sandbox:/mnt/data/exoplanet_habitability/analyze_feature_outliers_chunk3.py) |
| Reproducible clustering script | [Clustering analysis script]
(sandbox:/mnt/data/exoplanet_habitability/analyze_clusters_chunk3.py) |

9. CRISP-DM process mindmap

Exoplanet habitability drivers

- |
- | — 1. Business understanding COMPLETE
 - | | — Define flux-based HZ candidate
 - | | — Define moderate-period × cooler-star hypothesis
 - | | — Separate astrophysics from observation
 - | | — Define catalog-conditional estimand
- |
- | — 2. Data understanding COMPLETE
 - | | — Data dictionary and units
 - | | — Integrity checks
 - | | — Overall and method-specific missingness
 - | | — Target coverage
 - | | — Host and class imbalance
- |
- | — 2A. Raw visual baseline COMPLETE
 - | | — Period-temperature panels
 - | | — Flux-zone overlays
 - | | — Missingness chart
- |
- | — 3. Data preparation and feature engineering COMPLETE
 - | | — Nullable HZ targets
 - | | — Physics-derived flux
 - | | — Kepler-derived separation
 - | | — Log transformations
 - | | — Missingness policy
 - | | — Leakage-safe feature sets
- |
- | — 4. Feature selection COMPLETE
 - | | — Host-balanced correlations
 - | | — VIF and rank-deficiency audit
 - | | — Bootstrap marginal effects
 - | | — Core component representation
 - | | — Interaction-model specification
- |
- | — 4A. Physics-aware outlier analysis COMPLETE
 - | | — Giant and evolved hosts
 - | | — Compact remnants and hot subdwarfs
 - | | — Ultra-short-period planets
 - | | — High-mass and extremely wide companions
 - | | — Keep, annotate, and sensitivity-test policy
- |

└─ 5. Clustering	COMPLETE
└─ Mass-characterized cohort	
└─ Candidate-k diagnostics	
└─ Host-bootstrap stability	
└─ Four descriptive mass-period archetypes	
└─ Outlier-trim sensitivity	
└─ HZ overlap with uncertainty	
└─	
└─ 6. Classification and regression	NEXT
└─ Prevalence and simple-rule baselines	
└─ Interaction logistic regression	
└─ Regularized logistic regression	
└─ Random forest	
└─ Gradient boosting	
└─ Log-flux regression	
└─ Host-grouped validation	
└─ Calibration and threshold analysis	
└─	
└─ 7. Interpretation and recommendations	
└─ Coefficient and response-surface interpretation	
└─ SHAP and grouped permutation importance	
└─ Formal period × temperature interaction assessment	
└─ Transit versus RV versus imaging selection bias	
└─ Astrophysical conclusion	
└─ General-audience recommendation	

Next chunk

Chunk 4 will fit and evaluate the predictive models. It will begin with honest naive baselines and then compare interaction logistic regression, regularized logistic regression, random forest, gradient boosting, and log-insolation regression using host-grouped validation. Performance will be reported with precision–recall AUC, ROC AUC, balanced accuracy, recall, precision, Matthews correlation, Brier score, calibration, and host-bootstrap uncertainty.

The central comparison will test whether adding the **orbital-period × stellar-temperature interaction** improves out-of-host prediction beyond additive period and temperature terms, and whether that improvement changes after controlling for discovery method.

Standing requirement carried forward: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias using a staged, limited-compute CRISP-DM workflow.

Reply **continue** for Chunk 4.

PROMPT · SEP 01, 2026 09:03 AM

continue.

As we move into Chunk 4, please fit and evaluate only one model per chunk, starting with the naive baselines and the interaction logistic regression — the same restraint you already used for clustering. If a method needs more room (e.g. gradient boosting with hyperparameter search), split it across multiple chunks and I will say continue.

Given the reported-label cohort has only 26 HZ-positive hosts, please use repeated or leave-one-host-out grouped cross-validation rather than a single 5-fold split, and report every metric with host-bootstrap uncertainty intervals, the same way you did for the cluster HZ-overlap rates in Chunk 3 — not single point estimates.

Treat the $\log(\text{period}) \times \text{stellar-temperature}$ interaction test as the primary scientific comparison for this stage: report whether adding the interaction term improves out-of-host PR-AUC (or log-loss) over the additive $\log(\text{period}) + \log(\text{temperature})$ model, with a bootstrap or permutation-based significance check rather than relying on AIC alone. Continue using PR-AUC, recall, F1, and Matthews correlation as the primary metrics given the class imbalance — not raw accuracy.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow. Explain the mindmap of where we are and what's next at the end of the chunk.

CHATGPT · SEP 01, 2026 09:03 AM

I'll evaluate naive baselines, then compare additive and period-temperature interaction logistic models using repeated host-grouped validation, fold-contained preprocessing, and host-level uncertainty.

CHATGPT · SEP 01, 2026 09:07 AM

Every held-out prediction will exclude its entire host; threshold-free metrics remain primary, while F1, recall, and MCC use training-selected thresholds. Interaction evidence will use paired host-level permutation.

CHATGPT · SEP 01, 2026 09:16 AM

Both logistic specifications now share quadratic log-period curvature; only the interaction differs. Early results suggest a small interaction lift, but uncertainty overlaps zero, so interpretation remains cautious.

Exoplanet Habitability Drivers

Chunk 4 — Naive Baselines and the Period × Temperature Logistic Model

All calculations in this chunk use the uploaded `all_exoplanets_2021.csv`.
[filecite]turn2file0[L1-L4]

This chunk evaluates **one model family only**: L2-regularized logistic regression. The additive and interaction specifications are nested versions of that same model, allowing a direct test of whether the period–temperature interaction adds genuine out-of-host predictive information.

1. Scientific question for this stage

The targeted hypothesis is:

> Exoplanets at moderate orbital periods around cooler stars should be more likely to lie within the adopted habitable-zone flux interval, and the favorable orbital period should shift outward as stellar temperature increases.

This hypothesis has three distinct statistical components:

1. **Moderate period:** HZ probability should not increase forever with period. Very short periods are commonly too hot, while sufficiently long periods are too cold.
2. **Cooler-star association:** at a comparable period, cooler stars should generally permit lower irradiation.
3. **Period × temperature interaction:** the orbital period associated with temperate irradiation should depend on stellar temperature.

The third component is the primary formal comparison in this chunk.

2. Modeling cohort and effective sample size

The reported-flux core cohort contains:

Property	Count
Planets	366
Host systems	264
HZ planets	32
Hosts containing at least one HZ planet	26
Transit planets	306
Radial-velocity planets	60
Imaging planets	0

The ordinary planet-level HZ prevalence is:

$$\frac{32}{366} = 8.74\%$$

However, multiple planets sometimes share the same star. To prevent a multi-planet system from dominating the fit or evaluation, each host was assigned a total weight of one. Under this host-balanced estimand, the effective HZ prevalence is:

$$7.67\%$$

That value is the appropriate no-skill PR-AUC reference for this analysis.

Most importantly, repeated cross-validation does not turn 26 positive hosts into a large sample. Repetition reduces split instability; it does not create additional independent HZ systems.

3. Model specification

3.1 Transformations

Let:

$$\begin{aligned} & \backslash[\\ & z_P = \\ & z_{\left[\log_{10}(P_{\mathrm{days}})\right]}, \\ & \backslash] \end{aligned}$$

and:

$$\begin{aligned} & \backslash[\\ & z_T = \\ & z_{\left[\log_{10}(T_{\mathrm{eff}}/\mathrm{K})\right]}, \\ & \backslash] \end{aligned}$$

where standardization is fitted using training hosts only.

A quadratic period term was included because the binary non-HZ group contains both too-hot and too-cold planets. A purely linear period effect would incorrectly imply that ever-longer periods are always more favorable.

3.2 Additive comparator

The additive model is:

$$\begin{aligned} & \backslash[\\ & \operatorname{logit}\{P(\mathrm{HZ}=1)\} \\ & = \end{aligned}$$

$$\begin{aligned} &\beta_0 + \\ &\beta_1 z_P + \\ &\beta_2 z_P^2 + \\ &\beta_3 z_T. \end{aligned}$$

This model permits a moderate-period maximum, but assumes that the shape of the period relationship does not change with stellar temperature.

3.3 Interaction model

The interaction model adds exactly one term:

$$\begin{aligned} &\logit\{P(HZ=1)\} \\ &= \\ &\beta_0 + \\ &\beta_1 z_P + \\ &\beta_2 z_P^2 + \\ &\beta_3 z_T + \\ &\beta_4 z_P z_T. \end{aligned}$$

The two models therefore have identical:

- period main effects;
- period curvature;
- temperature main effect;
- preprocessing;
- regularization;
- validation splits.

Only the period $\times$ temperature cross-term differs.

An L2 ridge penalty with $(C=1)$ was fixed in advance. Avoiding a large hyperparameter search is appropriate given only 26 positive hosts.

Stellar radius, stellar mass, insolation, equilibrium temperature, and discovery method are intentionally not predictors in this targeted model. Radius and mass will enter the next, bias-aware physical model; insolation and equilibrium temperature would introduce target leakage.

4. Validation design

4.1 Why leave-one-host-out was not used

Leave-one-host-out validation would produce many test sets containing no HZ planet. PR-AUC, recall, and related metrics would then be undefined or uninformative within most folds.

Instead, I used:

$$\begin{aligned} & \backslash \\ & 15 \backslash \text{repeats} \times 6 \backslash \text{host-grouped folds} \\ & = \\ & 90 \backslash \text{outer test folds}. \\ & \backslash \end{aligned}$$

`StratifiedGroupKFold` ensured that:

- no host appeared in both training and testing;
- each planet received 15 out-of-host predictions;
- every outer fold contained both outcome classes;
- each outer test fold contained between three and six HZ-positive hosts, with a median of four.

4.2 Nested threshold selection

PR-AUC and log-loss do not require a classification threshold. Recall, precision, F1, and Matthews correlation do.

For every outer training fold, the decision threshold was selected using an additional:

$$\frac{1}{2} \times \text{repeats} \times \text{inner host-grouped folds}.$$

The threshold maximizing host-balanced F1 on the inner out-of-fold predictions was then applied to the untouched outer test hosts.

The median selected thresholds were:

Model	Median probability threshold
Additive logistic	0.224
Interaction logistic	0.237

These are sensibly below 0.5 because HZ candidates form a small minority.

4.3 Uncertainty estimation

The analysis used:

- 3,000 host-bootstrap samples for overall metrics;
- 2,500 host-bootstrap samples for method-stratified metrics;
- 500 host-bootstrap refits for coefficients and response curves;
- 100,000 host-level sign flips for the paired log-loss test.

Each bootstrap draw resampled complete host systems rather than individual planet rows.

5. Naive baselines

Two naive decision strategies are useful.

5.1 Prevalence-probability baseline

This baseline assigns approximately the training-set HZ prevalence to every planet.

Its host-balanced results are:

Metric	Estimate	[95% host-bootstrap interval]
PR-AUC	0.077	[0.050, 0.108]
Log-loss	0.271	[0.203, 0.349]
Brier score	0.071	[0.048, 0.097]
Recall at a 0.5 threshold	0	
F1	0	
Matthews correlation	0	

A conventional 0.5 cutoff labels every planet non-HZ.

5.2 Always-HZ screening rule

A second naive strategy labels every planet as an HZ candidate:

Metric	Estimate	[95% host-bootstrap interval]
Recall	1.000	[1.000, 1.000]
Precision	0.077	[0.048, 0.107]
F1	0.142	[0.092, 0.194]
Matthews correlation	0	

This baseline demonstrates why recall alone is insufficient. Perfect recall can be achieved by flagging everything, but the rule has almost no practical discrimination.

6. Out-of-host model performance

![Out-of-host logistic model evaluation]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_logistic_performance_and_interaction_test.png)

[Vector PDF of the model-evaluation figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_logistic_performance_and_interaction_test.pdf)

6.1 Primary metrics

Model	PR-AUC	Recall	Precision	F1	Matthews correlation
Prevalence baseline	0.077 [0.050, 0.108]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Additive logistic	0.497 [0.358, 0.695]	0.655 [0.459, 0.843]	0.448 [0.293, 0.613]	0.533 [0.374, 0.674]	0.496 [0.333, 0.648]
Interaction logistic	0.525 [0.380, 0.721]	0.728 [0.537, 0.906]	0.450 [0.297, 0.609]	0.557 [0.398, 0.693]	0.528 [0.371, 0.673]

Both logistic specifications are substantially more useful than the naive baseline.

For example, the interaction model's PR-AUC of 0.525 is nearly seven times the host-balanced no-skill level of 0.077. That does not imply that it is ready for operational target selection—the confidence interval remains broad—but it demonstrates real predictive information in period and stellar temperature.

The interaction model also raises recall while leaving estimated precision approximately unchanged. This is potentially useful for a scientific screening application, where failing to flag an HZ candidate may be more costly than inspecting additional false positives.

6.2 Proper probabilistic scores

Lower values are better:

Model	Log-loss	Brier score
Prevalence baseline	0.271 [0.203, 0.349]	0.071 [0.048, 0.097]
Additive logistic	0.153 [0.115, 0.195]	0.049 [0.035, 0.066]
Interaction logistic	0.149 [0.114, 0.191]	0.048 [0.034, 0.065]

The large improvement over the prevalence baseline shows that the model contributes probabilistic information, not merely a favorable threshold.

The difference between the two logistic specifications is much smaller.

7. Primary period × temperature interaction test

The scientifically relevant quantities are the paired out-of-host differences between the two nested models.

For loss metrics, the sign is defined so that a positive number means that the interaction model has lower—and therefore better—loss.

Comparison	Interaction improvement	95% host-bootstrap interval	Bootstrap probability improvement $\backslash(>0\backslash)$
PR-AUC	+0.0278	−0.0058 to +0.0761	0.933
Recall	+0.0729	0.0000 to +0.1958	0.951
Precision	+0.0020	−0.0578 to +0.0644	0.512
F1	+0.0241	−0.0363 to +0.1040	0.744
Matthews correlation	+0.0321	−0.0342 to +0.1216	0.783
Log-loss reduction	+0.00337	−0.00098 to +0.00802	0.932
Brier-score reduction	+0.00105	−0.00065 to +0.00287	0.881

The paired host-level sign-flip test for log-loss produced:

$\backslash$

$p_{\text{one-sided}}=0.076$

\]

and:

\[

$p_{\text{two-sided}}=0.152$.

\]

Reviewer-style conclusion

The interaction term has:

- the expected directional point estimate;
- a small improvement in PR-AUC;
- a small improvement in log-loss;
- a noticeable point increase in recall.

However, the 95% intervals for PR-AUC, F1, Matthews correlation, and log-loss improvement all include zero. The sign-flip result also does not cross the conventional 0.05 threshold in the pooled analysis.

Therefore:

> **The interaction result is suggestive but not conclusive.**

It would be inappropriate to claim that this dataset has definitively established an independent period × temperature interaction from reported flux alone.

8. What the fitted shape says about the hypothesis

![Period–temperature response curves and interaction map]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_a

nd_interaction_map.png)

[Vector PDF of the period–temperature response figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.pdf)

The coefficient analysis separates the three parts of the hypothesis.

| Standardized term | Ridge-logistic estimate | 95% host-bootstrap interval | Directional bootstrap result |

|---|---:|---:|---:|

| Period squared, β_P^2 | -0.604 | -0.651 to -0.091 | 98.8% of draws negative |

| Stellar temperature, β_T | -1.772 | -1.949 to -1.012 | 100% of draws negative |

| Period × temperature, $\beta_{P \times T}$ | +0.368 | -0.622 to +0.522 | 63.8% of draws positive |

These coefficients should be interpreted as regularized conditional associations, not as causal effect sizes.

8.1 Moderate period is better supported than “longer is always better”

The negative quadratic period term implies an inverted-U-shaped relationship:

$$\beta_P^2 < 0.$$

Within the model, HZ probability first rises as planets move away from extremely short, highly irradiated orbits, reaches a maximum, and then declines at sufficiently long periods.

This is consistent with the scientific meaning of **moderate orbital period**.

The result also corrects an important possible misconception from the raw medians: HZ planets have longer median periods than the overwhelmingly too-hot comparison group, but this does not mean arbitrarily long periods are favorable.

8.2 Cooler-star association is clear within this targeted model

The negative temperature coefficient indicates that, at the model's reference period, cooler host stars are associated with higher reported-HZ probability.

This direction is consistent with the raw exploratory analysis and with the basic irradiation physics.

It remains a catalog-conditional association. The labeled sample disproportionately contains certain well-characterized transit and RV systems.

8.3 The cross-term has the expected direction but weak precision

A positive period $\times$ temperature coefficient means that the period effect becomes more favorable at higher stellar temperature. Equivalently, the model's preferred period shifts toward longer orbits as stars become hotter.

The full-cohort point estimates place the maximum modeled probability at approximately:

Stellar temperature	Point-estimated peak period
---: ---:	
3,500 K	223 days
4,500 K	359 days
5,800 K	593 days

The ordering is in the expected direction.

However, the upper 95% bootstrap interval for every peak reached the 1,500-day boundary of the analysis grid. The exact maxima are therefore poorly identified.

These values must **not** be interpreted as physical habitable-zone boundaries because this deliberately restricted model omits:

- stellar radius;
- stellar mass;
- luminosity;
- orbital semi-major axis;

- discovery method.

The useful information is the direction of the shift, not the numerical peak period.

9. Detection-method stratification

The same pooled models were evaluated separately on transit and radial-velocity planets. No method-specific model was fitted in this chunk.

9.1 Transit subset

The transit subset contains:

- 306 planets;
- 232 hosts;
- 19 HZ planets;
- 16 HZ-positive hosts.

Metric	Additive	Interaction	Paired interaction change
---	---	---	---
PR-AUC	0.536 [0.345, 0.833]	0.568 [0.369, 0.869]	+0.032 [-0.020, +0.121]
Recall	0.757 [0.518, 0.986]	0.768 [0.526, 1.000]	+0.011 [0.000, +0.041]
F1	0.561 [0.345, 0.730]	0.556 [0.347, 0.725]	-0.005 [-0.050, +0.032]
Log-loss	0.107 [0.073, 0.142]	0.107 [0.073, 0.142]	improvement +0.00018 [-0.00315, +0.00393]

The one-sided host sign-flip result for the transit log-loss difference was:

\[
p=0.467.
\]

There is no convincing evidence that the interaction materially improves transit predictions

beyond additive period curvature and temperature.

9.2 Radial-velocity subset

The RV subset contains:

- 60 planets;
- 37 hosts;
- 13 HZ planets;
- 10 HZ-positive hosts.

	Metric	Additive	Interaction	Paired interaction change
	---	---	---	---
	PR-AUC	0.463 [0.294, 0.704]	0.487 [0.308, 0.735]	+0.024 [−0.026, +0.087]
	Recall	0.489 [0.227, 0.833]	0.667 [0.361, 1.000]	+0.178 [0.000, +0.452]
	F1	0.449 [0.214, 0.653]	0.536 [0.282, 0.743]	+0.087 [−0.059, +0.269]
	Log-loss	0.430 [0.269, 0.615]	0.409 [0.265, 0.578]	improvement +0.0219 [+0.0009, +0.0456]

For RV log-loss:

$$\begin{aligned} & \backslash \\ & p_{\{\text{one-sided}\}}=0.032, \\ & \backslash\!\!\!\backslash \\ & p_{\{\text{two-sided}\}}=0.064. \\ & \backslash \end{aligned}$$

A leave-one-host-out influence diagnostic kept the RV log-loss improvement positive regardless of which individual host was removed; the remaining-host improvement ranged from approximately 0.0166 to 0.0245.

This is more encouraging than the pooled result, but it must remain **exploratory** because:

- only 37 RV hosts are present;
- only 10 are HZ-positive;
- the method-stratified analysis was not the sole pre-specified primary test;
- the pooled model was evaluated by method rather than separately estimated within each

method;

- PR-AUC improvement remains uncertain even where log-loss improves.

The RV result may indicate that the interaction improves probability calibration more than rank ordering in that subset.

10. What this tells us about real physics versus selection bias

10.1 Evidence compatible with real astrophysics

The model recovers three physically sensible patterns:

- ultra-short periods are unfavorable;
- HZ probability has a moderate-period maximum rather than rising indefinitely;
- cooler stars are associated with higher HZ probability;
- the point-estimated favorable period moves outward for hotter stars.

These are consistent with the expected dependence of incident flux on stellar properties and orbital distance.

10.2 Evidence that the catalog process matters

The interaction behaves differently when the predictions are evaluated by discovery method:

- it adds essentially no log-loss skill in the transit subset;
- it adds suggestive log-loss skill in the much smaller RV subset;
- its PR-AUC benefit is uncertain in both;
- imaging cannot be evaluated at all.

Thus, the pooled relationship is not demonstrated to be method-invariant.

Transit surveys heavily populate the short-period region. RV surveys contribute a comparatively larger fraction of longer-period and higher-mass systems. Reported insolation is also available only for a selectively characterized portion of either population.

A repeated host bootstrap quantifies sampling uncertainty among the observed hosts. It does **not** correct the underlying probability that a planet was:

1. detected;
2. confirmed;
3. assigned a sufficiently complete stellar solution;
4. given a reported insolation value.

10.3 Imaging is an identification failure, not a negative result

There are no imaging planets in the reported-label model cohort.

Accordingly, this analysis says nothing reliable about the HZ rate or period–temperature interaction among directly imaged planets.

The correct conclusion is:

> The reported-flux data do not provide the support required for an imaging-specific comparison.

It would be wrong to translate the absence of labeled imaging planets into evidence that imaging discoveries do not occupy habitable-zone orbits.

11. Decision from this model stage

Supported reasonably well

- Orbital period contains substantial HZ-predictive information.
- The period relationship is curved: moderate periods are more favorable than either extreme.
- Cooler stellar hosts are associated with higher reported-HZ probability.
- Period and stellar temperature together strongly outperform a prevalence-only baseline.

Suggestive rather than established

- Adding an explicit period $\times$ temperature interaction improves pooled out-of-host prediction.
- The preferred period shifts outward for hotter stars.
- The interaction may be more useful for RV discoveries than transit discoveries.

Not established

- A discovery-method-independent interaction effect.
- Exact HZ-optimal orbital periods from temperature alone.
- An imaging-specific relationship.
- Population-level HZ occurrence rates.
- Causal feature effects.

Plain-language interpretation

> Planets are not more promising simply because they take longer to orbit. Very short orbits are usually too hot, while very long orbits can be too cold. The most promising period depends on the star: cooler stars can provide moderate heating at closer, faster orbits, while hotter stars require planets to orbit farther away. This broad pattern is visible in the data, but the additional statistical evidence that the period relationship changes with stellar temperature is still too uncertain to call definitive. The way planets were discovered and characterized remains an important part of the observed pattern.

12. Reproducibility artifacts

Figures

- [Model evaluation — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_logistic_performance_and_interaction_test.png)
- [Model evaluation — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_logistic_performance_and_interaction_test.pdf)
- [Period-temperature response and interaction map — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.png)
- [Period-temperature response and interaction map — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.pdf)

Results

- [Primary model metrics and host-bootstrap intervals]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_key_model_metrics.csv)
- [Paired interaction improvements]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_interaction_increment_host_bootstrap.csv)
- [Discovery-method-stratified performance]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_method_stratified_performance_host_bootstrap.csv)
- [Overall interaction log-loss sign-flip test]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_interaction_logloss_sign_flip_test.csv)
- [Method-specific sign-flip tests]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_method_sign_flip_tests.csv)
- [Interaction coefficient bootstrap summary]
(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_interaction_coefficient_summary.csv)

- [Peak-period uncertainty table]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_peak_period_by_temperature_host_bootstrap.csv)

- [Always-HZ naive baseline]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_always_hz_baseline_host_bootstrap.csv)

Validation audit and predictions

- [Aggregated repeated out-of-host predictions]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_repeated_group_cv_predictions_aggregated.csv)

- [Complete repeated out-of-host prediction records]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_repeated_group_cv_predictions_long.csv)

- [Outer-fold positive-host audit]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_outer_fold_audit.csv)

- [Nested threshold records]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_inner_cv_thresholds.csv)

- [Run metadata and exact design decisions]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_run_metadata.json)

Code

- [Interaction-logistic analysis module]

(sandbox:/mnt/data/exoplanet_habitability/analyze_interaction_logistic_chunk4_reviewed.py)

- [Presentation-chart rendering script]

(sandbox:/mnt/data/exoplanet_habitability/render_chunk4_reviewed_charts.py)

13. Updated CRISP-DM mindmap

Exoplanet habitability drivers

	└─ 1. Business understanding	COMPLETE
	└─ Flux-based HZ definition	
	└─ Moderate-period × cooler-star hypothesis	
	└─ Catalog-conditional estimand	
	└─ Detection-bias question	
	└─ 2. Data understanding	COMPLETE
	└─ Column dictionary and units	
	└─ Missingness by field and discovery method	
	└─ Target availability	
	└─ Host and class imbalance	
	└─ 3. Data preparation	COMPLETE
	└─ Nullable reported-HZ target	
	└─ Physics-derived sensitivity target	
	└─ Leakage controls	
	└─ Log transformations	
	└─ Host-grouped preprocessing	
	└─ 4. Feature, outlier, and cluster analysis	COMPLETE
	└─ Collinearity	
	└─ Bootstrap marginal relevance	
	└─ Physics-aware outlier treatment	
	└─ Mass-period clusters	
	└─ Cluster-HZ overlap	
	└─ 5. Predictive modeling	IN PROGRESS
	└─ 5A. Naive baselines	COMPLETE
	└─ Prevalence-probability baseline	
	└─ Always-HZ screening baseline	
	└─ 5B. Targeted interaction logistic	COMPLETE – THIS
	CHUNK	
	└─ Period curvature	
	└─ Stellar-temperature main effect	
	└─ Period × temperature interaction	
	└─ Repeated host-grouped validation	
	└─ Host-bootstrap intervals	
	└─ Transit versus RV stratification	

PROMPT · SEP 01, 2026 09:56 AM

continue.

As you add stellar radius and stellar mass to the period-temperature model in Chunk 5, please explicitly report how much the period $\times$ temperature interaction and the stellar-temperature main effect change once those collinear stellar variables are included — both the coefficient shift and any change in out-of-host PR-AUC/log-loss versus the Chunk 4 interaction model. Given the VIF results from Chunk 3 (temperature 7.8, radius 5.4, mass 11.2), this collinearity could dilute or destabilize the individual coefficients even if the joint stellar-structure effect remains real, so please interpret radius and mass jointly (e.g. a joint permutation-importance or likelihood-ratio test dropping all three stellar terms together) rather than reading each coefficient in isolation.

When you introduce discovery method as an observation-process variable afterward, keep it as a separate step within this same chunk family (still one conceptual change at a time), and use appropriate shrinkage/partial pooling given transit and RV are the only methods with enough labeled data — do not fit an unregularized per-method coefficient.

Continue using host-grouped repeated cross-validation, host-bootstrap intervals, and PR-AUC/recall/F1/MCC/log-loss as the primary metrics, consistent with Chunk 4.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow. Explain the mindmap of where we are and what's next at the end of the chunk.

CHATGPT · SEP 01, 2026 09:56 AM

I'll extend the host-grouped pipeline sequentially: first adding stellar radius and mass, then regularized discovery method. I'll quantify coefficient drift, predictive gains, and joint stellar-structure relevance.

CHATGPT · SEP 01, 2026 10:00 AM

A preliminary refit shows strong coefficient dilution: temperature shifts from -1.77 to -0.23 , while $\text{period} \times \text{temperature}$ falls from $+0.37$ to $+0.07$. I'm testing whether predictive performance actually changes.

CHATGPT · SEP 01, 2026 10:09 AM

Using identical host-held-out folds, radius and mass improve PR-AUC and screening metrics, though log-loss gains remain uncertain. Adding regularized transit/RV method slightly worsens PR-AUC and log-loss.

Exoplanet Habitability Drivers

Chunk 5 — Full Stellar-Structure Logistic Model and Discovery-Method Adjustment

This analysis continues from the uploaded `all_exoplanets_2021.csv`.
[filecite]turn2file0[L1-L4]

This chunk remains within **one model family: L2-regularized logistic regression**. Two controlled changes are evaluated sequentially:

1. Add stellar radius and stellar mass to the Chunk 4 period–temperature model.
2. Then add one regularized radial-velocity-versus-transit discovery-method contrast.

No tree model, boosting model, or new target is introduced here.

1. Executive findings

Adding stellar radius and stellar mass improves the full model's point performance:

- PR-AUC rises from **0.525 to 0.596**.
- Recall rises from **0.728 to 0.802**.
- F1 rises from **0.557 to 0.647**.
- Matthews correlation rises from **0.528 to 0.626**.
- Log-loss falls from **0.149 to 0.140**.

However, the paired uncertainty interval for the PR-AUC increase includes zero, as does the interval for the log-loss reduction. Thus, the incremental contribution of radius and mass beyond the Chunk 4 period–temperature model is **promising but not conclusive** at the present sample size.

The collinearity concern is real:

- the standardized stellar-temperature coefficient moves **87% toward zero** after radius and mass are added;
- the period × temperature coefficient shrinks by approximately **80%**.

This attenuation does **not** mean the stellar signal disappears. When temperature, radius, mass, and the period–temperature interaction are removed as one block, out-of-host PR-AUC falls by approximately **0.295**, and log-loss worsens by approximately **0.058**. The joint stellar-structure signal is strong even though individual attribution among correlated stellar variables is unstable.

Adding a ridge-shrunk discovery-method intercept does not improve prediction:

- PR-AUC falls slightly from **0.596 to 0.579**;
- log-loss rises slightly from **0.140 to 0.142**;
- the adjusted RV-versus-transit odds ratio is **0.94**, with a very wide interval of **0.30–2.38**.

The physical predictors already reproduce the large raw difference between transit and RV HZ prevalence. This does **not** mean detection bias is absent. It means that, within this selectively labeled cohort, detection-method bias appears primarily through **which physical systems enter each method's sample**, rather than through an additional method-specific intercept after those physical variables are included.

2. Cohort and effective statistical power

The modeling cohort remains unchanged:

Property Count
--- ---:
Planets 366
Host systems 264
HZ-positive planets 32
Hosts containing an HZ-positive planet 26
Transit planets 306
Transit HZ-positive hosts 16

Radial-velocity planets	60
RV HZ-positive hosts	10
Imaging planets	0

The effective inferential sample is much closer to **26 positive host systems** than to 366 independent planets.

Repeated validation stabilizes results across possible host partitions, but it does not generate new independent positive systems. Coefficient and method-specific intervals must consequently remain broad.

3. Controlled model sequence

All continuous variables are base-10 log-transformed where appropriate and standardized using training hosts only:

$$\begin{aligned} & \backslash[\\ z_P &= z[\log_{10}(P)] \\ & \backslash] \end{aligned}$$

$$\begin{aligned} & \backslash[\\ z_T &= z[\log_{10}(T_{\mathrm{eff}})] \\ & \backslash] \end{aligned}$$

$$\begin{aligned} & \backslash[\\ z_R &= z[\log_{10}(R_{\star})] \\ & \backslash] \end{aligned}$$

$$\begin{aligned} & \backslash[\\ z_M &= z[\log_{10}(M_{\star})]. \\ & \backslash] \end{aligned}$$

3.1 Chunk 4 interaction model

The reference model from the preceding chunk is:

$$\begin{aligned} \backslash[\\ \backslash\eta_{\{\mathrm{C4}\}} \\ = \\ \backslash\beta_{0+} \\ \backslash\beta_{1z_P+} \\ \backslash\beta_{2z_P^2+} \\ \backslash\beta_{3z_T+} \\ \backslash\beta_{4z_Pz_T}. \\ \backslash] \end{aligned}$$

The quadratic period term allows HZ probability to peak at a moderate period rather than increase monotonically.

3.2 Full stellar-structure model

The first controlled extension is:

$$\begin{aligned} \backslash[\\ \backslash\eta_{\{\mathrm{stellar}\}} \\ = \\ \backslash\eta_{\{\mathrm{C4}\}} \\ + \\ \backslash\beta_{5z_R+} \\ \backslash\beta_{6z_M}. \\ \backslash] \end{aligned}$$

No new interaction is introduced. This isolates what happens when stellar radius and mass are added to the existing period–temperature structure.

3.3 Method-adjusted model

The second controlled extension is:

$$\begin{aligned} & \backslash[\\ & \backslash\eta_{\backslash\mathrm{method}} \\ & = \\ & \backslash\eta_{\backslash\mathrm{stellar}} \\ & + \\ & \backslash\beta_{7z_{\backslash\mathrm{RV}}}, \\ & \backslash] \end{aligned}$$

where $\backslash(z_{\backslash\mathrm{RV}})$ is a standardized RV-versus-transit indicator.

This method term is:

- estimated with the same L2 ridge penalty as the physical coefficients;
- shrunk toward zero;
- a single intercept contrast;
- not an unregularized method coefficient;
- not a method-specific period or temperature slope.

No imaging coefficient is fitted because there are no reported-flux imaging planets in the cohort.

4. Validation and uncertainty protocol

The evaluation uses the same design as Chunk 4:

$$\begin{aligned} & \backslash[\\ & 15\backslash\text{ repeats}\backslash\times 6\backslash\text{ host-grouped folds}=90\backslash\text{ outer folds}. \\ & \backslash] \end{aligned}$$

Every host is kept entirely within training or testing in a given fold.

Each planet receives 15 out-of-host probability estimates, which are aggregated before final evaluation. Every host receives a total evaluation weight of one, so multi-planet systems do

not dominate the metrics.

For thresholded metrics, the exact training-only threshold selected for the Chunk 4 interaction model in each outer fold is frozen and applied to all models in that fold. This creates a controlled comparison:

> Changes in recall, F1, and Matthews correlation cannot be attributed to selecting a more favorable threshold for the new model.

PR-AUC and log-loss remain the more important incremental-comparison metrics because they do not depend on that threshold.

Uncertainty uses:

- **2,000 whole-host bootstrap resamples** for model metrics and paired changes;
- **500 whole-host bootstrap refits** for coefficients;
- **100,000 host-level sign flips** for paired out-of-host log-loss comparisons.

Bootstrap fitting weights were normalized to preserve the total sample weight in every resample. Without that correction, ridge regularization would become artificially stronger or weaker across bootstrap draws.

5. Overall out-of-host performance

![Model performance and incremental value]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/performance_and_incremental_value_final.png)

[Vector PDF of the model-performance figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/performance_and_incremental_value_final.pdf)

5.1 Ranking and thresholded screening metrics

All intervals below are 95% whole-host bootstrap intervals.

Model	PR-AUC	Recall	Precision	F1	Matthews correlation
Prevalence baseline	0.077 [0.048, 0.109]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Chunk 4 interaction	0.525 [0.387, 0.718]	0.728 [0.535, 0.900]	0.450 [0.295, 0.603]	0.557 [0.396, 0.689]	0.528 [0.364, 0.668]
Full stellar structure	0.596 [0.445, 0.793]	0.802 [0.626, 0.950]	0.542 [0.383, 0.690]	0.647 [0.499, 0.770]	0.626 [0.473, 0.757]
Method-adjusted full	0.579 [0.429, 0.783]	0.802 [0.626, 0.950]	0.534 [0.372, 0.684]	0.641 [0.489, 0.764]	0.620 [0.469, 0.750]

The full stellar-structure model has the strongest point estimates across all primary screening metrics.

Its PR-AUC is approximately:

$$\frac{0.596}{0.077} \approx 7.8$$

times the host-balanced no-skill level. That comparison establishes useful discrimination, although the absolute uncertainty remains substantial.

5.2 Probabilistic metrics

Lower values are better.

Model	Log-loss	Brier score
Prevalence baseline	0.271 [0.200, 0.350]	0.071 [0.047, 0.098]
Chunk 4 interaction	0.149 [0.112, 0.190]	0.048 [0.034, 0.065]
Full stellar structure	0.140 [0.102, 0.183]	0.044 [0.030, 0.060]
Method-adjusted full	0.142 [0.104, 0.186]	0.045 [0.030, 0.061]

The full stellar model is also best on both proper probabilistic scoring rules, but its incremental advantage over the Chunk 4 model is small compared with the much larger improvement over the naive baseline.

6. Stage A: adding stellar radius and mass

6.1 Paired out-of-host performance change

The same held-out hosts are used for both models. A positive change favors the full stellar-structure model.

Metric	Change after adding radius and mass	95% host-bootstrap interval
PR-AUC	+0.071	−0.015 to +0.168
Recall	+0.074	0.000 to +0.204
Precision	+0.092	−0.004 to +0.195
F1	+0.091	+0.005 to +0.186
Matthews correlation	+0.098	+0.007 to +0.199
Log-loss reduction	+0.0091	−0.0063 to +0.0233

The host-level log-loss sign-flip test gives:

$$p_{\{\mathrm{one-sided}\}}=0.123.$$

Interpretation

The point estimates are consistently favorable:

- better ranking;
- more HZ candidates recovered;
- higher precision;
- improved F1 and Matthews correlation;
- lower probabilistic loss.

The F1 and Matthews-correlation intervals remain above zero at the frozen operating threshold. However, the more fundamental PR-AUC and log-loss intervals include zero.

Therefore:

> Radius and stellar mass probably improve screening, but this dataset does not establish a precise incremental probability-model benefit beyond period and temperature alone.

This distinction is important. The full model appears better, but only 26 positive hosts are available to distinguish a modest improvement from split and sampling variation.

7. Coefficient redistribution under stellar collinearity

![Coefficient attenuation and joint stellar-block test]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.png)

[Vector PDF of the coefficient and stellar-block figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.pdf)

7.1 Stellar-temperature main effect

| Model | Standardized temperature coefficient |

---|---|

| Chunk 4 interaction | -1.772 [-2.265, -1.095] |

Full stellar structure	-0.225 [-0.918, +0.522]	
Method-adjusted full	-0.223 [-0.898, +0.515]	

Adding radius and mass changes the coefficient by:

\[	
\Delta\beta_T=	+1.547
\]	

with a 95% host-bootstrap interval of:

\[	
[+0.631, +2.387].	
\]	

Because the original estimate was negative, this represents an approximately:

\[	
87.3\%	
\]	

movement toward zero.

The direct temperature coefficient is therefore no longer separately identifiable with useful precision once radius and mass are present.

7.2 Period × temperature interaction

Model	Standardized interaction coefficient
--- ---:	
Chunk 4 interaction	+0.368 [-0.444, +0.792]
Full stellar structure	+0.074 [-0.798, +0.521]
Method-adjusted full	+0.082 [-0.777, +0.534]

Adding radius and mass changes the interaction coefficient by:

\[	
----	--

$$\Delta\beta_{P\times T}=-0.294$$

\]

with a 95% interval of:

\[

$$[-0.546,-0.040].$$

\]

That is approximately an:

\[

$$79.9\%$$

\]

reduction in the point coefficient.

The interaction's preferred direction remains slightly positive, but its magnitude becomes much smaller and its uncertainty remains broad.

8. Why coefficient attenuation is expected

Temperature, radius, and mass do not represent unrelated predictors.

For approximate stellar luminosity:

\[

$$L_{\star}\propto R_{\star}^2T_{\mathrm{eff}}^4.$$

\]

Stellar mass also enters the conversion between orbital period and orbital scale through Kepler's law:

\[

$$a^3\propto M_{\star}P^2.$$

\]

These features jointly trace the underlying stellar sequence and the irradiation environment. In the Chunk 3 cohort, they already showed high VIF values and strong pairwise correlations.

The coefficient bootstraps in the full model further demonstrate this competition:

\[

$\operatorname{corr}(\beta_T, \beta_R) \approx -0.74$

\]

and:

\[

$\operatorname{corr}(\beta_T, \beta_M) \approx -0.46$.

\]

When one coefficient becomes more negative in a bootstrap sample, another correlated coefficient can become less negative while preserving a similar predicted probability.

Ridge regularization deliberately spreads predictive information among correlated variables. Therefore, the coefficient shift should not be read as:

> “Temperature was important before, but becomes physically unimportant after adding radius and mass.”

The defensible interpretation is:

> **The data support a combined stellar-structure effect, but they do not contain enough independent variation to assign that effect cleanly to temperature, radius, or mass one variable at a time.**

9. Joint stellar-structure block test

Because individual stellar coefficients are unstable, the more appropriate question is:

> How much predictive information is lost when all stellar-structure terms are removed together?

The block contains:

- stellar effective temperature;
- stellar radius;
- stellar mass;
- period × temperature interaction.

The comparison retains the same curved period terms in both models.

This is an out-of-host **block-drop importance test** rather than a classical likelihood-ratio test. A standard likelihood-ratio result would not account adequately for:

- ridge penalization;
- grouped observations;
- repeated cross-validation;
- the intended out-of-host generalization target.

9.1 Joint block result

Metric	Full stellar model minus curved period-only model	95% interval
PR-AUC	+0.295	+0.119 to +0.473
Recall	+0.253	−0.058 to +0.557
Precision	+0.158	−0.005 to +0.319
F1	+0.195	−0.002 to +0.397
Matthews correlation	+0.221	−0.008 to +0.449
Log-loss reduction	+0.058	+0.022 to +0.100

The host-level sign-flip result for log-loss is:

$$p_{\mathrm{one-sided}}=0.00056.$$

After retaining the method indicator in both comparator models, the result is nearly unchanged:

- PR-AUC increase: **+0.283** [0.104, 0.472];
- log-loss reduction: **+0.055** [0.021, 0.093];
- one-sided host sign-flip $\backslash(p=0.00075\backslash)$.

9.2 Scientific interpretation

The joint stellar block contributes clear out-of-host information beyond orbital period alone.

This is the central result of the collinearity analysis:

> Collinearity weakens attribution to individual stellar variables, but it does not eliminate the predictive value of stellar structure as a group.

The stellar environment matters. What remains uncertain is how much unique credit should be assigned to temperature versus radius versus mass within this small and highly correlated labeled sample.

10. Consequences for the period $\times$ temperature hypothesis

The Chunk 4 model gave a suggestive positive interaction: hotter stars appeared to require longer favorable orbital periods.

After radius and mass are included, the explicit interaction coefficient becomes much smaller.

There are two plausible, non-exclusive interpretations.

Statistical explanation

Radius and mass absorb part of the same variation previously attributed to temperature and its interaction with period. Because the predictors are collinear, the Chunk 4 interaction may partly have represented omitted stellar structure.

Astrophysical explanation

Temperature alone is not stellar luminosity. Two stars with similar effective temperatures can differ in radius and therefore luminosity. Similarly, stellar mass changes the period–distance conversion.

A period × temperature model is consequently an incomplete approximation to the more general relationship:

$$\begin{aligned} & \propto \\ & R_{\text{star}}^2 T_{\text{eff}}^4 \\ & M_{\text{star}}^{-2/3} \\ & P^{-4/3}. \end{aligned}$$

Once radius and mass are available, less information needs to be carried by the period × temperature cross-term.

Current verdict

The evidence still supports the broad statement:

> Moderate periods are more favorable, and what counts as moderate depends on the host star.

The more specific claim that a standalone **period × temperature** coefficient has a stable, independently identifiable effect is not supported with precision after the correlated stellar variables are included.

That is an important refinement rather than a contradiction of the original hypothesis.

11. Stage B: adding discovery method with shrinkage

The reported-label sample contains only two discovery methods:

- transit;
- radial velocity.

A single standardized RV-versus-transit intercept contrast was therefore used. It shares the ridge penalty with all other terms and is pulled toward zero unless it contributes reproducible out-of-host information.

No separate method-specific slopes were fitted because:

- only 37 RV hosts are present;
- only 10 RV hosts contain an HZ-positive planet;
- unrestricted period $\times$ method and temperature $\times$ method coefficients would be weakly identified;
- imaging has no labeled support.

![Discovery-method adjustment]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.png)

[Vector PDF of the discovery-method figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.pdf)

12. Does discovery method improve the full physical model?

Metric	Full stellar structure	Method-adjusted full	Paired change from adding method
PR-AUC	0.596 [0.445, 0.793]	0.579 [0.429, 0.783]	-0.017 [-0.042, -0.002]
Recall	0.802 [0.626, 0.950]	0.802 [0.626, 0.950]	0.000 [0.000, 0.000]
Precision	0.542 [0.383, 0.690]	0.534 [0.372, 0.684]	-0.009 [-0.030, 0.000]
F1	0.647 [0.499, 0.770]	0.641 [0.489, 0.764]	-0.006 [-0.022, 0.000]
Matthews correlation	0.626 [0.473, 0.757]	0.620 [0.469, 0.750]	-0.006 [-0.021, 0.000]
Log-loss	0.140 [0.102, 0.183]	0.142 [0.104, 0.186]	Worsens by 0.0023 [0.0009, 0.0043]

The host-level sign-flip analysis finds that the small log-loss degradation is consistent across the paired held-out hosts. Its two-sided sign-flip result is approximately:

$$p=0.00001.$$

The effect is statistically consistent but practically small.

The interpretation is not that the method variable is harmful in a universal sense. Rather:

> A discovery-method intercept provides no additional generalizable information once period and the correlated stellar-structure measurements are available in this labeled cohort.

13. Raw versus physically adjusted method differences

The raw host-balanced HZ prevalences differ substantially:

Discovery method	Observed host-balanced HZ prevalence
Transit	5.6% [2.9%, 8.5%]
Radial velocity	20.3% [9.5%, 32.9%]

At first sight, RV planets appear several times more likely to be HZ candidates.

However, the full physical model—without receiving discovery method—predicts:

Discovery method	Observed HZ prevalence	Mean probability from physical model
Transit	5.6%	5.6% [4.2%, 7.2%]
Radial velocity	20.3%	21.3% [15.5%, 27.7%]

After adding method explicitly, the predictions are:

Discovery method	Mean probability from method-adjusted model
Transit	5.7% [4.2%, 7.3%]
Radial velocity	20.9% [15.2%, 27.2%]

The raw method difference is therefore almost entirely reproduced by the physical-feature distributions.

The adjusted RV-versus-transit contrast is:

$$\text{log-odds difference} = -0.063$$

with interval:

$$[-1.209, +0.866].$$

Expressed as an odds ratio:

$$\text{OR}_{\text{RV/Transit}} = 0.94$$

with interval:

\[
[0.30,2.38].
\]

Expressed as a standardized probability difference:

\[
P(HZ\mid RV)-P(HZ\mid Transit)
=
-0.003
\]

with interval:

\[
[-0.044,+0.043].
\]

There is no evidence of a stable residual method-intercept effect.

14. Method-specific predictive performance

The physical model's performance differs in important ways between methods.

Method	PR-AUC	Recall	F1	Matthews correlation	Log-loss
Transit	0.591 [0.397, 0.870]	0.845 [0.618, 1.000]	0.680 [0.473, 0.830]	0.672 [0.477, 0.825]	0.102 [0.070, 0.136]
Radial velocity	0.599 [0.372, 0.880]	0.733 [0.412, 1.000]	0.581 [0.315, 0.786]	0.464 [0.148, 0.737]	0.372 [0.226, 0.534]

The PR-AUC point estimates are similar, but the uncertainty is very wide.

The much higher RV log-loss indicates that probabilities are less precise for the RV subset, even though its ranking performance is comparable. This could arise from:

- only 37 RV hosts;
- different coverage of the period–stellar-property space;
- different target-characterization practices;
- heterogeneous mass and orbital information;
- selective availability of reported insolation.

The method intercept does not fix this calibration difference. A single intercept cannot represent complex method-dependent support, completeness, and measurement processes.

15. What “no added method coefficient” does—and does not—mean

It does mean

Within the reported-flux cohort, after controlling for period, temperature, radius, mass, and period curvature:

- knowing whether a planet was found by transit or RV does not improve out-of-host ranking or calibration;
- the raw transit–RV HZ-rate difference is explained by their different physical covariate distributions;
- an additional method intercept is unnecessary for this predictive model.

It does not mean

It does not demonstrate that transit and RV surveys are unbiased.

Discovery method affects:

- the periods that are detectable;
- the masses and radii that can be measured;

- which stars are monitored;
- the probability of follow-up characterization;
- whether insolation is eventually reported;
- which planets enter this Kaggle snapshot.

The method variable can therefore operate upstream by determining the observed feature distribution and target availability. Once we condition on the selected physical variables in the selected labeled sample, little residual intercept information may remain.

This distinction is analogous to covariate shift:

$$\begin{aligned} & \mathbb{P}(X \mid \text{Transit}) \\ & \neq \\ & \mathbb{P}(X \mid \text{RV}), \end{aligned}$$

even when:

$$\begin{aligned} & \mathbb{P}(\text{HZ} \mid X, \text{Transit}) \\ & \approx \\ & \mathbb{P}(\text{HZ} \mid X, \text{RV}) \end{aligned}$$

within the region where labels exist.

Imaging remains unidentified

No reported-flux imaging planet enters the model.

The absence of an imaging coefficient is not evidence of a zero imaging effect or zero imaging HZ prevalence. It is an explicit statement that this dataset has no labeled support for such an estimate.

16. Statistical-power assessment

Findings with relatively strong support

The following conclusions are supported by out-of-host block comparisons with host-level uncertainty:

- Stellar structure collectively adds predictive information beyond orbital period.
- A period-only description is inadequate.
- The full physical model substantially outperforms the naive prevalence baseline.
- Adding a simple discovery-method intercept does not improve the full physical model.
- Coefficient attribution among temperature, radius, and mass is unstable because of collinearity.

Findings that remain suggestive

- Radius and stellar mass improve performance beyond the Chunk 4 period-temperature model.
- Their point improvements in PR-AUC and log-loss are real in most bootstrap samples, but the 95% intervals include no improvement.
- The moderate-period relationship is better represented when more complete stellar structure is included.

Findings that are not established

- A separately identifiable, stable period $\times$ temperature coefficient after radius and mass are controlled.
- Unique causal effects for temperature, radius, or stellar mass.
- Method-specific period or temperature slopes.
- A population-level correction for transit or RV completeness.
- Any imaging-specific HZ relationship.

17. Plain-language interpretation

- > A planet's orbital period is not enough to determine whether it receives Earth-like heating. The star's properties matter collectively. Hotter, larger, and more massive stars alter where a temperate orbit would occur, and those stellar properties are so closely related that this dataset cannot cleanly say which one deserves most of the statistical credit.
- > Adding stellar size and mass improves the model's apparent ability to identify habitable-zone candidates, although there are too few independent positive systems to prove that the incremental improvement will generalize precisely.
- > Radial-velocity planets have a higher raw habitable-zone rate than transit planets in this labeled sample. But the physical model already predicts that difference because the two methods discover and characterize different kinds of systems. Merely adding the method's name does not improve prediction. This does not remove detection bias; it shows that the bias is expressed mainly through the kinds of planets and stars present in each observed sample.

The strongest general-audience conclusion at this stage is:

- > **Potentially temperate planets tend to occupy an intermediate orbital regime whose location depends on the complete structure of the host star—not on stellar temperature or orbital period in isolation.**

18. Reproducibility artifacts

Presentation figures

- [Performance and incremental value — PNG]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/performance_and_incremental_value_

final.png)

- [Performance and incremental value — vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/performance_and_incremental_value_final.pdf)

- [Coefficient redistribution and joint stellar effect — PNG]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.png)

- [Coefficient redistribution and joint stellar effect — vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.pdf)

- [Discovery-method adjustment — PNG]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.png)

- [Discovery-method adjustment — vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.pdf)

Statistical results

- [Model metrics with whole-host bootstrap intervals]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_model_metrics.csv)

- [Paired incremental comparisons]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_incremental_comparisons.csv)

- [Key coefficient estimates]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_coefficient_estimates.csv)

- [Paired coefficient shifts]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_coefficient_shifts.csv)

- [Method calibration results]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/method_calibration_host_bootstrap.csv)

- [Method-stratified performance]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/method_stratified_performance_host_bootstrap.csv)

- [Host-level sign-flip tests]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/logloss_sign_flip_tests.csv)

- [Aggregated repeated out-of-host predictions]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/repeated_group_cv_predictions_aggr

egated.csv)

- [Outer-fold validation audit]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/outer_fold_audit.csv)

- [Run metadata and exact design decisions]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/run_metadata.json)

Reproducible code

- [Core analysis module]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/analysis_core.py)

- [Repeated grouped-CV runner]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/run_cv.py)

- [Coefficient-bootstrap runner]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/run_coefficients.py)

- [Results-combination script]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/combine_results.py)

- [Presentation-chart renderer]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/render_final_charts.py)

19. Updated CRISP-DM mindmap

Exoplanet habitability drivers

	└─ 1. Business understanding	COMPLETE
	└─ Flux-based HZ-candidate definition	
	└─ Moderate-period × cooler-star hypothesis	
	└─ Catalog-conditional estimand	
	└─ Detection and characterization bias	
	└─ 2. Data understanding	COMPLETE
	└─ Column dictionary and units	
	└─ Missingness by field and method	
	└─ Target coverage	
	└─ Host-level effective sample size	
	└─ 3. Data preparation	COMPLETE
	└─ Nullable reported-HZ target	
	└─ Physics-derived sensitivity target	
	└─ Leakage prevention	
	└─ Log transformations	
	└─ Host-grouped preprocessing	
	└─ 4. Feature, outlier, and clustering analysis	COMPLETE
	└─ Collinearity audit	
	└─ Physics-aware outliers	
	└─ Mass-period clusters	
	└─ Cluster-HZ overlap	
	└─ 5. Predictive modeling	IN PROGRESS
	└─ 5A. Naive baselines	COMPLETE
	└─ 5B. Period-temperature ridge logistic	COMPLETE
	└─ Period curvature	
	└─ Temperature main effect	
	└─ Period × temperature interaction	
	└─ 5C. Full stellar-structure ridge logistic	COMPLETE – THIS
	CHUNK	
	└─ Add stellar radius	
	└─ Add stellar mass	
	└─ Quantify coefficient redistribution	
	└─ Joint stellar-block test	

		└ Repeated host-grouped validation	
		└ 5D. Regularized discovery-method adjustment	COMPLETE – THIS
CHUNK			
		└ One ridge-shrunk RV/transit intercept	
		└ No unrestricted method coefficients	
		└ No unsupported imaging coefficient	
		└ Physical covariate shift versus residual method effect	
		└ 5E. One nonlinear tree-based classifier	NEXT
		└ Random forest only	
		└ Constrained hyperparameter search	
		└ Identical host-grouped validation	
		└ Grouped permutation importance	
		└ Nonlinear period–stellar response surface	
		└ 5F. Gradient boosting	LATER
		└ 5G. Log–insolation regression	LATER
		└ 6. Interpretation and recommendations	
		└ Model-family comparison	
		└ SHAP or model-appropriate explanations	
		└ Real astrophysics versus selection effects	
		└ Robustness synthesis	
		└ General-audience conclusion	

Next chunk

Chunk 6 will fit one nonlinear model only: a constrained random-forest classifier. It will use the same reported-label cohort, host grouping, bootstrap uncertainty, and core physical features. Its purpose will be to test whether flexible nonlinear boundaries materially outperform the ridge-logistic model without relying on an explicitly specified period × temperature term. Interpretation will use grouped permutation importance so that correlated stellar variables are evaluated collectively as well as individually.

Standing requirement carried forward: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are

more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias using a staged, limited-compute CRISP-DM workflow.

Reply **continue** for Chunk 6.

continue.

For the random forest in Chunk 6, please constrain hyperparameters explicitly rather than running an open-ended search – with only 26 HZ-positive hosts, an unconstrained forest will likely memorize them. Compare a small number of pre-specified, deliberately shallow configurations (e.g. limited max depth, a minimum samples-per-leaf floor scaled to the positive class size) using the same nested host-grouped validation as before, rather than tuning against the outer test folds.

Use the Method-adjusted full logistic model from Chunk 5 as the benchmark to beat, not just the naive baseline. Report grouped permutation importance both individually per feature and jointly for the stellar-structure block (temperature, radius, mass together), consistent with how you handled their collinearity in Chunk 5. Also produce a 2D partial-dependence or response-surface view of period versus stellar temperature, the same way you did for the logistic interaction model, so we can see whether the forest recovers a similar moderate-period band on its own or finds a different structure.

Given how many model variants we've now run relative to the size of the labeled sample, if the random forest does not clearly outperform the Chunk 5 logistic model on out-of-host PR-AUC and log-loss, please say so plainly and treat that as a meaningful finding in itself (a nonlinear model doesn't help at this sample size) rather than continuing to add complexity. In that case, propose folding gradient boosting and the log-insolation regression into a single, more compact comparison chunk next, so we can move toward final synthesis rather than open-ended model expansion.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow. Explain the mindmap of where we are and what's next at the end of the chunk.

CHATGPT · SEP 01, 2026 10:29 AM

I'll evaluate a few pre-specified shallow forests with nested host-grouped validation against the frozen Chunk 5 logistic benchmark, then assess grouped importance and the period-temperature response surface.

CHATGPT · SEP 01, 2026 10:32 AM

The fixed design uses 15×6 outer folds and 2×4 inner folds across three depth/leaf configurations, with five inputs matching the logistic benchmark.

CHATGPT · SEP 01, 2026 10:41 AM

Five of fifteen repeated partitions are complete; the one-standard-error rule often rejects the deepest forest. Completed folds are checkpointed, and permutation importance preserves method and stellar covariance.

Exoplanet Habitability Drivers

Chunk 6 — Constrained Random Forest, Held-Out Feature Importance, and the Nonlinearity Test

All calculations in this chunk use the uploaded `all_exoplanets_2021.csv`.
[filecite turn2file0 L1-L4](#)

This stage evaluates **one model family only**: a deliberately shallow random-forest classifier. Its purpose is not to maximize an in-sample leaderboard score. It is to test whether a flexible nonlinear learner can discover a more useful period–stellar relationship than the regularized logistic model under strict out-of-host validation.

1. Executive conclusion

The random forest **does not outperform** the method-adjusted full logistic benchmark from Chunk 5.

Model	Out-of-host PR-AUC	Out-of-host log-loss
Method-adjusted ridge logistic	0.579 [0.429, 0.774]	0.142 [0.103, 0.184]
Nested shallow random forest	0.475 [0.332, 0.678]	0.162 [0.119, 0.207]

In the paired comparison on the same held-out hosts:

$$\Delta \text{PR-AUC}_{\text{RF-Logistic}} = -0.104$$

with a 95% whole-host bootstrap interval of:

$\backslash$
[-0.221,-0.003].
 $\backslash$

The forest's log-loss is higher by:

$\backslash$
0.019
 $\backslash$

with a 95% interval of:

$\backslash$
[0.001,0.036].
 $\backslash$

Only about **2.1%** of the paired host-bootstrap samples favored the forest on either PR-AUC or log-loss. A two-sided host-level sign-flip test for the log-loss difference gave:

$\backslash$
p=0.035,
 $\backslash$

with the observed difference favoring logistic regression.

Therefore:

> Added nonlinear flexibility does not help at this sample size. The physics-informed, regularized logistic model remains the better generalizing model.

The forest is not useless—it still substantially outperforms the naive prevalence baseline—but it does not clear the relevant benchmark.

2. Why this is a meaningful negative result

Machine-learning analyses often assume that a more flexible model should perform better. That assumption is especially dangerous here.

The labeled cohort contains:

Property	Count
Planets	366
Host systems	264
HZ-positive planets	32
HZ-positive host systems	26
Transit planets	306
Radial-velocity planets	60
Imaging planets	0

A random forest can represent:

- nonlinear thresholds;
- local pockets of elevated probability;
- high-order interactions;
- discontinuous decision boundaries.

But the model must infer all of those structures from only **26 independent HZ-positive hosts**. A highly flexible forest could easily model peculiarities of those systems rather than a general astrophysical relationship.

The result illustrates the bias–variance trade-off:

- logistic regression imposes a structured, smooth relationship;
- the forest makes weaker shape assumptions;
- the available data are not sufficient to estimate the more flexible shape reliably;
- the simpler model therefore generalizes better.

Repeated cross-validation reduces dependence on one fortunate split. It does not turn 26 positive hosts into hundreds of independent positive systems.

3. Capacity constraints fixed before evaluation

3.1 Input variables

The forest received five predictors:

$\log_{10}(P)$

$\log_{10}(T_{\mathrm{eff}})$

$\log_{10}(R_{\star})$

$\log_{10}(M_{\star})$

and a regular binary indicator for:

$\text{Radial velocity versus transit}$.

Unlike the logistic model, the forest was not given:

- a squared-period term;
- an explicit period $\times$ temperature term;
- stellar luminosity;

- equilibrium temperature;
- insolation flux;
- any HZ-derived feature.

The forest therefore had to discover curvature and interaction structure from the five raw transformed inputs.

3.2 Pre-specified candidate configurations

Only three configurations were allowed.

Configuration	Maximum depth	Minimum leaf size	Features considered at each split
Very shallow	2	At least 50% of positive training hosts, with absolute floor of 6	2
Shallow	3	At least 40% of positive training hosts, with absolute floor of 6	3
Moderate	4	At least 30% of positive training hosts, with absolute floor of 6	3

All configurations used:

- `criterion="log_loss"` ;
- bootstrap sampling;
- at most 80% of the outer training rows per tree;
- host-equal fitting weights;
- no class weighting;
- no unrestricted depth;
- no open-ended hyperparameter search.

The outer-fold models used 384 trees. The number of trees stabilizes the ensemble average; it does not increase individual-tree depth or permit smaller terminal leaves.

3.3 Why class weighting was not used

A balanced class weight can improve minority-class sensitivity, but it also changes the probability target by making the training procedure behave as though the classes were much closer to equally common.

That can produce probabilities that rank well but are badly calibrated to the observed HZ prevalence, harming log-loss.

Instead, the analysis used:

- host-equal sample weights;
- PR-AUC as the primary ranking metric;
- a threshold selected inside the inner grouped validation for recall, precision, F1, and MCC.

4. Nested host-grouped validation

The forest used the exact same 90 outer host splits used in Chunk 5:

\[
15\text{ repeats}\times6\text{ grouped folds}.
\]

No host appeared in both training and testing within a fold. Every outer test fold contained between three and six HZ-positive hosts.

Within every outer training set, configuration selection used:

\[
2\text{ repeats}\times4\text{ inner host-grouped folds}.
\]

The selection rule was conservative:

1. Calculate repeated inner out-of-host log-loss for all three configurations.
2. Identify the configuration with the lowest log-loss.
3. Calculate one standard error around that best inner result.
4. Select the **simplest** configuration within one standard error of the best.
5. Select the F1 operating threshold using only the inner out-of-host probabilities from that chosen configuration.

6. Refit the chosen configuration on the complete outer training set.
7. Evaluate it on the untouched outer hosts.

No outer-fold result influenced configuration or threshold selection.

The logistic benchmark uses its previously published Chunk 5 out-of-host predictions and training-only thresholds. Because the decision-threshold policies are valid but not identical, **PR-AUC and log-loss remain the decisive forest-versus-logistic comparisons.**

5. What the nested selection chose

![Random-forest performance and configuration selection]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_performance_selection.png)

[Vector PDF of the performance and selection figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_performance_selection.pdf)

The selected configurations were:

Configuration	Outer folds selected	Share of 90 folds
Very shallow	17	18.9%
Shallow	72	80.0%
Moderate	1	1.1%

For the commonly selected shallow forest:

- maximum depth was three;
- median outer-fold `min_samples_leaf` was nine;
- the leaf-size range was eight to ten;
- median training-only decision threshold was approximately 0.253.

5.1 Inner-CV comparison

Configuration	Median inner PR-AUC	Median inner log-loss
Very shallow	0.413	0.185
Shallow	0.451	0.161
Moderate	0.476	0.155

The moderate configuration had the best median point estimates, but its advantage was smaller than the inner host-level sampling uncertainty.

The one-standard-error rule consequently chose the shallower depth-three forest in 80% of the outer folds. The moderate depth-four forest was selected only once.

This is exactly the behavior desired from a conservative selection procedure:

> The method did not convert a small, uncertain improvement in inner-fold fit into a routine decision to use the most complex candidate.

6. Out-of-host performance

All intervals below are 95% whole-host bootstrap intervals based on 3,000 resamples.

Model	PR-AUC	Recall	Precision	F1	MCC	Log-loss
Prevalence baseline	0.077 [0.049, 0.107]	0.000	0.000	0.000	0.000	0.271 [0.201, 0.346]
Method-adjusted ridge logistic	0.579 [0.429, 0.774]	0.802 [0.631, 0.957]	0.534 [0.375, 0.688]	0.641 [0.490, 0.768]	0.620 [0.471, 0.754]	0.142 [0.103, 0.184]
Nested shallow random forest	0.475 [0.332, 0.678]	0.691 [0.497, 0.870]	0.535 [0.358, 0.709]	0.603 [0.432, 0.742]	0.571 [0.394, 0.721]	0.162 [0.119, 0.207]

6.1 What the forest does retain

The forest remains much better than the prevalence baseline:

- PR-AUC is 0.475 rather than 0.077;
- log-loss is 0.162 rather than 0.271;
- F1 is 0.603 rather than zero;
- MCC is 0.571 rather than zero.

It clearly detects genuine structure in orbital and stellar measurements.

6.2 What the forest loses relative to logistic regression

Relative to the method-adjusted logistic model, its point estimates show:

- lower PR-AUC;
- lower recall;
- approximately identical precision;
- lower F1;
- lower MCC;
- worse log-loss.

The forest is not trading calibration for superior ranking or vice versa. It is worse on both primary threshold-free criteria.

7. Paired forest-versus-logistic comparison

The paired host-bootstrap differences are:

Metric	Forest minus logistic	95% interval	Bootstrap probability forest is better
---	---:	---:	---:
PR-AUC	-0.104	-0.221 to -0.003	2.1%
Recall	-0.112	-0.298 to +0.064	9.5%

Precision	+0.001	-0.133 to +0.137	50.4%
F1	-0.038	-0.176 to +0.092	27.8%
MCC	-0.049	-0.200 to +0.093	24.5%
Log-loss increase	+0.019	+0.001 to +0.036	2.1% probability forest has lower loss

Separate confidence intervals for the two model estimates overlap. The paired intervals are more informative because every bootstrap draw compares predictions on the **same resampled hosts**.

The paired analysis indicates that:

- the logistic model's ranking advantage is unlikely to be explained solely by which hosts happened to enter the bootstrap draw;
- the forest's probability loss is also consistently worse;
- differences in thresholded F1 and MCC are less precisely estimated.

Given the number of model variants examined over the wider project, the exact sign-flip (p) -value should be treated as a diagnostic rather than an isolated discovery claim. The practically important finding is the coherent direction across PR-AUC, recall, F1, MCC, and log-loss.

8. Discovery-method stratification

8.1 Transit discoveries

The transit subset contains 306 planets and 16 HZ-positive hosts.

Model	PR-AUC	Recall	F1	MCC	Log-loss
---	---	---	---	---	---
Method-adjusted logistic	0.582 [0.397, 0.858]	0.845 [0.625, 1.000]	0.680 [0.487, 0.832]	0.672 [0.488, 0.827]	0.103 [0.071, 0.139]
Random forest	0.510 [0.325, 0.795]	0.812 [0.575, 1.000]	0.643 [0.421, 0.807]	0.633 [0.416, 0.797]	0.117 [0.081, 0.159]

The paired PR-AUC difference is:

$$\sqrt{-0.072}$$

with a wide interval:

$$\sqrt{[-0.247, +0.069]}.$$

Thus, the transit ranking difference is not estimated precisely.

The log-loss result is clearer. The forest's transit log-loss is higher by approximately:

$$\sqrt{0.0145}$$

with an interval of:

$$\sqrt{[0.0030, 0.0268]}.$$

The forest does not improve the transit probability model.

8.2 Radial-velocity discoveries

The radial-velocity subset contains only 60 planets, 37 hosts, and 10 HZ-positive hosts.

Model	PR-AUC	Recall	F1	MCC	Log-loss
---	---	---	---	---	---
Method-adjusted logistic	0.589 [0.363, 0.878]	0.733 [0.388, 1.000]	0.552 [0.290, 0.770]	0.425 [0.109, 0.711]	0.383 [0.229, 0.560]
Random forest	0.458 [0.264, 0.735]	0.489 [0.205, 0.819]	0.503 [0.232, 0.738]	0.381	

[0.086, 0.682] | 0.428 [0.264, 0.621] |

The paired PR-AUC change is:

\[
-0.132
\]

with a 95% interval of:

\[
[-0.275,-0.019].
\]

The forest performs particularly poorly at ranking RV planets in this sample.

The RV log-loss difference is less precise because there are only ten positive hosts. Although the forest's point log-loss is worse, the paired interval includes no difference.

8.3 Detection-bias interpretation

The forest underperforms in both method subsets; its shortfall is not produced by one method alone.

However, none of these comparisons corrects survey completeness. They remain conditional on:

- the planet having been detected;
- the planet having been confirmed;
- period and stellar properties being available;
- reported insolation having been assigned.

There are still no imaging planets in the reported-label cohort. An imaging-specific comparison remains unidentified.

9. Held-out grouped permutation importance

Permutation importance was calculated strictly on held-out outer-fold planets.

For each outer fold:

- physical features were shuffled within discovery method;
- the transit/RV composition was therefore held approximately constant;
- temperature, radius, and mass were additionally shuffled together as one stellar-structure block;
- the joint shuffle used the same donor row for all three stellar features, preserving their internal covariance;
- 12 permutation replicates were generated per feature group per outer fold.

Uncertainty was estimated using 1,500 whole-host bootstrap resamples.

![Grouped permutation importance]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_grouped_permutation_importance.png)

[Vector PDF of the permutation-importance figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_grouped_permutation_importance.pdf)

9.1 Importance estimates

| Feature or feature block | PR-AUC decrease after permutation | Log-loss increase after permutation |

---|---:|---:|

| Orbital period | **0.235** [0.059, 0.446] | **0.091** [0.049, 0.140] |

| Stellar structure jointly | **0.197** [0.055, 0.376] | **0.036** [0.012, 0.062] |

| Stellar radius individually | 0.041 [-0.012, 0.112] | 0.0067 [-0.0030, 0.0165] |

| Stellar effective temperature individually | 0.023 [-0.017, 0.068] | 0.0055 [-0.0016, 0.0127] |

| Stellar mass individually | 0.017 [-0.027, 0.070] | 0.0059 [-0.0024, 0.0140] |

| Discovery method | 0.002 [−0.008, 0.015] | 0.0005 [−0.0005, 0.0016] |

9.2 Orbital period is the strongest individual predictor

Destroying the held-out period information reduces PR-AUC by approximately 0.235 and increases log-loss by approximately 0.091.

This is a large effect relative to the forest’s original:

$$\begin{aligned} & \backslash[\\ & \text{PR}\text{-}\text{AUC}=0.475 \\ & \backslash] \end{aligned}$$

and:

$$\begin{aligned} & \backslash[\\ & \text{log-loss}=0.162. \\ & \backslash] \end{aligned}$$

The forest therefore relies heavily on orbital period.

9.3 Stellar structure matters jointly

The joint temperature–radius–mass permutation also produces a substantial and well-supported decline:

$$\begin{aligned} & \backslash[\\ & \Delta \text{PR}\text{-}\text{AUC}=-0.197 \\ & \backslash] \end{aligned}$$

and:

$$\begin{aligned} & \backslash[\\ & \Delta \text{log-loss}=+0.036. \\ & \backslash] \end{aligned}$$

This independently reproduces the main collinearity conclusion from Chunk 5:

> **The stellar environment contains useful predictive information as a block, even though the data cannot assign that information cleanly to temperature, radius, or mass individually.**

9.4 Individual stellar importances are not reliable attribution measures

The individual temperature, radius, and mass intervals all include zero.

That does not contradict the strong joint-block result. The variables can substitute for one another:

- a tree that loses temperature may still split on radius;
- a tree that loses radius may still use stellar mass;
- all three trace correlated stellar structure.

Individual permutation importances are therefore not additive and should not be read as isolated physical effects.

9.5 Discovery method contributes almost nothing conditionally

Permuting the transit/RV feature changes PR-AUC and log-loss by almost zero.

This agrees with Chunk 5: once the physical feature distribution is available, the discovery-method label adds little residual predictive information.

This does **not** imply an unbiased catalog. It means the selection process is expressed mainly through:

$\mathbb{P}(X \mid \text{discovery method}),$

the kinds of systems entering each method's sample, rather than through a large residual method coefficient conditional on those observed physical properties.

10. Period-temperature response surface

A standard partial-dependence surface would independently alter stellar temperature while retaining each planet's original radius and mass. Because those stellar quantities are strongly correlated, that procedure could create unrealistic synthetic stars.

I therefore used a local conditional response surface.

At each temperature:

1. identify the 90 nearest observed rows in log stellar temperature;
2. retain their observed stellar radius, stellar mass, and discovery method;
3. assign host-equal Gaussian-kernel weights;
4. vary orbital period across the grid;
5. average the predictions over those locally plausible stellar configurations.

The final display forest uses the modal nested configuration:

- maximum depth three;
- minimum leaf size 11 on the full cohort;
- three candidate features per split;
- 1,024 trees for visual stability.

The surface is descriptive. Model performance remains based exclusively on the repeated out-of-host predictions.

![Random-forest and logistic period-temperature response surfaces]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_period_temperature_response_surface.png)

[Vector PDF of the response-surface figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_period_temperature_r

esponse_surface.pdf)

11. Does the forest recover a moderate-period band?

11.1 What it recovers

The forest learns several scientifically sensible structures without receiving an explicit interaction term:

- very short periods have low predicted HZ probability;
- probability rises over an intermediate period range;
- cooler stellar environments have higher probabilities;
- orbital period and stellar structure both matter;
- the highest-probability region is not at the shortest or longest edge of the period grid.

It therefore independently supports the broad statement:

> A moderate orbital regime is more favorable than an ultra-short orbit, and the stellar environment matters.

11.2 What it does not recover well

The forest does not produce a smooth diagonal ridge in which the favorable period steadily moves outward with stellar temperature.

Its conditional peak is approximately:

- 174 days around 3,000–4,000 K;
- 225 days from roughly 4,500 K upward.

By comparison, the logistic benchmark produces a smoother outward movement across the

plotted temperature range.

The exact peak periods are not physical HZ boundaries. They depend on:

- the selected cohort;
- conditional averaging;
- correlated stellar variables;
- the forest's shallow split structure;
- sparse positive support at warmer temperatures and long periods.

The relevant comparison is the **shape**:

- the forest produces broad, block-like plateaus;
- the logistic model produces a smoother period-temperature structure;
- the forest's preferred period changes relatively little with temperature.

11.3 Where the forest disagrees with logistic regression

The difference panel indicates that the forest tends to:

- assign higher probabilities than logistic regression in a localized moderate-period, intermediate-temperature region;
- assign lower probabilities over much of the long-period region around approximately 5,000–6,000 K;
- compress the probability surface into a smaller number of broad steps.

This suggests that the forest is fitting a few local concentrations of positives but not reliably ordering the rarer warmer-star, longer-period HZ systems.

That interpretation is consistent with its:

- lower PR-AUC;
- lower RV PR-AUC;
- lower recall;
- worse log-loss.

12. Statistical-power assessment

Relatively well supported

- The random forest contains real predictive information relative to the naive baseline.
- Orbital period is its strongest individual held-out predictor.
- Temperature, radius, and stellar mass matter collectively.
- The forest does not outperform the regularized logistic benchmark.
- Discovery method contributes little conditional importance after physical features are present.
- A smooth, high-resolution nonlinear interaction surface cannot be estimated reliably from 26 positive hosts.

Suggestive

- The forest identifies a broad moderate-period region.
- The cooler-star portion of the surface is more favorable.
- A localized intermediate-period and intermediate-temperature pocket may contain extra structure.

That local pocket should be treated cautiously because shallow tree thresholds can move when only a few positive hosts change folds.

Not established

- A stable forest-derived optimal period at each temperature.
- A complex high-order interaction beyond the physics-informed logistic structure.
- A method-specific nonlinear HZ boundary.
- An imaging-specific response surface.
- Population-level HZ occurrence rates.

13. Model-selection decision

Random forest decision: do not promote

The random forest will not be selected as the primary HZ classifier.

It fails the two pre-specified benchmark criteria:

1. It does not improve out-of-host PR-AUC.
2. It does not improve out-of-host log-loss.

It also does not produce a more convincing period–temperature surface.

The best point-performing model developed so far remains the **full stellar-structure ridge logistic model without the unnecessary method intercept**, which achieved:

$$\text{PR-AUC}=0.596$$

and:

$$\text{log-loss}=0.140.$$

The requested method-adjusted logistic benchmark also clearly outperforms the forest.

Why this is scientifically useful

The result strengthens, rather than weakens, the project.

It indicates that:

- the dominant physical pattern is relatively low dimensional;
- period and collective stellar structure contain most of the usable signal;
- the explicit physics-informed shape is more data-efficient than unconstrained local partitioning;
- additional model complexity should not be adopted merely because it is available.

The appropriate conclusion is not “random forests are poor models.” It is:

> For this small, selectively labeled exoplanet cohort, a shallow forest cannot estimate nonlinear structure more reliably than a regularized model informed by the expected astrophysical relationship.

14. Plain-language conclusion

> The random forest confirms that orbital period and the star’s overall physical structure are important. It also recognizes that extremely short orbits are generally unfavorable and that cooler stellar environments contain more promising candidates.

> But it does not predict new host systems as reliably as the simpler logistic model. With only 26 host stars contributing positive examples, the forest divides the data into broad, unstable regions rather than learning a precise temperature-dependent habitable-zone band.

> The practical lesson is that a more complicated algorithm is not automatically a better scientific model. In this dataset, the simpler model generalizes better and gives a more coherent representation of how the favorable orbital period changes with the host star.

15. Reproducibility artifacts

Presentation figures

- [Performance and nested configuration selection — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_performance_selection.png)
- [Performance and nested configuration selection — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_performance_selection.pdf)
- [Grouped permutation importance — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_grouped_permutation_importance.png)
- [Grouped permutation importance — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_grouped_permutation_importance.pdf)
- [Period–temperature response surfaces — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_period_temperature_response_surface.png)
- [Period–temperature response surfaces — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_period_temperature_response_surface.pdf)

Statistical results

- [Model metrics with whole-host bootstrap intervals]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/model_performance_host_bootstrap.csv)
- [Paired forest-versus-logistic comparison]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_vs_logistic_paired_host_bootstrap.csv)
- [Host-level log-loss sign-flip test]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_vs_logistic_logloss_sign_flip_test.csv)
- [Method-stratified performance]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/method_stratified_performance_host_bootstrap.csv)

- [Configuration-selection frequency]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/configuration_selection_frequency.csv)

- [Inner-CV configuration diagnostics]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/configuration_inner_cv_summary.csv)

- [Held-out grouped permutation importance]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/heldout_grouped_permutation_importance_host_bootstrap.csv)

- [Conditional response-surface grid]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/period_temperature_conditional_response_surface.csv)

- [Response-surface peak-period summary]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/period_temperature_response_peak_periods.csv)

Validation records and code

- [Aggregated repeated out-of-host forest predictions]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_repeated_group_cv_predictions_aggregated.csv)

- [Outer-fold validation audit]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_outer_fold_audit.csv)

- [Run metadata and exact design decisions]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/run_metadata.json)

- [Nested random-forest analysis script]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/run_random_forest_chunk6.py)

- [Bootstrap, importance, and surface-analysis script]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/analyze_random_forest_results.py)

- [Presentation-chart rendering script]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/render_chunk6_charts.py)

- [Packaged key artifacts]

(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/chunk6_random_forest_key_artifacts.zip)

16. Updated CRISP-DM mindmap

Exoplanet habitability drivers

└─ 1. Business understanding	COMPLETE
└─ Flux-based HZ-candidate definition	
└─ Moderate-period × cooler-star hypothesis	
└─ Catalog-conditional estimand	
└─ Detection and characterization bias	
└─ 2. Data understanding	COMPLETE
└─ Data dictionary and units	
└─ Missingness by field and method	
└─ Target availability	
└─ Host-level effective sample size	
└─ 3. Data preparation	COMPLETE
└─ Nullable reported-HZ target	
└─ Physics-derived sensitivity target	
└─ Leakage-safe feature sets	
└─ Log transformations	
└─ Host-grouped preprocessing	
└─ 4. Feature, outlier, and clustering analysis	COMPLETE
└─ Collinearity audit	
└─ Physics-aware outlier handling	
└─ Mass-period clusters	
└─ Cluster-HZ overlap	
└─ 5. Predictive modeling	IN PROGRESS
└─ 5A. Naive baselines	COMPLETE
└─ 5B. Period-temperature ridge logistic	COMPLETE
└─ Period curvature	
└─ Temperature association	
└─ Period × temperature interaction	
└─ 5C. Full stellar-structure ridge logistic	COMPLETE
└─ Radius and stellar mass	
└─ Coefficient redistribution	
└─ Joint stellar-block importance	
└─ 5D. Discovery-method adjustment	COMPLETE

- | | | — Ridge-shrunk RV/transit intercept
- | | | — Physical covariate shift
- | | | — No supported imaging coefficient
- | |
- | | — 5E. Constrained random forest COMPLETE – THIS
- CHUNK
- | | | — Three pre-specified shallow configurations
- | | | — Nested host-grouped selection
- | | | — Paired comparison with logistic benchmark
- | | | — Held-out grouped permutation importance
- | | | — Conditional period-temperature surface
- | | | — No benefit from added nonlinear flexibility
- | |
- | | — 5F. Compact final model comparison NEXT
 - | | — One tightly constrained gradient-boosting classifier
 - | | — No broad hyperparameter search
 - | | — Continuous log-insolation regression cross-check
 - | | — MAE, RMSE, R^2 , and HZ-boundary calibration
 - | | — Same host-grouped evaluation
 - | | — Stop model expansion after this comparison
- |
- | — 6. Final synthesis and recommendation
 - | — Compare all model families
 - | — Consolidate feature importance
 - | — Quantify the period-stellar interaction
 - | — Separate astrophysics from selection effects
 - | — State limitations and research priorities
 - | — General-audience conclusion

Next chunk

Chunk 7 will combine the remaining modeling work into one compact comparison. It will evaluate one tightly constrained gradient-boosting classifier and one continuous log-insolation regression analysis, without an open-ended search. The regression will test whether modeling flux directly produces a more stable scientific conclusion than thresholding it into HZ versus non-HZ. After that comparison, the project will move to final synthesis rather than adding further algorithms.

Standing requirement carried forward: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias using a staged, limited-compute CRISP-DM workflow.

Reply **continue** for Chunk 7.

continue with the compact Chunk 7 as planned: one tightly constrained gradient-boosting classifier and the log-insolation regression cross-check, no open-ended hyperparameter search.

One correction before this runs: Chunk 5 concluded the best-performing model overall is the "Full stellar structure" ridge logistic (no discovery-method term), since adding the method intercept did not improve PR-AUC or log-loss. Chunk 6 benchmarked the random forest against "Method-adjusted full" instead. Please use "Full stellar structure" as the primary benchmark for gradient boosting (and for any future comparison), and note in passing that the conclusion versus the forest does not change under the corrected benchmark, so this is a precision fix, not a reversal of Chunk 6.

Apply the same discipline used for the random forest: pre-specify a small number of shallow, regularized configurations (limited depth, limited learning rate, a minimum leaf/child-weight floor scaled to the positive host count) and select via the same one-standard-error nested rule — no broad grid or Bayesian search. Use the identical 90 outer host-grouped folds so all model families remain directly comparable. For the log-insolation regression, report MAE, RMSE, and R^2 under host-grouped validation, plus how often the regression's predicted flux and the classifier's HZ label agree on hot/HZ/cold zone membership, and state plainly whether regressing the continuous flux gives a more stable or more ambiguous picture than the binary threshold.

After this chunk, move directly to final synthesis in research-paper format, in chunks, as the last stage — do not introduce further algorithms or an ensembling step even if gradient boosting also underperforms the logistic benchmark. If gradient boosting does underperform, treat that the same way as Chunk 6: a real, reportable finding that the physics-informed logistic structure is more data-efficient than flexible tree ensembles at this sample size, not a reason to keep searching for a model that wins.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow. Explain the mindmap of where we are and what's next at the end of the chunk.

CHATGPT · SEP 01, 2026 11:11 AM

I'll preserve the exact 90 host-grouped folds, benchmark against the full-stellar ridge model, and test only three pre-specified shallow boosting configurations before the continuous log-flux cross-check.

CHATGPT · SEP 01, 2026 11:14 AM

The boosting ladder is fixed: one stump and two depth-two models at learning rates 0.02–0.03, with positive-host-scaled leaf floors and one-standard-error selection by inner grouped log-loss.

CHATGPT · SEP 01, 2026 11:23 AM

The one-standard-error rule selected boosted stumps in all 90 folds, rejecting added interaction capacity. Meanwhile, continuous-flux slopes closely match irradiation physics, with markedly smaller out-of-host uncertainty.

Exoplanet Habitability Drivers

Chunk 7 — Constrained Gradient Boosting and Continuous-Insolation Regression

All results in this chunk were calculated from the uploaded `all_exoplanets_2021.csv`.
[filecite]turn2file0[L1-L4]

This is the final modeling chunk. It evaluates:

1. one tightly constrained gradient-boosting classifier; and
2. one ridge regression of continuous log insolation.

No additional algorithms, ensembles, or open-ended searches are introduced.

1. Benchmark correction

The correct primary classifier benchmark is the **Full stellar structure ridge logistic model without a discovery-method term**, because Chunk 5 found that adding the method intercept slightly worsened both PR-AUC and log-loss.

Its repeated out-of-host performance was:

$$\text{PR}\text{-}\text{AUC}=0.596$$

and:

$$\text{log-loss}=0.140.$$

Chunk 6 compared the random forest with the method-adjusted logistic model instead. Repeating that comparison against the corrected benchmark strengthens, rather than reverses, the conclusion:

Corrected random-forest comparison Random forest minus full stellar ridge
--- ---:
PR-AUC -0.122 [-0.235, -0.019]
Log-loss +0.021 [+0.004, +0.040]

The random forest remains inferior. This is a precision correction, not a change in the substantive result.

2. Cohort and validation design

The labeled core cohort remains:

Property Count
--- ---:
Planets 366
Host systems 264
HZ-positive planets 32
HZ-positive host systems 26
Transit discoveries 306
Radial-velocity discoveries 60
Imaging discoveries 0

The exact same outer test memberships from Chunks 5 and 6 were reused:

$$\backslash[$$

$$15 \backslash \text{repeats} \times 6 \backslash \text{host-grouped folds}$$

$$=$$

$$90 \backslash \text{outer folds}.$$

$$\backslash]$$

Every planet received 15 predictions made by a model that had not seen any planet from that host.

All reported overall intervals use 3,000 whole-host bootstrap resamples. Method-specific results use 2,000 whole-host resamples. Paired log-loss comparisons use 100,000 host-level sign flips.

The automated audit confirmed:

- all 366 rows received exactly 15 gradient-boosting predictions;
- all 90 outer folds matched the earlier fold definitions;
- no host crossed a training–test boundary;
- each outer test fold contained three to six HZ-positive hosts;
- no model probability or regression prediction was missing.

3. Constrained gradient-boosting design

3.1 Features

For a like-for-like comparison with the best physical logistic model, gradient boosting received only:

$$\begin{aligned} & \backslash[\\ & \backslash\log_{10}(P) \\ & \backslash] \end{aligned}$$
$$\begin{aligned} & \backslash[\\ & \backslash\log_{10}(T_{\mathrm{eff}}) \\ & \backslash] \end{aligned}$$
$$\begin{aligned} & \backslash[\\ & \backslash\log_{10}(R_{\star}) \\ & \backslash] \end{aligned}$$

$$\log_{10}(M_{\star}).$$

It did not receive:

- reported or derived insolation;
- equilibrium temperature;
- an engineered period $\times$ temperature interaction;
- discovery method;
- discovery facility;
- any target-derived variable.

The tree learner therefore had to discover any nonlinear period relationship or stellar interaction from the physical inputs alone.

3.2 Pre-specified capacity ladder

Only three configurations were permitted.

Configuration	Depth	Learning rate	Trees	Minimum leaf rule	Subsample
Very conservative stumps	1	0.03	120	50% of positive training hosts	0.80
Shallow, low rate	2	0.03	100	45% of positive training hosts	0.80
Shallow, moderate rate	2	0.05	80	35% of positive training hosts	0.75

For every model fit:

$$n_{\mathrm{leaf,min}} = \max \left[7, \left\lceil f \times n_{\mathrm{positive\ training\ hosts}} \right\rceil \right]$$

\right\rceil
\right].
\]

Thus, a leaf could not be constructed around one or two unusual HZ systems.

There was:

- no unrestricted depth;
- no Bayesian search;
- no large hyperparameter grid;
- no early stopping against outer-test data;
- no outer-fold-driven model choice.

3.3 Nested model selection

Within every outer training set, all three configurations were evaluated using:

\[
2\ \text{\textit{repeats}}\times4\ \text{\textit{inner host-grouped folds}}.
\]

The configuration with minimum inner host-balanced log-loss was identified, after which the **simplest configuration within one standard error of that minimum** was selected.

The classification threshold was also chosen from inner out-of-host predictions only.

4. What the one-standard-error rule selected

![Classifier performance and constrained capacity selection]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.png)

[Vector PDF of the classifier figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.pdf)

| Configuration | Lowest point inner loss | Selected by one-SE rule |

|---|---:|---:|

| Depth-1 stumps | 2 of 90 folds | **86 of 90 folds** |

| Depth-2, learning rate 0.03 | 53 of 90 folds | 4 of 90 folds |

| Depth-2, learning rate 0.05 | 35 of 90 folds | 0 of 90 folds |

The deeper configurations frequently had the lowest numerical inner loss. However, their gains were usually smaller than host-level sampling uncertainty.

The one-standard-error rule consequently chose depth-one decision stumps in 95.6% of outer folds.

This matters scientifically. An additive ensemble of decision stumps cannot represent a genuine period $\times$ temperature interaction. Depth-two trees can, but the inner validation process almost never found sufficient evidence to justify that extra capacity.

The nested result is therefore:

> **The labeled sample supports nonlinear main effects, but it does not provide stable validation evidence for a more complex tree-derived interaction surface.**

The median training-only probability threshold was approximately 0.15, reflecting the minority prevalence and the emphasis on HZ recall.

5. Out-of-host classifier performance

All cells below report the estimate followed by a 95% whole-host bootstrap interval.

| Model | PR-AUC | Recall | Precision | F1 | MCC | Log-loss |

|---|---:|---:|---:|---:|---:|---:|

| Prevalence baseline | 0.077 [0.049, 0.107] | 0 | 0 | 0 | 0 | 0.271 [0.201, 0.346] |

| **Full stellar ridge logistic** | **0.596 [0.447, 0.789]** | **0.802 [0.631, 0.957]** | **0.542 [0.379, 0.698]** |
0.647 [0.496, 0.775] | **0.626 [0.476, 0.761]** | **0.140 [0.102, 0.181]** |
 | Constrained random forest | 0.475 [0.332, 0.678] | 0.691 [0.497, 0.870] | 0.535 [0.358, 0.709] |
 0.603 [0.432, 0.742] | 0.571 [0.394, 0.721] | 0.162 [0.119, 0.207] |
 | Constrained gradient boosting | 0.366 [0.243, 0.548] | 0.786 [0.600, 0.945] | 0.337 [0.215,
 0.465] | 0.472 [0.325, 0.599] | 0.456 [0.310, 0.586] | 0.183 [0.134, 0.234] |

Gradient boosting remains substantially better than the prevalence baseline, so it is learning real signal.

It nevertheless performs worse than the ridge-logistic benchmark on:

- ranking;
- precision;
- F1;
- Matthews correlation;
- log-loss;
- Brier score.

Its recall is close to the logistic model's recall, but it obtains that recall by flagging many more non-HZ planets.

6. Paired comparison with the primary benchmark

Because the models predict the same held-out hosts, paired differences are more informative than comparing two separate confidence intervals.

For discrimination metrics, positive values would favor boosting. For log-loss, the table reports the amount by which boosting is worse.

Metric	Gradient boosting minus full stellar ridge	95% host-bootstrap interval
PR-AUC	-0.231	-0.364 to -0.121
Recall	-0.016	-0.259 to +0.231
Precision	-0.206	-0.334 to -0.086

F1	**-0.176**	**-0.324 to -0.036**
Matthews correlation	**-0.170**	**-0.342 to -0.005**
Log-loss increase	**+0.042**	**+0.020 to +0.067**

None of the 3,000 paired host-bootstrap samples gave gradient boosting a higher PR-AUC than the logistic benchmark. None gave it a lower log-loss.

The host-level sign-flip test for the log-loss disadvantage produced:

$\backslash$
 $p_{\{\mathrm{one-sided}\}}=0.00005.$
 $\backslash$

The magnitude and consistency across metrics are more important than the isolated p-value:

> **Gradient boosting clearly fails to beat the full stellar-structure ridge model.**

This is not merely a failure to show superiority. The paired estimates indicate a material predictive disadvantage.

7. Discovery-method-stratified evaluation

Gradient boosting underperforms within both discovery methods, so its overall result is not caused solely by one method dominating the pooled cohort.

Discovery method	Model	PR-AUC	Log-loss
Transit	Full stellar ridge	0.591 [0.397, 0.861]	0.102 [0.069, 0.139]
Transit	Gradient boosting	0.383 [0.222, 0.624]	0.135 [0.092, 0.182]
Radial velocity	Full stellar ridge	0.599 [0.371, 0.869]	0.372 [0.226, 0.544]
Radial velocity	Gradient boosting	0.379 [0.210, 0.635]	0.469 [0.288, 0.689]

Among transit planets, both models recover approximately 84.5% of the host-balanced HZ cases, but boosting's lower precision reduces F1 from approximately 0.680 to 0.496.

Among RV planets:

- logistic recall is approximately 0.733;
- boosting recall is approximately 0.689;
- logistic F1 is approximately 0.581;
- boosting F1 is approximately 0.425.

The RV intervals remain wide because only ten RV hosts contribute positive labels.

No imaging comparison is possible because no imaging discovery has the reported-flux label and complete core feature set.

8. Held-out grouped permutation importance

Permutation importance was evaluated only in held-out outer folds.

To avoid confusing method composition with physical importance:

- orbital period was shuffled within discovery method;
- stellar quantities were shuffled by host within discovery method;
- temperature, radius, and mass were also moved together as one physically correlated block;
- eight permutations were performed per outer fold;
- intervals use 600 whole-host bootstrap resamples.

![Gradient-boosting grouped permutation importance]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_grouped_permutation_importance.png)

[Vector PDF of the importance figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_grouped_permutation_importance.pdf)

| Feature or block | PR-AUC decrease | Log-loss increase |

|---|---:|---:|

| Orbital period | 0.127 [-0.063, 0.314] | **0.0677 [0.0272, 0.1140]** |

Stellar structure jointly	**0.108 [0.027, 0.217]**	**0.0266 [0.0086, 0.0478]**
Stellar radius individually	0.070 [0.013, 0.160]	0.0051 [approximately 0, 0.0115]
Stellar mass individually	0.062 [0.006, 0.134]	0.0073 [−0.0011, 0.0172]
Stellar temperature individually	0.057 [−0.001, 0.130]	0.0069 [−0.0022, 0.0170]

Two conclusions survive the model's disappointing overall performance.

Orbital period matters

Removing held-out period information increases log-loss by approximately 0.068. Period is the most important individual predictor for the boosting model.

Stellar structure matters collectively

Moving temperature, radius, and mass together reduces PR-AUC and worsens log-loss substantially.

The block result is stronger and more interpretable than any individual stellar importance because the three variables can substitute for one another. This agrees with the ridge-logistic analysis:

> **The star's collective structure is reproducibly important, but attribution to temperature, radius, or mass individually is unstable.**

9. Does boosting discover the period–temperature relationship?

![Gradient-boosting versus ridge-logistic period–temperature surfaces]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_period_temperature_surface.png)

[Vector PDF of the period–temperature surface]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_period_temperature_surface.pdf)

The response surfaces use local conditional averaging over observed radius–mass configurations near each stellar temperature. This avoids constructing obviously unrealistic stars by varying temperature independently while holding unrelated radius and mass values fixed.

9.1 Structure learned by boosting

The modal depth-one boosting model finds:

- low probability at ultra-short periods;
- a broad increase over intermediate periods;
- lower probabilities in hotter stellar environments;
- a broad maximum at approximately 143 days.

Thus, boosting independently recovers the general idea that an intermediate orbital regime is more favorable than an ultra-short orbit.

9.2 Structure it does not learn

The location of the boosting maximum remains close to 143 days across nearly the entire temperature range.

It does not recover a stable diagonal ridge in which the preferred period moves progressively outward as stellar temperature increases.

That is consistent with the nested configuration result: depth-one stumps were selected in 86 of 90 folds, and stumps cannot represent a period $\times$ temperature interaction.

The full stellar ridge model shows a smoother outward-moving maximum, while the boosting surface is composed of broad additive steps.

Exact peak periods in either surface should not be interpreted as physical HZ boundaries.

They depend on:

- a small, selected labeled cohort;
- conditional averaging;
- correlated stellar measurements;
- regularization;
- the available period–temperature support.

The defensible finding is:

> Boosting finds a broad moderate-period window and a cooler-star main effect, but it does not provide additional evidence for a stable nonlinear period × temperature interaction.

10. Continuous log-insolation regression

Binary classification throws away information. A planet receiving 1.74 Earth flux and one receiving 100 Earth flux are both labeled non-HZ, despite being physically very different.

The cross-check therefore models:

$$\begin{aligned} & \backslash[\\ & y = \log_{10}(S/S_{\oplus}). \\ & \backslash] \end{aligned}$$

A ridge regression with fixed $\alpha=1$ was fitted using:

$$\begin{aligned} & \backslash[\\ & \log_{10}(P), \quad \\ & \log_{10}(T_{\mathrm{eff}}), \quad \\ & \log_{10}(R_{\star}), \quad \\ & \log_{10}(M_{\star}). \\ & \backslash] \end{aligned}$$

It used the exact same 90 outer host-grouped folds. The comparator assigned each outer-test

planet the mean log flux of its training hosts.

The physical approximation predicts:

$$\begin{aligned} & \propto \\ & S \\ & R_{\star}^2 \\ & T_{\mathrm{eff}}^4 \\ & M_{\star}^{-2/3} \\ & P^{-4/3}. \end{aligned}$$

Taking logarithms gives a linear relationship:

$$\begin{aligned} & \log S \\ & = \\ & C \\ & + \\ & 2 \log R_{\star} \\ & + \\ & 4 \log T_{\mathrm{eff}} \\ & - \\ & \frac{2}{3} \log M_{\star} \\ & - \\ & \frac{4}{3} \log P. \end{aligned}$$

![[Continuous-insolation regression cross-check]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.png)

[Vector PDF of the regression cross-check]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.pdf)

11. Continuous regression performance

Metric	Physics-form ridge regression	Training-mean baseline
MAE	0.0300 dex [0.0240, 0.0369]	0.936 dex [0.862, 1.011]
RMSE	0.0624 dex [0.0478, 0.0764]	1.141 dex [1.062, 1.219]
$\langle R^2 \rangle$	0.9970 [0.9954, 0.9982]	-0.0067 [-0.0247, -0.0058]

An MAE of 0.030 dex corresponds to a typical multiplicative error of approximately:

$$\sqrt{10^{0.030}} \approx 1.072,$$

or about 7%.

The RMSE corresponds to a multiplicative factor of approximately:

$$\sqrt{10^{0.062}} \approx 1.154.$$

Predictions are also stable across the 15 repeated outer fits. The median planet-level standard deviation across repeat predictions is only:

$$0.00284 \text{ dex}.$$

11.1 Method-specific regression results

Method	MAE	RMSE	$\langle R^2 \rangle$
---	---	---	---

| Transit | 0.0289 dex [0.0225, 0.0352] | 0.0581 [0.0436, 0.0708] | 0.9970 [0.9955, 0.9983] |
 | Radial velocity | 0.0354 dex [0.0160, 0.0615] | 0.0813 [0.0237, 0.1250] | 0.9919 [0.9790, 0.9993] |

The physical reconstruction works very well in both labeled method subsets, although RV uncertainty is wider.

12. Learned flux exponents

Term	Fitted exponent	95% host-bootstrap interval	Approximate physical exponent
---	---	---	---
Orbital period	-1.338	-1.353 to -1.325	-1.333
Stellar temperature	+4.031	+3.868 to +4.258	+4.000
Stellar radius	+1.863	+1.705 to +1.914	+2.000
Stellar mass	-0.550	-0.638 to -0.429	-0.667

The period and temperature exponents are extraordinarily close to the expected irradiation scaling.

The radius and mass estimates differ modestly from the idealized exponents. Likely contributors include:

- ridge shrinkage;
- catalog rounding;
- heterogeneous source references;
- reported flux sometimes being calculated from a directly tabulated semi-major axis rather than one inferred from period and mass;
- multiple-star or unusual-host systems;
- measurement error in stellar quantities.

This regression supplies the clearest confirmation that the catalog obeys the expected irradiation relationship.

It is crucial, however, to interpret that correctly.

Reported insolation in an exoplanet catalog is commonly calculated from stellar luminosity and orbital information closely related to these same predictors. The regression is therefore partly reconstructing the archive's calculation.

Consequently:

> **The R^2 of 0.997 is an excellent physical and data-consistency result, but it is not independent evidence that the model has discovered habitability from first principles.**

13. Hot/HZ/cold zone recovery from predicted flux

The continuous prediction was converted back into the three requested zones:

$\backslash$
 $S < 0.35$: $\backslash$ `\text{too cold}`
 $\backslash$

$\backslash$
 $0.35 \leq S \leq 1.75$: $\backslash$ `\text{HZ}`
 $\backslash$

$\backslash$
 $S > 1.75$: $\backslash$ `\text{too hot}`.
 $\backslash$

Zone metric Estimate [95% host-bootstrap interval]
--- ---:
Three-zone accuracy 0.9867 [0.9735, 0.9981]
Three-zone macro recall 0.9523 [0.8776, 0.9993]
HZ recall 0.9506 [0.8422, 1.000]
HZ precision 0.8850 [0.7424, 1.000]
HZ F1 0.9166 [0.8187, 0.9872]
HZ Matthews correlation 0.9101 [0.8081, 0.9863]

The row-level confusion pattern was:

Observed zone	Predicted too cold	Predicted HZ	Predicted too hot
--- ---	---	---	---
Too cold	16	1	0
HZ	0	31	1
Too hot	0	2	315

Only four of the 366 planets cross a zone boundary:

Planet	Reported flux	Reported zone	Predicted flux	Predicted zone
--- ---	---	---	---	---
GJ 3323 b	2.58	Too hot	1.25	HZ
Kepler-1704 b	0.34	Too cold	0.66	HZ
Kepler-436 b	1.69	HZ	2.60	Too hot
Kepler-439 b	1.83	Too hot	1.65	HZ

Three lie close to a classification boundary. GJ 3323 b shows a larger disagreement and merits a catalog-component consistency review.

14. Agreement between flux regression and the primary classifier

A binary classifier cannot distinguish “too hot” from “too cold.” Therefore, an exact three-class agreement statistic between the two models is not defined.

The appropriate comparison has two parts:

1. compare the regression’s HZ/non-HZ implication with the logistic classifier’s HZ/non-HZ label;
2. examine whether logistic positives fall in the regression’s too-cold, HZ, or too-hot zone.

14.1 Binary HZ agreement

Agreement metric	Estimate [95% interval]
------------------	-------------------------

|---|---:|

| Overall binary HZ agreement | **0.9386 [0.9093, 0.9636]** |

| HZ-set Jaccard overlap | **0.5227 [0.3692, 0.6787]** |

| Regression-predicted HZ share | 0.0824 [0.0527, 0.1151] |

| Logistic-classified HZ share | 0.1134 [0.0785, 0.1506] |

The 93.9% overall agreement is partly driven by the large number of non-HZ planets. Positive-set overlap is more modest: the weighted Jaccard index is approximately 52%.

14.2 Three-zone composition of the logistic labels

The following are host-balanced row percentages:

| Logistic classifier label | Regression too cold | Regression HZ | Regression too hot |

|---|---:|---:|---:|

| Logistic non-HZ | 1.3% | 1.7% | 97.0% |

| Logistic HZ | 24.9% | **59.3%** | 15.9% |

The logistic model's HZ screen is deliberately broader than the exact flux-regression interval:

- it is tuned to recover HZ candidates under severe class imbalance;
- its operating threshold is not identical to the 0.35–1.75 flux boundaries;
- it cannot distinguish the two physically different non-HZ states.

This explains why it achieves high HZ recall while having lower positive-zone agreement than the nearly deterministic flux reconstruction.

15. Is continuous regression more stable than binary classification?

Predictively: yes

The continuous regression is far more stable:

- R^2 is approximately 0.997;
- repeat-to-repeat variation is extremely small;
- three-zone accuracy is approximately 98.7%;
- 31 of 32 reported HZ planets remain HZ after out-of-host flux prediction.

It uses the full continuous ordering of the target rather than discarding that information at two thresholds.

Scientifically: stable, but less independent

The regression gives a clearer picture of the irradiation law:

- flux decreases strongly with period;
- flux rises steeply with stellar temperature;
- stellar radius and mass modify the period–flux relationship;
- the fitted exponents closely match the expected physical equation.

But this clarity is partly built into the catalog target. Reported insolation is itself usually calculated from closely related stellar and orbital quantities.

The binary model is noisier because it asks the more difficult screening question at two sharp boundaries and is evaluated on only 26 positive hosts.

The correct synthesis is:

> Regressing continuous flux gives a much more stable and physically coherent account of stellar irradiation, but it does not provide more independent evidence of actual habitability. It is best treated as a physics-consistency cross-check for the binary HZ analysis.

16. Astrophysics versus detection bias

16.1 What appears astrophysical

Across model families, the most reproducible physical findings are:

- orbital period is a major predictor;
- very short periods are generally unfavorable;
- the relationship is not “longer is always better”;
- stellar temperature, radius, and mass matter jointly;
- cooler and less luminous stellar environments permit temperate flux at shorter orbital periods;
- the physical flux scaling is accurately reconstructed from period and stellar structure.

16.2 What remains affected by observation

The labeled sample contains only transit and radial-velocity discoveries.

The two methods populate different regions of feature space:

- transit observations heavily sample shorter periods;
- RV observations supply a larger fraction of longer-period systems in this labeled cohort;
- reported flux exists only for a selected, well-characterized subset;
- RV probability calibration remains less precise;
- imaging contributes no reported-flux model cases.

Gradient boosting underperforms within both transit and RV subsets, so its poor performance is not merely a pooling artifact.

Nevertheless, neither boosting nor regression corrects the probability that a system was:

- surveyed;
- detected;
- confirmed;
- followed up;

- assigned complete stellar parameters;
- assigned a reported insolation value.

The finding from Chunk 5 remains the most precise bias statement:

> The raw transit–RV HZ-rate difference is largely reproduced by their different physical covariate distributions. A method label adds little after those features are known, but the upstream catalog-selection process remains substantial.

16.3 Imaging remains unidentified

No imaging planet can enter the reported-target classifier or regression.

The analysis therefore cannot estimate:

- imaging-specific HZ prevalence;
- an imaging-specific period–temperature interaction;
- comparative classifier performance for imaging discoveries.

This is lack of labeled support, not evidence that directly imaged planets are non-HZ.

17. Final model decision

Primary classifier

Retain the:

> Full stellar-structure ridge logistic model without discovery method

as the principal HZ screening model.

It has the best out-of-host combination of:

- PR-AUC;
- recall;
- precision;
- F1;
- Matthews correlation;
- log-loss;
- interpretability;
- data efficiency.

Physical cross-check

Retain the:

> Ridge regression of log reported insolation

as a physical reconstruction and target-consistency analysis.

It should not replace the classifier in claims about independent habitability prediction.

Models not promoted

Do not promote:

- constrained random forest;
- constrained gradient boosting;
- method-adjusted logistic regression.

All three fail to improve on the full stellar ridge benchmark.

Model expansion stops here

No gradient-boosting escalation, ensembling, or additional algorithm search is justified.

The negative ensemble results are themselves scientifically reportable:

> **With only 26 positive host systems, physics-informed regularization is more data-efficient than flexible tree ensembles.**

18. Plain-language conclusion

> Planets are most likely to receive temperate levels of starlight when their orbital period is appropriate for the size, mass, and temperature of their star. Very short orbits are generally too hot, but simply choosing the longest orbit is not the answer because distant planets can become too cold.

> The data consistently show that orbital period and the star's overall structure matter. Cooler stars can provide moderate heating at shorter periods, whereas hotter and more luminous stars require more distant, slower orbits.

> More complicated machine-learning models did not improve predictions. With only 26 host systems supplying positive examples, tree ensembles were less reliable than a simpler model whose structure reflects the expected astrophysics.

> Predicting the continuous stellar flux was extremely accurate, but that is largely because catalog insolation is calculated from the same kinds of orbital and stellar measurements. It confirms that the physical relationship is internally consistent; it does not prove that the planets are actually habitable.

19. Reproducibility artifacts

Presentation figures

- [Classifier performance and capacity selection — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.png)
- [Classifier performance and capacity selection — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.pdf)
- [Gradient-boosting grouped permutation importance — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_grouped_permutation_importance.png)
- [Gradient-boosting grouped permutation importance — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_grouped_permutation_importance.pdf)
- [Period-temperature response surfaces — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_period_temperature_surface.png)
- [Period-temperature response surfaces — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_period_temperature_surface.pdf)
- [Continuous-insolation regression cross-check — PNG]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.png)
- [Continuous-insolation regression cross-check — vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.pdf)

Key statistical tables

- [Classifier performance with host-bootstrap intervals]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_model_performance_host_bootstrap.csv)
- [Paired classifier comparisons]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_paired_model_contrasts_host_bootstrap.csv)
- [Method-stratified classifier performance]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_method_stra

tified_performance.csv)

- [Nested boosting configuration results]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_configuration_summary.csv)

- [Held-out grouped permutation importance]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/gb_heldout_grouped_permutation_importance_host_bootstrap.csv)

- [Log-insolation regression metrics]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_metrics_host_bootstrap.csv)

- [Flux-zone classification metrics]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_zone_metrics_host_bootstrap.csv)

- [Regression coefficient bootstrap results]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_coefficient_bootstrap.csv)

- [Classifier-regression HZ agreement]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_regression_hz_agreement_host_bootstrap.csv)

- [Four zone disagreements]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_flux_regression_zone_disagreements.csv)

Audit, metadata, and package

- [Validation and prediction QA checks]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_qa_checks.csv)

- [Verified run metadata]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_run_metadata_verified.json)

- [Compact analysis summary]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_analysis_summary.json)

- [Verified Chunk 7 artifact package]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_verified_key_artifacts.zip)

20. Updated CRISP-DM mindmap

Exoplanet habitability drivers

	└─ 1. Business understanding	COMPLETE
	├─ Define radiative HZ candidate	
	├─ Define moderate-period × cooler-star hypothesis	
	├─ Separate habitability from irradiation	
	└─ Define catalog-conditional estimand	
	└─ 2. Data understanding	COMPLETE
	├─ Data dictionary and units	
	├─ Missingness and target coverage	
	├─ Method-dependent characterization	
	└─ Host-level effective sample size	
	└─ 3. Data preparation	COMPLETE
	├─ Cleaning and integrity checks	
	├─ Nullable HZ labels	
	├─ Physics-derived sensitivity variables	
	├─ Leakage controls	
	└─ Host-grouped preprocessing	
	└─ 4. Exploratory and structural analysis	COMPLETE
	├─ Raw visual baseline	
	├─ Collinearity and feature selection	
	├─ Physics-aware outlier analysis	
	├─ Mass-period clustering	
	└─ Cluster-HZ overlap	
	└─ 5. Predictive modeling	COMPLETE
	└─ 5A. Naive baselines	COMPLETE
	└─ 5B. Period-temperature logistic	COMPLETE
	├─ Moderate-period curvature	
	└─ Suggestive period × temperature interaction	
	└─ 5C. Full stellar-structure ridge logistic	COMPLETE
	├─ Best-performing classifier	
	├─ Joint stellar-block importance	
	└─ Collinearity-aware interpretation	
	└─ 5D. Discovery-method adjustment	COMPLETE

			— No incremental method-intercept value	
			— Bias expressed mainly through covariate shift	
			— 5E. Constrained random forest	COMPLETE
			— Does not beat the ridge benchmark	
			— 5F. Constrained gradient boosting	COMPLETE – THIS
CHUNK				
			— Three pre-specified configurations	
			— Nested one-standard-error selection	
			— Stumps selected in 86 of 90 folds	
			— Does not beat the ridge benchmark	
			— 5G. Continuous log-insolation regression	COMPLETE – THIS
CHUNK				
			— Same 90 outer host folds	
			— MAE, RMSE, and R^2	
			— Physical-exponent recovery	
			— Hot/HZ/cold boundary analysis	
			— Stable but partly deterministic cross-check	
			— 6. Model expansion	CLOSED
			— No further algorithms	
			— No ensembling	
			— No search for a manufactured “winning” model	
			— 7. Research-paper synthesis	NEXT
			— Part 1: Title, abstract, introduction, and methods	
			— Part 2: Results and figure narrative	
			— Part 3: Discussion, detection bias, and limitations	
			— Part 4: Final conclusions and general-audience summary	

The next chunk will begin the **final research-paper synthesis, Part 1: title, abstract, scientific motivation, hypotheses, data, and methods**, without introducing any new models.

Standing requirement carried into the synthesis: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias using a staged, limited-compute CRISP-DM

workflow.

Reply **continue** for Final Synthesis, Part 1.

PROMPT · SEP 01, 2026 11:49 AM

continue with Final Synthesis, Part 1.

Since this synthesis will also be excerpted directly into a public Medium article and a GitHub README (not just submitted as a chat transcript), please write the abstract and introduction so they work as standalone text for a first-year master's data science class with no physics or astronomy background. Assume the reader knows CRISP-DM, cross-validation, class imbalance, regularization, and feature importance well, but has never heard of insolation flux, equilibrium temperature, stellar spectral type, or Kepler's third law. Every physics term must be introduced with a plain-language analogy the first time it appears (e.g., explain "habitable zone" as "the orbital distance where a planet gets roughly Earth-like sunlight – not so close it's roasted, not so far it's frozen," not as a flux-ratio inequality). Do not lead with equations. Lead with the concrete, surprising findings framed as a data science story: the moderate-period × cooler-star pattern, the joint-stellar-block result despite unstable individual coefficients, the discovery-method covariate-shift finding, and the negative results for random forest and gradient boosting versus the simpler logistic model.

In the methods section, keep the full rigor (host-grouped validation, bootstrap methodology, leakage controls, the reported-versus-derived flux distinction) but write it for the same data-science-fluent, physics-naïve reader – explain **why** each astrophysical quantity was included in data-science terms (e.g., "stellar temperature, radius, and mass are three measurements of the same underlying thing, so they behave like collinear features" rather than assuming familiarity with stellar structure). Push equations (Kepler's law, the flux-scaling relation, the fitted exponents) into a clearly separated, optional "for the physics-curious" callout rather than the main narrative flow, so a reader can skip it without losing the argument.

Please also explicitly state, in one paragraph a data science student could quote directly, what this analysis does and does not claim about actual planetary habitability versus radiative flux screening, since that distinction needs to survive intact into a general-audience article title and framing.

Remember the standing requirement: determine which stellar and orbital features predict habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the real astrophysical relationship from transit, radial-velocity, and imaging selection bias, using a staged, limited-compute CRISP-DM workflow.

CHATGPT · SEP 01, 2026 11:49 AM

I'll turn the completed analysis into standalone paper prose: findings-first abstract, accessible introduction, explicit claim boundaries, rigorous methods, and a skippable physics callout.

continue with Final Synthesis, Part 1.

Since this synthesis will also be excerpted directly into a public Medium article and a GitHub README (not just submitted as a chat transcript), please write the abstract and introduction so they work as standalone text for a first-year master's data science class with no physics or astronomy background. Assume the reader knows CRISP-DM, cross-validation, class imbalance, regularization, and feature importance well, but has never heard of insolation flux, equilibrium temperature, stellar spectral type, or Kepler's third law. Every physics term must be introduced with a plain-language analogy the first time it appears (e.g., explain "habitable zone" as "the orbital distance where a planet gets roughly Earth-like sunlight — not so close it's roasted, not so far it's frozen," not as a flux-ratio inequality). Do not lead with equations. Lead with the concrete, surprising findings framed as a data science story: the moderate-period $\times$ cooler-star pattern, the joint-stellar-block result despite unstable individual coefficients, the discovery-method covariate-shift finding, and the negative results for random forest and gradient boosting versus the simpler logistic model.

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Predicting Radiative Habitable-Zone Membership from Stellar and Orbital Features

A CRISP-DM Study of 4,575 Confirmed Exoplanets

Suggested public-facing subtitle:

Why a physics-informed logistic model outperformed random forests and gradient boosting — and what that does not prove about alien life

Abstract

This study asks a deliberately narrower question than “Which planets are habitable?” It asks which stellar and orbital measurements predict whether a detected exoplanet receives roughly Earth-like levels of starlight. That orbital region is commonly called the **habitable zone**: the “Goldilocks” distance where a planet is not so close to its star that it is roasted and not so far away that it is frozen. Receiving the right amount of light is only one prerequisite for habitability, but it is a measurable and useful screening target.

We analyzed a 2021 catalog of 4,575 confirmed exoplanets using the CRISP-DM framework. The primary target was based on **insolation flux**, meaning the amount of starlight reaching a planet relative to the amount Earth receives. Only 370 planets had a reported flux value, and only 32 met the adopted habitable-zone criterion. Those 32 planets belonged to just 26 independent host-star systems, making class imbalance and grouped validation central methodological concerns.

The main predictive pattern was not “longer orbital periods are better.” Instead, habitable-zone candidates occupied a moderate-period region whose location depended on the host star. Cooler stars can provide moderate heating at relatively short periods, while hotter or larger stars generally require more distant, slower orbits. A regularized logistic model using orbital period, stellar temperature, stellar radius, and stellar mass achieved an out-of-host precision-recall area under the curve of 0.596, compared with a no-skill level of approximately 0.077.

Stellar temperature, radius, and mass behaved like strongly collinear measurements of a common underlying stellar structure. Their individual coefficients were unstable: after radius and mass were added, the temperature coefficient moved 87% toward zero and the explicit period-by-temperature interaction shrank by about 80%. Yet removing the stellar variables as a group reduced out-of-host PR-AUC by approximately 0.295 and worsened log-loss by 0.058. The correct interpretation is therefore that **stellar structure matters collectively, even when the data cannot assign stable credit to one stellar measurement at a time.**

Discovery method introduced substantial covariate shift. Transit surveys, which detect the small dimming caused when a planet crosses in front of its star, populated different regions of feature space from radial-velocity surveys, which detect the star's back-and-forth wobble under a planet's gravitational pull. The raw habitable-zone rate was much higher among radial-velocity discoveries, but a discovery-method indicator added no predictive value once orbital and stellar features were included. This suggests that the observed method difference arose mainly because the methods sampled different kinds of systems, not because the method label itself contained residual physical information.

More flexible models did not help. A constrained random forest reached a PR-AUC of 0.475, and constrained gradient boosting reached 0.366; both also produced worse log-loss than the simpler logistic model. By contrast, continuous regression reconstructed reported insolation with $(R^2=0.997)$, but this result was partly deterministic because catalog flux is calculated from closely related stellar and orbital measurements.

The principal conclusion is that a low-dimensional, physics-informed, regularized model was more data-efficient than flexible tree ensembles at this sample size. Orbital period and collective stellar structure predict radiative habitable-zone membership, but the explicit temperature–period interaction remains suggestive rather than independently conclusive, and the catalog does not support claims about actual habitability or unbiased planetary occurrence rates.

1. Introduction

1.1 A data-science problem hidden inside an astronomy catalog

An exoplanet is simply a planet orbiting a star other than the Sun. Its **host star** is the star it travels around, just as the Sun is Earth's host star. Thousands of such planets have now been detected, but they form an unusually difficult machine-learning dataset.

The difficulty is not primarily the number of records. A table with several thousand rows is small by modern machine-learning standards. The more important challenges are structural:

- many scientifically valuable columns are missing for most planets;
- the missingness depends on how each planet was discovered;
- several planets can orbit the same star, creating grouped observations;
- the positive class is rare;
- the most physically meaningful predictors are highly correlated;
- the target can be nearly reconstructed from some of the predictors, creating leakage risks;
- the catalog represents detected and characterized planets, not a random sample of all planets.

This makes the problem an instructive example of why model evaluation cannot be separated from domain understanding.

A random row split, global median imputation, and a large hyperparameter search could easily produce an impressive-looking score while answering the wrong scientific question. The analysis therefore followed a staged CRISP-DM process: define a narrow target, audit how it was measured, understand the selection mechanism, construct leakage-safe feature sets, validate by host system, and increase model complexity only when repeated grouped validation justified it.

1.2 What is being predicted?

The target is based on **insolation flux**. Insolation flux is a planet's sunlight budget: how much energy from its star reaches the planet, expressed relative to Earth's sunlight budget. A value of one means that the planet receives approximately the same amount of stellar energy per unit area that Earth receives from the Sun.

This quantity is central because a planet's surface cannot remain temperate if it receives an extreme amount of energy or almost none at all. The analysis defined a radiative habitable-zone candidate as a planet receiving between 0.35 and 1.75 times Earth's flux.

In plain language, this is a deliberately broad “not obviously roasted and not obviously frozen” screen.

That definition does not directly measure:

- whether the planet is rocky or gaseous;
- whether it has an atmosphere;
- whether that atmosphere traps heat;
- whether liquid water is present;
- whether the climate is stable;
- whether the star produces destructive flares;
- whether any form of life exists.

The target is therefore best understood as **radiative suitability**, not habitability itself.

Quote-ready scope statement

> **This analysis does not determine whether any exoplanet is actually habitable or inhabited. It predicts whether a detected planet receives a broadly Earth-like amount of stellar energy—a first-pass radiative screening condition. A planet can fall inside that light-and-heat range and still be a gas giant, lack an atmosphere, experience extreme radiation, or be otherwise unsuitable for life. Conversely, atmospheric effects could make some planets outside the adopted range scientifically interesting. The results should therefore be described as predictions of habitable-zone candidacy, not discoveries of habitable worlds.**

1.3 Why “moderate orbital period” depends on the star

A planet's **orbital period** is the length of its year: the time needed to complete one orbit

around its star.

At first glance, one might expect longer periods to be more favorable because longer-period planets are usually farther from their stars and therefore receive less heat. But that relationship is not monotonic.

A planet with an extremely short year is often very close to its star and intensely heated. As the orbital period grows, irradiation generally falls into a more moderate range. If the period becomes sufficiently long, however, the planet may receive too little energy and become too cold.

The expected pattern is therefore closer to an inverted U:

very short period	moderate period	very long period
usually too hot	→ potentially temperate	→ potentially too cold

The location of that moderate region depends on the host star.

Stellar effective temperature is a standardized measure of how hot the star's visible surface is. It can be understood as a numerical version of the star's color: cooler stars look redder, while hotter stars look whiter or bluer.

A cooler star emits less energy per unit surface area, so a planet can orbit relatively close to it without being overheated. A hotter star generally requires a planet to orbit farther away.

Temperature is not the entire story. **Stellar radius** measures the physical size of the star, much as a radius measures the size of a ball. A large star can emit much more total energy than a smaller star even when their surface temperatures are similar. **Stellar mass** measures the amount of matter in the star and affects how orbital period translates into orbital distance.

The scientific hypothesis is therefore conditional:

> Planets with moderate orbital periods should be more likely to fall within the radiative habitable zone, but the meaning of "moderate" should change with the temperature, size, and mass of the host star.

This is naturally a feature-interaction problem.

1.4 The first surprising result: the interaction is visible but difficult to isolate

The raw data showed the expected broad geometry. Habitable-zone candidates appeared at shorter periods around cooler stars and at longer periods around warmer stars.

A logistic model using period and temperature also produced a period-by-temperature interaction in the expected direction. Its predicted favorable period moved outward as stellar temperature increased.

However, the interaction's uncertainty interval included zero. It was suggestive rather than conclusive.

The interpretation became more subtle after stellar radius and mass were added. The explicit temperature coefficient and period-by-temperature interaction both shrank sharply. A superficial coefficient reading might suggest that temperature ceased to matter.

That would be incorrect.

Temperature, radius, and mass behave like three collinear measurements of the same underlying stellar system. In ordinary stars:

- hotter stars tend to be larger;
- hotter stars tend to be more massive;
- larger stars tend to be more luminous;
- mass changes how a given orbital period maps to physical distance.

The model therefore had several correlated ways to represent similar information. Regularization redistributed the predictive signal among them.

When the stellar variables were tested jointly, their contribution was clear. Removing the complete stellar block—temperature, radius, mass, and the interaction—caused a large decline in out-of-host performance.

This leads to an important data-science lesson:

> When domain variables are strongly collinear, unstable individual coefficients do not imply that the feature family is unimportant. Grouped or block-level importance can be much more scientifically meaningful than assigning credit one variable at a time.

1.5 Discovery method is part of the data-generating process

Exoplanets in the catalog were not discovered through one uniform measurement process.

The **transit method** detects a small, repeated dip in a star's brightness when a planet crosses in front of it. A useful analogy is a tiny insect passing across a distant lamp: the lamp becomes slightly dimmer at regular intervals. This method is especially effective for planets that orbit frequently and happen to cross our line of sight.

The **radial-velocity method** detects the small back-and-forth motion of a star caused by the gravitational pull of an orbiting planet. The star and planet behave like two unequal dancers moving around a shared balance point. This technique is particularly informative about planetary mass and can detect systems that do not transit.

Direct imaging attempts to separate the faint light of a planet from the overwhelming glare of its star, similar to trying to photograph a firefly beside a searchlight. It tends to find young, bright, massive planets at wide separations.

These techniques do not sample the same planet population.

In machine-learning terms, discovery method creates **covariate shift**:

$$\begin{aligned} & \mathbb{P}(X \mid \text{transit}) \\ & \neq \\ & \mathbb{P}(X \mid \text{radial velocity}) \\ & \neq \\ & \mathbb{P}(X \mid \text{imaging}). \end{aligned}$$

\]

The observed features have different distributions depending on the measurement channel.

This distinction mattered in the analysis. Radial-velocity discoveries had a much higher raw habitable-zone rate than transit discoveries among the small subset with reported flux. Yet adding a regularized discovery-method indicator to the full physical model did not improve out-of-host PR-AUC or log-loss.

The physical model already reproduced the method-level prevalence difference because the transit and radial-velocity samples contained different orbital and stellar systems.

This does not mean discovery bias vanished. It means the method effect was expressed primarily through which cases appeared in the sample, rather than through an additional intercept after the observed physical variables were known.

A comparable imaging analysis was not possible. No directly imaged planet had both a reported flux label and the complete core predictor set. That absence is a support limitation, not evidence that imaged planets cannot occupy habitable-zone orbits.

1.6 The second surprising result: simpler modeling won

The project deliberately increased model flexibility in stages.

The strongest classifier was not a random forest or a gradient-boosting ensemble. It was a regularized logistic model with a small, domain-informed feature set:

- log orbital period;
- curved period effect;
- stellar temperature;
- stellar radius;
- stellar mass.

Its out-of-host PR-AUC was 0.596.

A constrained random forest achieved 0.475. Constrained gradient boosting achieved 0.366. Both also produced worse log-loss.

These were not uncontrolled default models. Their depth, leaf size, learning rate, and configuration search were deliberately restricted, and configuration selection was nested inside the host-grouped validation.

The negative result is therefore informative:

> Flexible tree ensembles could learn broad period and stellar effects, but with only 26 positive host systems they could not estimate a more reliable decision surface than the structured logistic model.

This is a practical example of data efficiency. The logistic model encoded a small amount of justified structural bias: a smooth period relationship and a limited interaction hypothesis. The tree models had to infer that structure from sparse positive examples.

In this setting, the simpler model generalized better because it asked less of the data.

1.7 Study objectives

The analysis addressed four connected questions.

First, which orbital and stellar features predict whether a detected exoplanet lies in the adopted radiative habitable zone?

Second, does the evidence support the hypothesis that a moderate orbital period around a cooler star is more favorable than an extremely short or long period?

Third, does an explicit period-by-temperature interaction improve predictions once stellar radius and mass are included?

Fourth, how much of the observed relationship reflects physical irradiation, and how much reflects the different regions of feature space sampled by transit, radial-velocity, and imaging discoveries?

These questions were answered using prediction, uncertainty analysis, feature-block tests, unsupervised clustering, method stratification, and a continuous-flux cross-check.

2. Methods

2.1 Study design and CRISP-DM structure

The analysis followed the six standard CRISP-DM phases:

| CRISP-DM phase | Application in this study |

|---|---|

| Business understanding | Define a narrow radiative screening question rather than claiming actual habitability |

| Data understanding | Audit all fields, units, missingness, discovery-method composition, grouping, and target coverage |

| Data preparation | Preserve unknown targets, avoid broad imputation, construct physics-based sensitivity variables, and prevent leakage |

| Modeling | Progress from baselines to regularized logistic regression and then to constrained tree ensembles |

| Evaluation | Use repeated host-grouped validation, whole-host bootstrap intervals, proper scoring rules, and paired model comparisons |

| Deployment or communication | Produce reproducible tables, publication-quality figures, and a plain-language distinction between flux screening and habitability |

The staged design was important because the main modeling limitation was not compute. It was information: only 26 host systems contributed positive reported-flux examples.

2.2 Data source and observational unit

The source table was the uploaded `all_exoplanets_2021.csv`, a Kaggle snapshot derived from the NASA Exoplanet Archive. `filecite{turn2file0L1-L4}`

It contained:

- 4,575 planet rows;
- 3,392 unique host systems;
- 23 columns;
- 11 discovery methods;
- discovery years from 1989 through 2021.

Each row represented a planet, but planets orbiting the same star were not treated as statistically independent.

They share:

- the same stellar temperature;
- the same stellar radius and mass;
- the same distance from Earth;
- much of the same observing history;
- correlated chances of follow-up characterization.

`Planet Host` was therefore retained as a grouping key rather than used as a predictive feature.

2.3 Outcome definitions

Reported-flux outcome

The primary target used the catalog's reported **insolation flux**, the planet's starlight exposure relative to Earth.

A planet was labeled:

- too cold when its flux was below 0.35 Earth flux;
- habitable-zone candidate from 0.35 through 1.75;
- too hot above 1.75.

For binary classification, the middle category was positive and the two outer categories were negative.

Missing reported flux remained an unknown target. It was never converted into a non-HZ label.

Only 370 planets had reported flux:

- 32 were HZ candidates;
- 338 were outside the adopted range;
- 4,205 had no reported label.

Four labeled planets lacked stellar temperature, leaving a primary core cohort of:

- 366 planets;
- 264 host systems;
- 32 positive planets;
- 26 positive host systems.

Derived-flux outcome

A second flux value was calculated from stellar and orbital measurements where possible.

This **derived flux** extended coverage to approximately 4,013 planets and was useful for:

- checking physical consistency;
- visualizing broader catalog structure;
- comparing cluster overlap;
- examining a small number of imaging systems.

It was not treated as an independent target for feature discovery.

Derived flux is calculated from the same types of features the model would use as predictors. Training a classifier to predict that derived label and then declaring those predictors important would be a form of circular reasoning.

The distinction was therefore maintained throughout:

```
reported flux → primary observational target  
derived flux  → physics-based sensitivity and consistency target
```

2.4 Predictor definitions in data-science terms

Orbital period

Orbital period is the planet's year length. It was log-transformed because the catalog spans extremely short orbits and exceptionally wide systems over many orders of magnitude.

A squared period term was added to the logistic models so that the relationship could rise and then fall. This represented the expectation that both very short and very long periods may be unfavorable.

Stellar effective temperature

This is a continuous numerical measure of stellar surface temperature. It contains more information than **spectral type**, which is a categorical label grouping stars broadly by color and temperature.

Spectral type can be understood as a coarse binning of a continuous temperature

measurement. Because it was almost 80% missing and substantially redundant with effective temperature, it was excluded from the primary models.

Stellar radius

Stellar radius measures the star's physical size. From a modeling perspective, it helps distinguish stars that have similar surface temperatures but very different total energy output.

Stellar mass

Stellar mass affects the gravitational relationship between a planet's year length and its physical orbital distance. Two planets with the same period can orbit at somewhat different distances if their stars have different masses.

Equilibrium temperature

Equilibrium temperature is a simplified theoretical thermostat reading: the approximate temperature a planet would have if it absorbed and reradiated stellar energy in a specified idealized way, without a realistic atmosphere or greenhouse effect.

It was excluded from the primary HZ classifier because:

- it was almost 80% missing;
- it is closely related to the flux target;
- including it would make the model partly predict one thermal estimate from another.

Semi-major axis

The **semi-major axis** is the characteristic size of an orbit—roughly the planet's average orbital distance for a nearly circular path.

It was missing for approximately 40% of the catalog and strongly overlaps with information

already contained in orbital period and stellar mass. It was excluded from the primary period-based model but used when constructing the derived-flux sensitivity measure.

Planet mass and eccentricity

Planet mass was missing for more than half the catalog, and its measurement meaning differed across discovery methods. Orbital **eccentricity**, which measures how stretched an orbit is compared with a circle, was missing for almost two-thirds of planets.

Neither was globally imputed into the core classifier.

Planet mass was retained for a separate clustering analysis among mass-characterized planets. Eccentricity was limited to sensitivity and descriptive work.

2.5 Missing-data policy

No catalog-wide statistical imputation was applied before validation.

This decision followed from three observations.

First, missingness was strongly discovery-method dependent. For example, radial-velocity planets usually had mass and eccentricity measurements, while many transit planets did not. Filling those values from the pooled catalog would blend measurement regimes.

Second, some missing fields were almost indistinguishable from target proxies. Imputing equilibrium temperature would introduce a noisy reconstruction of the outcome rather than an independent predictor.

Third, the core reported-label cohort was nearly complete for the selected predictors. Dropping four planets from one host avoided inventing the central temperature feature while preserving all 32 positive planets.

The primary model therefore required no statistical imputation.

When variables such as distance or apparent brightness were considered in observation-process sensitivity models, imputation was fitted on training hosts only and accompanied by missingness indicators. Those indicators were interpreted as catalog-characterization signals, not astrophysical causes.

2.6 Leakage controls

Several plausible-looking predictors were deliberately withheld.

The models predicting reported HZ membership did not use:

- reported insolation flux;
- the HZ label itself;
- physics-derived flux;
- equilibrium temperature;
- the Earth-equivalent orbital-period calculation;
- any feature directly constructed from the target boundaries.

This prevented the model from receiving either the answer or a near-deterministic restatement of it.

Identifiers such as planet name and host name were also excluded from the predictive matrix. Host name was used only for grouping.

The analysis further avoided fitting preprocessors globally. Scaling, threshold selection, and model selection occurred within training hosts for each validation split.

2.7 Outlier treatment

Extreme values were not assumed to be errors.

The catalog contained real and scientifically distinct populations:

- **giant or evolved stars**, which are much larger than the Sun;
- **compact stellar remnants**, such as white dwarfs or pulsar systems;
- **ultra-short-period planets** with years shorter than one Earth day;
- very massive companions near the planet–brown-dwarf boundary;
- exceptionally wide directly imaged companions.

Deleting or winsorizing such cases would replace genuine astrophysical systems with artificial values.

The analysis instead:

- created explicit physical flags;
- retained the observations in the master table;
- used logarithmic transformations;
- annotated unusual regimes in visualizations;
- performed sensitivity analyses excluding selected regimes where appropriate.

Ultra-short-period planets remained in the primary model because they are valid and informative examples of highly irradiated transit discoveries.

Giant and compact-star systems were interpreted separately because their evolutionary histories differ from those of ordinary main-sequence stars.

2.8 Unsupervised clustering

A clustering analysis examined broad planet populations in log planet mass and log orbital period.

Only planets with measured mass and period were included. Features were robustly scaled, and several candidate cluster counts were compared using silhouette separation and host-bootstrap stability.

A four-cluster descriptive solution separated:

- low-mass, short-period planets;
- intermediate-mass, moderate-period planets;
- hot or warm giant planets;
- long-period giant planets.

These labels were intentionally approximate.

The catalog did not contain planet radius, so it could not cleanly separate common physical categories such as super-Earths and sub-Neptunes. A conventional Earth Similarity Index was also not calculated because planet radius was absent and mass and equilibrium temperature were highly incomplete.

HZ overlap by cluster was estimated with whole-host bootstrap intervals rather than naive row-level proportions.

2.9 Model sequence

Model complexity increased only after simpler alternatives were evaluated.

Baselines

A prevalence-probability baseline assigned every planet approximately the training-set HZ prevalence.

An always-HZ rule provided a second reference for understanding recall under severe class imbalance.

Targeted logistic model

The first logistic specification included:

- log orbital period;
- period curvature;
- log stellar temperature.

The interaction specification added:

- log period $\times$ log stellar temperature.

This isolated the primary scientific comparison.

Full stellar-structure logistic model

The next model added:

- log stellar radius;
- log stellar mass.

L2 regularization was used because the stellar variables were highly collinear.

A separate method-adjusted version added one ridge-shrunk radial-velocity-versus-transit indicator. Unregularized method-specific coefficients were not fitted.

Random forest

Three shallow random-forest configurations were specified in advance. Maximum depth and minimum leaf size were deliberately constrained relative to the number of positive training hosts.

Gradient boosting

Three shallow, low-learning-rate boosting configurations were also specified in advance. A

one-standard-error rule strongly favored decision stumps, indicating that the data did not justify interaction-capable tree depth.

Continuous regression

A ridge regression predicted log reported insolation from the four physical predictors. This analysis preserved the continuous target and served as a physical reconstruction check rather than an independent habitability model.

No ensemble of the fitted models was constructed, and model expansion stopped after the pre-specified comparisons.

2.10 Host-grouped repeated cross-validation

A random row split would allow one planet from a multi-planet system into training while another planet from the same star entered testing. Because their stellar features are nearly identical, this would inflate apparent generalization.

All validation therefore grouped rows by host star.

The primary design used:

\[
15\ \text{\textit{repeats}}\times6\ \text{\textit{grouped folds}}=90\ \text{\textit{outer test folds}}.
\]

Every planet received 15 predictions from models that had seen no planet belonging to its host system.

Leave-one-host-out validation was considered but not used as the principal evaluation design. Most single-host test sets would contain no positive example, making PR-AUC, recall, and F1 undefined or uninformative at the fold level.

Repeated grouped folds provided:

- strict host separation;
- positive examples in each outer fold;
- repeated exposure to alternative train–test partitions;
- directly comparable test memberships across model families.

2.11 Nested model and threshold selection

Hyperparameter or configuration choices were made inside each outer training set using additional host-grouped inner folds.

The inner process selected:

- the allowed tree configuration;
- the classification threshold used for recall, precision, F1, and MCC.

Thresholds were not assumed to be 0.5. With a rare positive class, a 0.5 threshold would often label nearly every case negative.

For the constrained tree models, the one-standard-error rule was used:

1. identify the configuration with the lowest inner log-loss;
2. estimate the uncertainty around that inner result;
3. choose the simplest configuration whose performance lay within one standard error of the best.

This favored parsimony when the apparent benefit of added depth was smaller than host-level sampling variation.

2.12 Evaluation metrics

Raw accuracy was not a primary metric because an always-negative classifier would be correct for more than 90% of the labeled planets while detecting no HZ candidates.

The principal metrics were:

- **PR-AUC**, which evaluates positive-class ranking under class imbalance;
- **recall**, the proportion of HZ candidates recovered;
- **precision**, the proportion of predicted candidates that were actually labeled HZ;
- **F1**, the harmonic balance of precision and recall;
- **Matthews correlation coefficient**, a balanced summary of the binary confusion matrix;
- **log-loss**, which evaluates the quality of predicted probabilities;
- **Brier score**, another proper probabilistic scoring rule.

For continuous flux regression, the metrics were:

- mean absolute error;
- root mean squared error;
- $\sqrt{R^2}$;
- hot/HZ/cold zone agreement after converting predicted flux back to the adopted boundaries.

2.13 Uncertainty and statistical-power treatment

All primary uncertainty intervals resampled complete host systems.

Whole-host bootstrap sampling preserved:

- within-system dependence;
- multi-planet structure;
- the effective number of independent positive hosts.

Depending on the analysis, 1,500 to 4,000 bootstrap samples were used.

Paired model comparisons resampled the same hosts for both models. This was more

informative than comparing two independent confidence intervals because it directly measured which model performed better on the same systems.

Host-level sign-flip tests evaluated paired differences in log-loss without relying on ordinary row-level p-values.

The analysis treated several results as suggestive rather than conclusive when:

- intervals included no effect;
- positive-host counts were small;
- the finding appeared only in one discovery-method subset;
- the test followed several related model comparisons.

Repeated validation and bootstrapping quantify uncertainty among the observed hosts. They do not correct the unknown probability that a planet was discovered, confirmed, characterized, and assigned a flux value.

2.14 Collinearity-aware feature importance

Individual feature importance can be misleading when predictors substitute for one another.

Temperature, radius, and stellar mass were therefore evaluated in two ways:

1. individually, for descriptive comparison;
2. jointly as a stellar-structure block.

For logistic regression, a block-drop comparison measured how much out-of-host performance deteriorated when all stellar terms were removed together.

For random forest and gradient boosting, grouped permutation importance moved temperature, radius, and mass together in held-out data while preserving their within-block relationship.

The joint analysis was treated as the primary feature-importance result.

This avoided the incorrect conclusion that one stellar variable was unimportant merely because another correlated variable could replace it.

2.15 Detection-bias analysis

Discovery method was treated primarily as an observation-process variable rather than a physical cause.

Bias was examined through:

- discovery-method-specific missingness;
- method-specific feature distributions;
- method-stratified model performance;
- comparison of models with and without a regularized method indicator;
- permutation importance of the method variable;
- observed versus physically adjusted method-level HZ prevalence.

Transit and radial velocity were the only methods with enough reported-flux labels for quantitative comparison.

Direct imaging was retained in descriptive plots and derived-flux sensitivity work, but no imaging coefficient or performance estimate was manufactured from unsupported data.

The analysis therefore distinguishes:

- **physical conditional relationships**, such as how period and stellar structure relate to flux among labeled planets;
- **catalog-selection effects**, such as which systems each discovery method detects and characterizes;
- **population occurrence**, which cannot be estimated without survey-completeness information absent from this table.

Optional callout: for the physics-curious

The main argument does not require the equations below. They show why the selected predictors form a physically coherent feature set.

From stellar size and temperature to total output

A star's total energy output relative to the Sun can be approximated by:

$$\left[\frac{L_{\text{star}}}{L_{\text{odot}}} \approx \left(\frac{R_{\text{star}}}{R_{\text{odot}}} \right)^2 \left(\frac{T_{\text{eff}}}{5772 \text{ K}} \right)^4 \right]$$

A larger emitting surface increases output with the square of radius. Surface temperature enters to the fourth power, so relatively small temperature differences can matter substantially.

From year length to orbital distance

Kepler's third law links a planet's year length to its orbital distance. In plain language, longer years usually correspond to wider orbits, but the conversion also depends on the mass of the host star:

$$\left[a_{\text{AU}} \approx \left[\left(\frac{M_{\text{star}}}{M_{\text{odot}}} \right) \left(\frac{P}{1 \text{ year}} \right)^2 \right]^{1/3} \right]$$

From stellar output and distance to received flux

The planet's received flux is approximately:

$$\frac{S}{S_{\oplus}} = \frac{L_{\star}/L_{\odot}}{(a/\mathrm{AU})^2}.$$

Combining the relationships gives:

$$S \propto \frac{R_{\star}^2 T_{\mathrm{eff}}^4}{M_{\star}^{-2/3} P^{-4/3}}.$$

The continuous ridge regression recovered exponents close to those expected values:

Predictor	Fitted exponent	Approximate physical value
Orbital period	−1.338	−1.333
Stellar temperature	+4.031	+4.000
Stellar radius	+1.863	+2.000
Stellar mass	−0.550	−0.667

This close agreement explains the regression's R^2 of 0.997. It also explains why the regression result is partly deterministic: the reported flux field was itself constructed from closely related physical measurements.

3. Part 1 synthesis

The study design establishes four principles that govern the interpretation of the later results.

First, the outcome is radiative habitable-zone candidacy, not actual habitability.

Second, the relevant orbital relationship is conditional. A “moderate period” cannot be defined without considering the host star.

Third, stellar temperature, radius, and mass must be interpreted collectively because their individual statistical effects are diluted by collinearity.

Fourth, discovery method affects the observed sample before modeling begins. A weak residual method coefficient does not imply an unbiased catalog.

Within those constraints, the central data-science result is already clear:

> A physics-informed, regularized model extracted more reproducible out-of-host signal than flexible tree ensembles because the labeled data contained too few independent positive host systems to support a highly adaptive decision boundary.

Supporting analysis files

- [Prepared dataset with engineered fields and no global statistical imputation]
(sandbox:/mnt/data/exoplanet_habitability/exoplanets_prepared_no_statistical_imputation.csv)
)
- [Best-model and classifier comparison]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_model_performance_host_bootstrap.csv)
- [Joint stellar-block and incremental model comparisons]
(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_incremental_comparisons.csv)
- [Discovery-method calibration analysis]
(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/method_calibration_host_bootstrap.csv)
- [Continuous-flux regression results]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_metrics_host_bootstrap.csv)

- [Validation and prediction quality checks]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_qa_checks.csv)

CRISP-DM and paper-synthesis mindmap

Exoplanet habitability drivers

└─ CRISP-DM analysis

- | └─ 1. Business understanding COMPLETE
- | └─ 2. Data understanding COMPLETE
- | └─ 3. Data preparation COMPLETE
- | └─ 4. Exploratory, outlier, and cluster analysis COMPLETE
- | └─ 5. Predictive modeling COMPLETE
- | └─ 6. Evaluation and model selection COMPLETE

└─ Research-paper synthesis

- | └─ Part 1. Abstract, introduction, and methods COMPLETE – THIS CHUNK
 - | └─ Findings-first abstract
 - | └─ Physics-naïve introduction
 - | └─ Quote-ready claim boundary
 - | └─ Full data-science methodology
 - | └─ Optional physics equations
- | └─ Part 2. Results NEXT
 - | └─ Raw-data visual findings
 - | └─ Missingness and selection structure
 - | └─ Feature and outlier findings
 - | └─ Clustering results
 - | └─ Classifier comparisons
 - | └─ Interaction and stellar-block results
 - | └─ Continuous-flux cross-check
- | └─ Part 3. Discussion and limitations
 - | └─ Astrophysical interpretation
 - | └─ Transit versus RV covariate shift
 - | └─ Imaging underidentification
 - | └─ Target circularity
 - | └─ Statistical-power limits
- | └─ Part 4. Conclusions and public summary
 - | └─ Research conclusions
 - | └─ Recommendations for future data collection
 - | └─ General-audience summary
 - | └─ Medium and GitHub-ready takeaways

Next: Final Synthesis, Part 2 will present the results in research-paper form, organized around the publication-quality figures and the uncertainty-qualified statistical findings.

Standing requirement: determine which stellar and orbital features predict radiative habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the physical relationship from transit, radial-velocity, and imaging selection effects using a staged, limited-compute CRISP-DM workflow.

PROMPT · SEP 01, 2026 12:12 PM

continue with Final Synthesis, Part 2.

Keep the same voice established in Part 1 — plain-language first, with any deeper statistical or physics detail (e.g. VIF values, exact bootstrap replicate counts) available but not blocking the main narrative for a reader with no physics background.

For each major result (raw period-temperature pattern, missingness/selection structure, feature and outlier findings, clustering, classifier comparison, stellar-block interaction, discovery-method covariate shift, continuous-flux cross-check), lead with one plain-language sentence stating the finding before presenting the supporting numbers or figure, the same way the abstract led with findings.

Since we have roughly 15 figures across all chunks and only a handful can reasonably go in a Medium article, please flag your top 4-5 recommended figures for the public article specifically — the ones that are both visually clear to a non-specialist and carry the most scientific weight (my instinct is the period-temperature response surface, the classifier performance comparison, the collinearity/joint-stellar-block figure, and the discovery-method covariate-shift figure, but confirm or override that based on what actually reads clearly without caption-heavy explanation).

Continue reporting every finding with its uncertainty interval and treating suggestive results (e.g. the period $\times$ temperature interaction) as suggestive, not conclusive, consistent with Part 1's framing.

Standing requirement: determine which stellar and orbital features predict radiative habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the physical relationship from transit, radial-velocity, and imaging selection effects using a staged, limited-compute CRISP-DM workflow.

CHATGPT · SEP 01, 2026 12:12 PM

I'll consolidate the completed analyses into a findings-first Results section, verify key uncertainty intervals, and select the 4–5 strongest public-facing figures without adding new models.

CHATGPT · SEP 01, 2026 12:15 PM

The evidence hierarchy is clear: stellar structure is robust, the explicit period–temperature interaction remains suggestive, and simpler ridge modeling outgeneralizes both tree ensembles.

Final Synthesis, Part 2

Results

All results below were computed from the uploaded `all_exoplanets_2021.csv`.
[filecite]turn2file0[L1-L4]

Unless stated otherwise, uncertainty ranges are **95% whole-host bootstrap intervals**. These intervals resample complete host-star systems rather than individual planets, so several planets orbiting the same star do not masquerade as independent evidence. Exact catalog counts and missingness percentages are descriptive properties of this dataset and therefore are reported without sampling intervals.

3. Results

3.1 The raw data already show a moderate-period, cooler-star pattern

Habitable-zone candidates were concentrated at longer—but not arbitrarily long—orbital periods and around cooler stars, although the strength of the period pattern differed by discovery method.

The central raw-data visualization plots orbital period, or planet “year length,” against host-star temperature. Orbital period is shown on a logarithmic scale because the catalog spans periods from fractions of a day to millions of days.

Gray points represent planets without reported stellar-flux values. Colored points represent:

- planets receiving too little stellar energy;
- planets inside the adopted radiative habitable zone;

- planets receiving too much stellar energy.

Recommended public Figure 1

![Raw orbital period versus stellar temperature, faceted by discovery method]
(sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.png)

[Vector PDF]
(sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.pdf)

Among planets with reported flux, the descriptive medians were:

Group	Median orbital period	Median stellar temperature
--- ---	---	---
Habitable-zone candidates	32.4 days	3,466 K
Planets outside the adopted zone	6.7 days	5,055 K

These medians are useful for orientation, but the host-bootstrap rank comparisons provide a more reliable summary.

For the pooled reported-label sample:

- orbital period had a Cliff's delta of **+0.623** $\text{ } \left(\left[+0.453, +0.793 \right] \right)$;
- stellar temperature had a Cliff's delta of **-0.579** $\text{ } \left(\left[-0.735, -0.395 \right] \right)$.

Cliff's delta is a rank-separation measure. The positive period value means that a randomly selected HZ candidate was substantially more likely to have a longer period than a randomly selected non-HZ planet. The negative temperature value means that the HZ candidate was substantially more likely to orbit the cooler star.

The method-specific results were less uniform:

Discovery method	Period rank separation	Temperature rank separation
--- ---	---	---
Transit	+0.748 $\text{ } \left(\left[+0.496, +0.928 \right] \right)$	-0.577 $\text{ } \left(\left[-0.744, -0.362 \right] \right)$

| Radial velocity | $+0.123 \text{ } \backslash [-0.197, +0.470] \backslash$ | $-0.502 \text{ } \backslash [-0.847, -0.027] \backslash$ |

The period difference was pronounced among transit discoveries but statistically unresolved within the smaller radial-velocity subset. The radial-velocity estimate should not be interpreted as evidence that orbital period is irrelevant; its interval is wide because only ten radial-velocity host systems contributed positive cases.

The scatter plot also demonstrates that there is no universal “good period.” Around the coolest stars, some candidates occupy orbits lasting only days or tens of days. Around warmer stars, the candidate region shifts toward longer years.

The raw geometry therefore supports the broad hypothesis:

> A moderate orbital period is favorable only relative to the kind of star being orbited.

It does not, by itself, prove a precisely estimated period-by-temperature interaction.

3.2 The apparent sample size collapses once target availability is considered

Although the source table contains 4,575 planets, the primary habitability-screening question is supported by only 370 reported-flux labels and 26 independent positive host systems.

The most important missing field was the outcome itself. Reported insolation flux—the planet’s received starlight relative to Earth—was absent for 4,205 planets.

Recommended public Figure 5

![Raw-data missingness by column]

(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.png)

[Vector PDF](sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.pdf)

The highest missingness rates were:

Field Missing
Insolation flux 91.9%
Equilibrium temperature 79.8%
Stellar spectral type 79.7%
Orbital eccentricity 62.7%
Planet mass 56.2%
Orbital semi-major axis 39.6%
Stellar metallicity 30.0%
Stellar mass 16.0%
Stellar radius 9.8%
Stellar effective temperature 7.6%
Orbital period 3.5%

These percentages were not treated as missing-at-random noise. They reflected how exoplanets are observed and followed up.

For example:

- radial-velocity discoveries usually had planet-mass and eccentricity estimates;
- transit discoveries frequently had precise periods but lacked mass unless additional observations were obtained;
- directly imaged planets often had wide-separation information but poorly constrained periods;
- microlensing planets had sparse stellar characterization in this snapshot.

Reported flux was available only for two discovery methods:

Method Total planets Planets with reported flux Reported HZ candidates
Transit 3,444 310 19
Radial velocity 899 60 13
Imaging 54 0 0 observed
Microlensing 120 0 0 observed

After excluding four labeled planets missing stellar temperature, the primary modeling cohort

contained:

- 366 planets;
- 264 host systems;
- 32 positive planets;
- 26 positive host systems;
- 306 transit planets;
- 60 radial-velocity planets;
- no imaging planets.

This is why broad median imputation was rejected. Imputing fields with 50–90% missingness would not recover randomly lost measurements; it would create synthetic values across fundamentally different observing regimes.

The target’s missingness also limits the scientific estimand. The models predict:

$$\mathbb{E}\left[\frac{P(\text{reported HZ candidate} \mid \text{observed features and catalog inclusion})}{P(\text{reported HZ candidate})}\right]$$

not the prevalence of HZ planets among all planetary systems.

3.3 Orbital period supplied distinct information, while the stellar variables largely described the same underlying system

Orbital period behaved as an independent predictor, but stellar temperature, radius, and mass were so closely related that their individual statistical contributions could not be cleanly separated.

In the host-balanced correlation analysis, orbital period had almost no marginal correlation with the stellar measurements:

Feature pair	Host-balanced Spearman correlation
--- ---	---

	Period–temperature		0.00	
	Period–radius		0.00	
	Period–stellar mass		–0.02	

By contrast, the stellar variables were strongly related:

	Feature pair		Host-balanced Spearman correlation	
	---		---	
	Temperature–stellar mass		0.98	
	Radius–stellar mass		0.98	
	Temperature–radius		0.95	

Their variance inflation factors reinforced the same finding:

	Feature		VIF	
	---		---	
	Log orbital period		1.0	
	Log stellar radius		5.4	
	Log stellar temperature		7.8	
	Log stellar mass		11.2	

The orbital-period VIF indicates that period contributed a genuinely separate axis of information. The stellar variables instead behaved like several correlated measurements of one latent concept: how energetic and structurally large the star is.

Adding calculated luminosity or surface gravity to the same model would have made the feature matrix algebraically redundant. Those derived quantities were therefore used for physical checks rather than entered alongside all of their component variables.

The result is a familiar data-science problem: predictive information can be real at the feature-family level even when individual coefficients are unstable.

3.4 The moderate-period effect was supported, but the explicit period-by-temperature interaction remained suggestive

The model reliably favored an intermediate orbital range over either extreme, but it could not estimate a unique temperature-dependent shift with high precision.

The targeted logistic model included:

- log orbital period;
- a squared period term;
- stellar temperature;
- a period-by-temperature interaction.

The squared period coefficient was:

$\sqrt{-0.604 \pm [-0.651, -0.091]}$

Its negative direction implies a hill-shaped relationship: modeled HZ probability increases away from the shortest, usually overheated orbits, peaks at an intermediate range, and eventually declines at sufficiently long periods.

This result supports “moderate period” more clearly than it supports “longer is always better.”

Adding the explicit period-by-temperature interaction produced:

Comparison	Interaction-model improvement
PR-AUC	+0.0278 $\sqrt{[-0.0058, +0.0761]}$
Log-loss reduction	+0.00337 $\sqrt{[-0.00098, +0.00802]}$
Recall	+0.0729 $\sqrt{[0.0000, +0.1958]}$
F1	+0.0241 $\sqrt{[-0.0363, +0.1040]}$

The point estimates favored the interaction model, and approximately 93% of the host-bootstrap samples showed some PR-AUC or log-loss improvement. Nevertheless, both 95% intervals included zero. The one-sided host sign-flip result for log-loss was $(p=0.076)$.

The correct wording is therefore:

> The explicit period-by-temperature interaction had the expected direction and modest

apparent predictive value, but the reported-label sample was too small to establish it conclusively.

The interaction weakened further when stellar radius and mass entered the model. That does not overturn the underlying physical idea. It shows that stellar temperature alone had been acting as an incomplete proxy for a broader stellar-structure relationship.

[Full period–temperature response and interaction figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.png)

[Vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.pdf)

3.5 The individual stellar coefficients weakened, but the stellar variables remained strongly important as a group

Once stellar radius and mass were included, the model could no longer assign stable credit to temperature alone—even though removing all stellar information caused a large loss of predictive skill.

The standardized stellar-temperature coefficient changed from:

$$\begin{aligned} & \backslash[\\ & -1.772\backslashquad[-2.265,-1.095] \\ & \backslash] \end{aligned}$$

in the period–temperature model to:

$$\begin{aligned} & \backslash[\\ & -0.225\backslashquad[-0.918,+0.522] \\ & \backslash] \end{aligned}$$

after adding stellar radius and mass.

The paired coefficient shift was:

$$\begin{aligned} & \backslash[\\ & +1.547\backslashquad[+0.631,+2.387], \\ & \backslash] \end{aligned}$$

an approximately 87% movement toward zero.

The period-by-temperature coefficient similarly changed from:

$$\begin{aligned} & \backslash[\\ & +0.368\backslashquad[-0.444,+0.792] \\ & \backslash] \end{aligned}$$

to:

$$\begin{aligned} & \backslash[\\ & +0.074\backslashquad[-0.798,+0.521]. \\ & \backslash] \end{aligned}$$

Its paired change was:

$$\begin{aligned} & \backslash[\\ & -0.294\backslashquad[-0.546,-0.040], \\ & \backslash] \end{aligned}$$

an approximately 80% reduction in the point estimate.

Read individually, those coefficients look unstable. Read jointly, the conclusion is very different.

Recommended public Figure 3

![Coefficient redistribution and joint stellar-structure effect]
(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.png)

[Vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.pdf)

Removing temperature, radius, mass, and the period-by-temperature term as one stellar-structure block caused:

| Out-of-host block-drop result | Change |

|---|---:|

| PR-AUC | **-0.295** \([-0.473,-0.119]\) |

| Log-loss | **+0.058** \([+0.022,+0.100]\) |

Equivalently, adding the stellar block to a curved period-only model raised PR-AUC by 0.295 and reduced log-loss by 0.058.

This block-level effect was much more stable than any one stellar coefficient. The same pattern appeared in the tree models: permuting temperature, radius, and mass together damaged held-out performance much more than permuting any one of them separately.

The substantive result is therefore:

> The host star's collective structure is a major predictor, but this dataset cannot reliably decide how much unique credit belongs to temperature, radius, or mass.

This is the most important feature-importance lesson in the project.

3.6 The most extreme observations were usually real astrophysical populations, not generic data errors

Giant stars, compact stellar remnants, ultra-short-period planets, and extremely wide companions were retained because they represented meaningful observing regimes rather than obvious corrupted rows.

The physics-aware screen identified:

Outlier or unusual-regime flag	Planets
Giant or evolved stellar hosts	192
Compact remnants or hot subdwarfs	18
Orbital periods below one day	114
Planet masses above 13 Jupiter masses	74
Periods above 100,000 days	4

These regimes were strongly associated with discovery method:

- 176 of 192 giant-host planets were radial-velocity discoveries;
- 106 of 114 ultra-short-period planets were transit discoveries;
- all four extremely wide-period cases were imaging discoveries.

Thus, deleting the tails would also have selectively deleted particular detection channels.

All ten ultra-short-period planets with reported-flux labels were outside the HZ. The same was true for all 100 ultra-short-period planets with a derived-flux classification. These exact catalog counts support the interpretation that ultra-short periods are usually associated with excessive irradiation.

However, those planets were not removed from the primary classifier. They provide legitimate information about the too-hot region and about the population sampled by transit surveys.

Giant and compact-star systems were retained in the master table but separately annotated because their evolutionary histories differ from those of ordinary stars. The primary reported-label cohort contained no giant-host HZ positives, so giant systems could not be responsible for the core classifier's result.

No general winsorization was applied. Log transformations, explicit flags, and sensitivity subsets preserved the real physical diversity of the sample.

[Physics-aware outlier figure]

(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.png)

[Vector PDF](sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.pdf)

3.7 Clustering separated overheated giants from longer-period populations, but it could not identify one definitive “most habitable” planet type

The hot, short-period giant-planet cluster showed essentially no HZ overlap, while the available reported labels were too sparse to rank the longer-period clusters reliably.

Clustering used only observed planet mass and orbital period. It did not use HZ status to create the groups.

The mass-characterized clustering cohort contained:

- 1,850 planets;
- 1,384 host systems;
- 886 transit discoveries;
- 899 radial-velocity discoveries.

A four-cluster solution was selected as a scientifically interpretable refinement:

Cluster	Planets	Median mass	Median period
Low-mass, short-period	441	6.5 Earth masses	8.0 days
Intermediate-mass, moderate-period	255	22.5 Earth masses	57.3 days
Hot or warm giants	589	305 Earth masses	3.7 days
Long-period giants	565	826 Earth masses	871 days

The four-cluster solution had a median host-bootstrap adjusted Rand index of:

$$\sqrt{0.944} \pm [0.697, 0.993]$$

The wider lower bound indicates that the boundary between the low-mass and intermediate-mass clusters was not perfectly stable. The hot-giant and long-period-giant groups were more distinct.

![[Mass-period clusters and HZ overlap]
(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.png)

[Vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.pdf)

Reported-flux HZ overlap was:

Cluster	Reported HZ rate
---	---
Low-mass, short-period	8.7% $\backslash$ [2.8%, 15.3%]
Intermediate-mass, moderate-period	20.0% $\backslash$ [4.2%, 37.9%]
Hot or warm giants	0.0% $\backslash$ [0.0%, 3.0%]
Long-period giants	18.2% $\backslash$ [0.0%, 44.4%]

The near-zero hot-giant result was the clearest cluster finding.

The apparent difference between the intermediate-mass and long-period groups was not reliable. Their intervals were broad and substantially overlapping because the cluster analysis retained only:

- 243 reported-flux planets;
- 18 reported HZ positives;
- 13 independent positive host systems.

Using the broader physics-derived flux target produced:

Cluster	Derived HZ rate
---	---
Low-mass, short-period	3.7% $\backslash$ [1.4%, 6.3%]
Intermediate-mass, moderate-period	15.0% $\backslash$ [9.3%, 21.3%]
Hot or warm giants	0.0% $\backslash$ [0.0%, 0.45%]
Long-period giants	19.4% $\backslash$ [15.9%, 23.3%]

The derived result was more precise but partly built into the variables: both the clusters and the derived flux depended strongly on orbital period.

It also does not imply that long-period giant planets are Earth-like. A giant planet can receive moderate starlight while still lacking a solid surface.

The defensible conclusion is:

> Very short-period giant planets rarely receive moderate stellar flux, but this dataset cannot identify a uniquely superior terrestrial-like cluster because planet radius is absent and reported labels are sparse.

3.8 The physics-informed logistic model outperformed both tree ensembles

The simplest scientifically structured classifier generalized best; extra nonlinear flexibility reduced rather than improved out-of-host performance.

The primary benchmark was the full stellar-structure ridge-logistic model without a discovery-method term.

Its predictors were:

- log orbital period;
- orbital-period curvature;
- stellar temperature;
- stellar radius;
- stellar mass;
- the targeted period-by-temperature term.

The same 90 repeated host-held-out folds were used for all model families.

Recommended public Figure 2

![Classifier comparison and constrained boosting selection]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.png)

[Vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.pdf)

d_selection.pdf)

The out-of-host results were:

Model	PR-AUC	Recall	Precision	F1	MCC	Log-loss
Prevalence baseline	0.077	$\backslash([0.049,0.107])$	0	0	0	0.271
Full stellar ridge logistic	0.596	$\backslash([0.447,0.789])$	0.802	$\backslash([0.631,0.957])$	0.542	$\backslash([0.379,0.698])$
Constrained random forest	0.475	$\backslash([0.332,0.678])$	0.691	$\backslash([0.497,0.870])$	0.535	$\backslash([0.358,0.709])$
Constrained gradient boosting	0.603	$\backslash([0.432,0.742])$	0.571	$\backslash([0.394,0.721])$	0.162	$\backslash([0.119,0.207])$

The paired comparisons against the same held-out hosts were even clearer:

Flexible model versus full stellar logistic	PR-AUC difference	Log-loss increase
Random forest	-0.122	$\backslash([-0.235,-0.019])$
Gradient boosting	-0.231	$\backslash([-0.364,-0.121])$

Both flexible models were inferior on both threshold-free primary metrics.

The nested model-selection procedure also resisted unnecessary complexity:

- the random forest selected a maximum depth of three in 80% of outer folds;
- gradient boosting selected depth-one decision stumps in 86 of 90 folds.

Depth-one boosting can learn nonlinear main effects but cannot represent a true feature interaction within an individual tree. The validation process therefore found insufficient evidence to justify even modest interaction-capable tree depth.

This negative result is scientifically useful:

> **With only 26 positive host systems, the data supported a smooth, low-dimensional physical structure more reliably than a flexible collection of local tree splits.**

The result does not imply that tree ensembles are generally unsuitable for exoplanet science. It

means they were less data-efficient for this particular target and sample.

3.9 Orbital period was the strongest individual feature, while stellar structure was the strongest correlated feature block

Across model families, orbital period provided the clearest individual signal, and temperature, radius, and mass provided a comparably important joint signal.

In held-out random-forest permutation analysis, destroying the orbital-period information caused:

- a PR-AUC decrease of **0.235** $\backslash[0.059,0.446]\backslash$;
- a log-loss increase of **0.091** $\backslash[0.049,0.140]\backslash$.

Permuting stellar temperature, radius, and mass together caused:

- a PR-AUC decrease of **0.197** $\backslash[0.055,0.376]\backslash$;
- a log-loss increase of **0.036** $\backslash[0.012,0.062]\backslash$.

The individual stellar-feature intervals were much less decisive because each stellar variable could partly substitute for the other two.

Gradient boosting reproduced the same qualitative pattern:

- period permutation increased log-loss by **0.0677** $\backslash[0.0272,0.1140]\backslash$;
- joint stellar permutation increased log-loss by **0.0266** $\backslash[0.0086,0.0478]\backslash$.

The consistency across ridge-logistic block-drop tests and held-out tree permutation tests strengthens the central feature conclusion:

> Orbital period and collective stellar structure are the two reproducible sources of predictive information.

A ranking that declared temperature, radius, or mass individually “most important” would overstate what these data can identify.

3.10 The raw transit–radial-velocity difference was largely explained by covariate shift

Radial-velocity planets appeared more likely to be HZ candidates in the raw labeled sample, but the physical model reproduced that difference without needing the discovery-method label.

The observed host-balanced HZ prevalences were:

Method	Observed HZ prevalence
Transit	5.6% $\backslash$ [2.9%, 8.5%]
Radial velocity	20.3% $\backslash$ [9.5%, 32.9%]

Taken literally, this might suggest that radial velocity somehow discovers more habitable planets.

But the full physical model, without being told the discovery method, predicted:

Method	Mean predicted HZ probability
Transit	5.6% $\backslash$ [4.2%, 7.2%]
Radial velocity	21.3% $\backslash$ [15.5%, 27.7%]

The physical feature distributions therefore accounted for almost the entire raw method difference.

Recommended public Figure 4

! [Discovery-method adjustment and covariate shift]
(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.png)

[Vector PDF]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.pdf)

Adding one ridge-shrunk radial-velocity-versus-transit indicator caused:

| Metric change from adding discovery method | Estimate |

|---|---:|

| PR-AUC | $-0.017 \text{ } \backslash[-0.042,-0.002]\backslash$ |

| Log-loss | Worsened by $0.0023 \text{ } \backslash[0.0009,0.0043]\backslash$ |

The adjusted radial-velocity-versus-transit odds ratio was:

$\backslash$

$0.94 \backslash \text{quad}[0.30,2.38].$

$\backslash$

The corresponding standardized probability difference was:

$\backslash$

$-0.003 \backslash \text{quad}[-0.044,+0.043].$

$\backslash$

There was therefore no stable residual method-intercept effect after accounting for the observed physical variables.

This result must be interpreted carefully.

It does **not** show that the catalog is unbiased. Transit and radial-velocity surveys still differ in:

- the stars they monitor;
- the periods to which they are sensitive;
- the planet masses or sizes they can detect;
- their follow-up measurements;
- the probability that a reported flux enters the catalog.

The result shows that selection bias was expressed mainly as a shift in the observed

predictors:

$\backslash$
 $P(X \mid \text{transit})$
 $\neq$
 $P(X \mid \text{radial velocity})$,
 $\backslash$

rather than as a large additional method effect after conditioning on those predictors.

Direct imaging could not be assessed quantitatively because no imaging discovery had a reported-flux label in the core cohort.

The method conclusion is therefore:

> Discovery method changes which planetary systems appear in the dataset, but its label adds little once the selected system's orbital and stellar measurements are known.

3.11 Continuous flux regression produced a much more stable physical picture—but a less independent test

Predicting the continuous amount of received starlight was extremely accurate, largely because the target is constructed from the same physical quantities used as predictors.

Binary HZ classification discards a great deal of information. It assigns the same non-HZ label to:

- a planet just outside the warm boundary;
- an ultra-hot planet receiving hundreds of times Earth's flux;
- a distant planet receiving almost no stellar energy.

A ridge regression therefore predicted the continuous logarithm of reported flux from:

- orbital period;
- stellar temperature;

- stellar radius;
- stellar mass.

Its out-of-host performance was:

Regression metric	Estimate
MAE	0.0300 dex $\backslash$ [0.0240,0.0369] $\backslash$
RMSE	0.0624 dex $\backslash$ [0.0478,0.0764] $\backslash$
R^2	0.9970 $\backslash$ [0.9954,0.9982] $\backslash$

An MAE of 0.030 dex corresponds to a typical multiplicative error of approximately 7%.

The fitted slopes closely tracked the expected physical relationship between stellar output, orbital distance, and received flux. Converting the predicted continuous flux back into too-cold, HZ, and too-hot categories produced:

Zone metric	Estimate
Three-zone accuracy	98.67% $\backslash$ [97.35%,99.81%] $\backslash$
Three-zone macro recall	95.23% $\backslash$ [87.76%,99.93%] $\backslash$
HZ recall	95.06% $\backslash$ [84.22%,100%] $\backslash$
HZ precision	88.50% $\backslash$ [74.24%,100%] $\backslash$
HZ F1	91.66% $\backslash$ [81.87%,98.72%] $\backslash$

Only four planets crossed a hot/HZ/cold boundary between reported and out-of-host predicted flux.

![[Continuous-insolation regression cross-check
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.png)]]

[[Vector PDF
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.pdf)]]

The flux regression and the primary classifier agreed on binary HZ status for:

$$\sqrt{93.86\% \pm [90.93\%, 96.36\%]}$$

of the host-balanced observations.

However, agreement on the positive sets was lower. Their weighted HZ-set Jaccard overlap was:

$$\sqrt{0.523 \pm [0.369, 0.679]}.$$

The classifier was designed as a high-recall screen and therefore flagged a broader candidate set than the exact continuous-flux boundaries.

Most importantly, the regression's spectacular accuracy is partly circular. Catalog insolation is commonly calculated from closely related orbital and stellar quantities. The regression is therefore reconstructing an existing physical calculation, not independently discovering which planets support life.

The result should be summarized as:

> Continuous regression gives the most stable description of stellar irradiation, but the binary classifier gives the more independent test of whether a small set of measured features generalizes to unseen host systems. Neither establishes actual habitability.

4. Integrated answers to the research questions

Which features predict radiative habitable-zone membership?

Orbital period and the host star's collective structure were the two reproducible predictive components.

Orbital period was the strongest individual feature. Stellar temperature, radius, and mass were important jointly, but their high collinearity made individual attribution unstable.

Planet mass, eccentricity, equilibrium temperature, and spectral type were not required for the primary model:

- planet mass and eccentricity were too incomplete and method-dependent;
- equilibrium temperature was a near-target proxy;
- spectral type was highly incomplete and redundant with continuous stellar temperature.

Are moderate-period planets around cooler stars more likely to be HZ candidates?

Broadly yes, but “moderate” must be defined relative to the complete host-star system rather than by one universal period interval.

The negative period-curvature result supported an intermediate orbital range. Cooler stars were associated with HZ candidates, and the targeted interaction moved the model’s favorable range toward longer periods for hotter stars.

However, the interaction’s incremental PR-AUC and log-loss intervals included zero. After radius and mass were included, its coefficient shrank substantially.

The evidence therefore supports:

> moderate period × host-star structure

more strongly than it supports:

> a precisely estimated period × temperature coefficient.

How much does discovery method affect the result?

Discovery method strongly affected the feature distribution and target availability, but a method indicator added no residual predictive skill after the physical features were

known.

Transit and radial-velocity discoveries represented different regions of orbital and stellar feature space. Imaging and microlensing lacked the labeled support needed for comparable modeling.

This is evidence of covariate shift, not evidence that selection bias is absent.

Which model should be retained?

The full stellar-structure ridge-logistic model should be retained as the primary classifier.

It had the strongest combination of:

- PR-AUC;
- recall;
- F1;
- Matthews correlation;
- log-loss;
- interpretability;
- data efficiency.

Both constrained tree ensembles underperformed it on the same held-out hosts. No further algorithm search or ensembling is justified.

5. Recommended figures for the public Medium article

I would slightly revise the proposed figure set. The full response-surface triptychs are scientifically useful, but they require long captions and can be mistaken for literal physical habitable-zone boundaries. The raw faceted scatter plot communicates the hypothesis, the actual data support, and the missing imaging labels more honestly and intuitively.

Core four figures

| Rank | Figure | Why it belongs in the article |

|---:|---|---|

| **1** | [Raw period–temperature plot by discovery method]

(sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.png) | Best opening figure. It shows the moderate-period/cooler-star pattern in the observations while simultaneously exposing the scarcity of labels and lack of imaging support. |

| **2** | [Classifier comparison and capacity selection]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.png) | Carries the central machine-learning finding: the physics-informed logistic model beats both tree ensembles, and nested validation rejects added capacity. |

| **3** | [Coefficient shift and joint stellar-block effect]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.png) | Communicates the most valuable data-science lesson: unstable individual coefficients can coexist with a strong joint correlated-feature effect. |

| **4** | [Discovery-method covariate-shift result]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.png) | Separates raw method prevalence from conditional prediction and supports the detection-bias discussion directly. |

Fifth figure when space allows

| Rank | Figure | Why it earns the optional slot |

|---:|---|---|

| **5** | [Raw missingness by column]

(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.png) | Extremely readable for a general audience and explains why the analysis is based on 26 positive hosts rather than “4,575 planets.” It also motivates the no-global-imputation decision. |

Better suited to the GitHub README or technical appendix

| Figure | Placement rationale |

|---|---|

| [Period–temperature interaction response]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.png) | Scientifically central but visually dense. A cropped version of the middle response-surface panel could work in a longer article; the full triptych belongs in technical documentation. |

| [Mass–period clusters and HZ overlap]

(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.png) |

Strong supplementary result, but the cluster labels require caveats about absent planet radius and sparse reported HZ labels. |

| [Continuous-flux regression cross-check]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.png) | Visually compelling, but its circularity caveat is essential. It works best after readers understand the distinction between prediction and physical reconstruction. |

| [Physics-aware outlier landscape]

(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.png) | Useful for the methods appendix and teaching material on domain-informed outlier treatment, but secondary to the main scientific result. |

| [Gradient-boosting versus logistic response surfaces]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_period_temperature_surface.png) | Best used when discussing why flexible models did not recover a stable interaction. It is less suitable as the primary visualization of the astrophysical pattern. |

For an article limited to exactly four figures, the first four above form the strongest and most coherent narrative:

Observed pattern

↓

Why individual stellar coefficients are unstable

↓

Why the simpler model wins

↓

How detection method changes the observed population

6. Results-section conclusion

The completed results support a qualified but coherent answer.

A planet's year length is the strongest individual predictor of whether it receives moderate stellar energy, but period alone is insufficient. The host star's temperature, size, and mass collectively determine where a moderate-energy orbit can occur. The data support an intermediate-period region and a cooler-star association, while the explicit period-by-temperature interaction remains suggestive rather than conclusively isolated.

The observed transit–radial-velocity difference is mostly a difference in the kinds of systems included in each sample. Once the physical measurements are known, discovery method adds little residual predictive value. Imaging remains underidentified because the catalog contains no directly imaged planets with the primary reported-flux target.

The best-performing classifier is the full stellar-structure ridge-logistic model. Random forest and gradient boosting add flexibility but lose out-of-host skill, demonstrating that, with only 26 positive host systems, scientifically structured regularization is more data-efficient than adaptive tree ensembles.

Continuous flux regression reconstructs the irradiation calculation with near-perfect accuracy, but that stability is partly deterministic and must not be confused with independent evidence of actual planetary habitability.

7. Research-paper synthesis mindmap

Exoplanet habitability drivers

└─ CRISP-DM analysis

- └─ Business understanding COMPLETE
- └─ Data understanding COMPLETE
- └─ Data preparation COMPLETE
- └─ Exploratory and structural analysis COMPLETE
- └─ Predictive modeling COMPLETE
- └─ Evaluation and model selection COMPLETE

└─ Research-paper synthesis

- └─ Part 1. Abstract, introduction, and methods COMPLETE
- └─ Part 2. Results COMPLETE – THIS CHUNK
 - └─ Raw period-temperature pattern
 - └─ Missingness and target support
 - └─ Collinearity and joint stellar effect
 - └─ Moderate-period interaction test
 - └─ Physics-aware outliers
 - └─ Mass-period clustering
 - └─ Classifier comparison
 - └─ Discovery-method covariate shift
 - └─ Continuous-flux cross-check
 - └─ Public-article figure selection
- └─ Part 3. Discussion and limitations NEXT
 - └─ Astrophysical interpretation
 - └─ Why the interaction weakened
 - └─ Selection bias versus causal effects
 - └─ Imaging underidentification
 - └─ Target circularity
 - └─ Small-positive-host uncertainty
 - └─ External validity and future data needs
- └─ Part 4. Conclusions and recommendations
 - └─ Formal research conclusions
 - └─ Practical modeling recommendations
 - └─ Future observing and data priorities
 - └─ General-audience summary
 - └─ Medium and GitHub-ready takeaways

Next: Final Synthesis, Part 3 will interpret the findings, separate physical explanation from catalog selection, discuss statistical and astrophysical limitations, and define which conclusions are robust, suggestive, or unsupported.

Standing requirement: determine which stellar and orbital features predict radiative habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the physical relationship from transit, radial-velocity, and imaging selection effects using a staged, limited-compute CRISP-DM workflow.

PROMPT · SEP 01, 2026 12:32 PM

continue with Final Synthesis, Part 3.

In addition to the planned discussion and limitations content, please add one explicit subsection titled something like "claims the Medium article must not make" — a short, direct list of overclaims this analysis cannot support (e.g. "these are habitable planets," "this proves the interaction effect," "this estimates how common habitable planets are," "radial-velocity surveys find more habitable worlds") paired with the correct, defensible phrasing for each. This is meant as a practical guardrail for writing the public article, not just an academic caveat section.

Keep the same robust/suggestive/unsupported framing used throughout, and make sure the imaging-underidentification and target-circularity points are stated in language a data-science reader (not an astronomer) will immediately understand as "missing support" and "leakage-adjacent," respectively, consistent with the accessible framing from Parts 1 and 2.

Standing requirement: determine which stellar and orbital features predict radiative habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the physical relationship from transit, radial-velocity, and imaging selection effects using a staged, limited-compute CRISP-DM workflow.

CHATGPT · SEP 01, 2026 12:32 PM

I'll frame Part 3 around physical interpretation, selection bias, statistical power, and public-writing guardrails, ending with a robust/suggestive/unsupported claim map and updated synthesis mindmap.

Final Synthesis, Part 3

Discussion, Limitations, and Public-Article Guardrails

This discussion interprets the analysis of the uploaded `all_exoplanets_2021.csv` catalog.
[filecite]turn2file0[L1-L4]

The conclusions below are conditional on the observed catalog and, more narrowly, on the 366 planets from 264 host systems that had both a reported flux label and the complete core predictors. Only 26 host systems contributed a positive habitable-zone example.

4. Discussion

4.1 The main finding is a conditional orbital pattern, not a universal “best period”

A planet’s orbital period is informative only when interpreted in the context of its host star.

The raw data and fitted models reject two oversimplified rules:

- “Shorter periods are better.”
- “Longer periods are better.”

Extremely short orbits usually place a planet very close to its star, where it receives excessive energy. Moving outward initially makes the energy budget more moderate. Moving sufficiently far outward eventually produces too little energy.

This creates an intermediate or moderate-period region rather than a one-directional relationship. In the targeted logistic model, the negative period-curvature coefficient was:

$$\begin{aligned} & \backslash \\ & -0.604 \pm [-0.651, -0.091]. \\ & \backslash \end{aligned}$$

That interval remained below zero in the host bootstrap, giving reasonably strong evidence for a hill-shaped relationship within the labeled catalog.

But the location of the hill is not fixed. A cooler star emits less energy than a hotter or larger star, so its temperate region can occur at a shorter orbital period. A more energetic star generally requires a wider orbit and a longer planetary year.

The most defensible scientific statement is therefore:

> Radiative habitable-zone candidates tend to occupy an intermediate orbital regime whose location depends on the host star’s overall physical structure.

That statement is better supported than either “long-period planets are more habitable” or “a specific number of days defines a habitable orbit.”

4.2 The broad period–star relationship is stronger than the explicit interaction coefficient

The data support the idea that the favorable period changes with the star, but they do not estimate a standalone period-by-temperature interaction with high precision.

In the targeted period–temperature model, adding the interaction improved out-of-host PR-AUC by:

$$\begin{aligned} & \backslash \\ & +0.0278 \pm [-0.0058, +0.0761] \\ & \backslash \end{aligned}$$

and reduced log-loss by:

$$\backslash$$

$0.00337 \pm [-0.00098, +0.00802]$.

$\backslash$

The point estimates favored the interaction, and approximately 93% of host-bootstrap samples showed improvement. However, the 95% intervals included zero, and the one-sided host sign-flip result for log-loss was $\backslash(p=0.076\backslash)$.

The correct classification is therefore **suggestive**, not conclusive.

The interaction weakened further when stellar radius and mass entered the model:

- the interaction coefficient fell from $\backslash(+0.368\backslash)$ to $\backslash(+0.074\backslash)$;
- the paired shift was $\backslash(-0.294\backslash)$ $\backslash[[-0.546, -0.040]\backslash)$;
- the point magnitude decreased by approximately 80%.

This attenuation has a sensible explanation. Temperature alone is an incomplete description of a star's total energy output. Radius determines how much emitting surface the star has, and mass influences how orbital period maps to physical distance. Once those variables are included, less information has to be carried by the period-by-temperature term.

This does not mean the physical interaction disappeared. It means the more specific statistical interaction:

period $\times$ temperature

was partly functioning as a proxy for the broader relationship:

period $\times$ complete stellar structure

A larger and more diverse labeled sample would be needed to separate those components reliably.

4.3 Collinearity changed attribution, not the existence of the stellar signal

The model could not decide which stellar measurement deserved the most credit, but it clearly benefited from knowing the star's collective structure.

Stellar temperature, radius, and mass were highly correlated:

- temperature–mass correlation: 0.98;
- radius–mass correlation: 0.98;
- temperature–radius correlation: 0.95.

Their variance inflation factors were correspondingly high:

- temperature: 7.8;
- radius: 5.4;
- stellar mass: 11.2.

When radius and mass were introduced, the standardized temperature coefficient moved from:

$$\begin{bmatrix} -1.772 \\ -2.265, -1.095 \end{bmatrix}$$

to:

$$\begin{bmatrix} -0.225 \\ -0.918, +0.522 \end{bmatrix}.$$

That was an approximately 87% movement toward zero.

A naive coefficient interpretation would say:

> “Temperature appeared important until radius and mass were controlled, so temperature

was probably not important after all.”

That conclusion would be misleading. The predictors can replace one another statistically because they encode overlapping information.

The appropriate test removed all stellar-structure information as a block. Doing so reduced out-of-host PR-AUC by:

$$\begin{matrix} \backslash \\ 0.295 \backslash \text{quad} [0.119, 0.473] \\ \backslash \end{matrix}$$

and worsened log-loss by:

$$\begin{matrix} \backslash \\ 0.058 \backslash \text{quad} [0.022, 0.100]. \\ \backslash \end{matrix}$$

The same pattern appeared in the held-out tree-model permutation analyses. Moving temperature, radius, and mass together damaged performance much more reliably than moving one stellar feature at a time.

The correct interpretation is:

> Stellar structure matters strongly at the group level, while unique attribution to temperature, radius, or mass remains unstable.

This is not merely an astronomy-specific lesson. It is a general feature-importance principle: when predictors are strongly correlated, the scientific unit of interpretation may need to be a feature block rather than a single column.

[Coefficient redistribution and joint stellar-block figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.png)

4.4 The simpler model won because it used the available information more efficiently

The tree ensembles were flexible enough to fit more complicated boundaries, but the labeled sample was too small to estimate those boundaries reliably on unseen host systems.

The best-performing classifier was the full stellar-structure ridge-logistic model:

$$\begin{aligned} & \backslash[\\ & \text{PR}\text{-AUC}=0.596\text{quad}[0.447,0.789] \\ & \backslash] \end{aligned}$$

and:

$$\begin{aligned} & \backslash[\\ & \text{log-loss}=0.140\text{quad}[0.102,0.181]. \\ & \backslash] \end{aligned}$$

The constrained random forest achieved:

$$\begin{aligned} & \backslash[\\ & \text{PR}\text{-AUC}=0.475\text{quad}[0.332,0.678] \\ & \backslash] \end{aligned}$$

and:

$$\begin{aligned} & \backslash[\\ & \text{log-loss}=0.162\text{quad}[0.119,0.207]. \\ & \backslash] \end{aligned}$$

On the same held-out hosts, the forest's PR-AUC was lower by:

$$\begin{aligned} & \backslash[\\ & 0.122\text{quad}[0.019,0.235], \\ & \backslash] \end{aligned}$$

while its log-loss was worse by:

$$\begin{aligned} & \backslash[\\ & 0.021\backslashquad[0.004,0.040]. \\ & \backslash] \end{aligned}$$

Gradient boosting performed still worse:

$$\begin{aligned} & \backslash[\\ & \text{PR}\backslash\text{AUC}=0.366\backslashquad[0.243,0.548] \\ & \backslash] \end{aligned}$$

and:

$$\begin{aligned} & \backslash[\\ & \backslash\text{log-loss}=0.183\backslashquad[0.134,0.234]. \\ & \backslash] \end{aligned}$$

Relative to the logistic model, boosting lost:

$$\begin{aligned} & \backslash[\\ & 0.231\backslashquad[0.121,0.364] \\ & \backslash] \end{aligned}$$

in PR-AUC and added:

$$\begin{aligned} & \backslash[\\ & 0.042\backslashquad[0.020,0.067] \\ & \backslash] \end{aligned}$$

to log-loss.

These models were not allowed to grow without constraint. Their depths and leaf sizes were deliberately restricted, and model choice occurred inside nested host-grouped validation. Even then, the validation procedure repeatedly rejected extra capacity:

- the forest selected a shallow depth-three configuration in most folds;
- boosting selected depth-one decision stumps in 86 of 90 folds.

A decision stump can represent a threshold in one feature but cannot express a within-tree interaction. The fact that stumps were usually preferred indicates that the training data did not justify more complex local interactions.

The result is best understood through **inductive bias**. The logistic model assumed a smooth, low-dimensional structure informed by the physical problem. The tree models had to infer the same structure from only 26 positive host systems.

The model-selection conclusion is:

> At this sample size, a structured regularized model was more data-efficient than flexible tree ensembles.

This result does not show that logistic regression is universally superior or that tree models should never be used in exoplanet research. With a larger, more uniformly labeled sample, flexible methods might identify departures from the smooth physical pattern that this analysis could not estimate.

[Classifier comparison figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.png)

4.5 Discovery method created covariate shift before modeling began

Transit and radial-velocity discoveries showed different raw HZ rates mainly because the two methods populated different regions of orbital and stellar feature space.

The host-balanced observed HZ prevalences were:

- transit: 5.6% $\backslash$ [2.9%, 8.5%];
- radial velocity: 20.3% $\backslash$ [9.5%, 32.9%].

It would be tempting to conclude that radial-velocity surveys are better at finding habitable planets.

The physical model was not given the discovery-method label, yet it predicted:

- 5.6% $\backslash[4.2\%,7.2\%]\backslash$ average HZ probability for the transit subset;
- 21.3% $\backslash[15.5\%,27.7\%]\backslash$ for the radial-velocity subset.

The physical covariates almost completely reproduced the raw prevalence difference.

Adding a ridge-shrunk discovery-method indicator did not improve the model:

- PR-AUC changed by $\backslash(-0.017)\backslash\backslash[-0.042,-0.002]\backslash$;
- log-loss worsened by 0.0023 $\backslash[0.0009,0.0043]\backslash$;
- the adjusted RV-versus-transit odds ratio was 0.94 $\backslash[0.30,2.38]\backslash$.

This is consistent with **covariate shift**:

```
Discovery method
      ↓
Which stars and orbital periods are detectable
      ↓
Which systems enter the characterized catalog
      ↓
Observed feature distribution and reported HZ rate
```

The method changes the cases available to the model. Once those cases' orbital and stellar measurements are known, the method label contributes little extra information.

That does not make the catalog unbiased. The selection mechanism occurs upstream:

- transit detection favors geometrically aligned systems and is especially effective for repeatedly orbiting planets;
- radial velocity has different sensitivity to mass and orbital period;
- follow-up resources determine which systems receive detailed stellar measurements;
- reported flux is not populated uniformly across all detected planets.

Host-grouped cross-validation prevents one kind of leakage—putting sibling planets from the

same star in both training and testing. It does not undo this upstream selection process.

The defensible conclusion is:

> **Discovery method affects the distribution of systems in the catalog, even though its label adds little conditional prediction once the observed physical features are included.**

[Discovery-method adjustment figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.png)

4.6 Imaging is a missing-support problem, not a negative scientific result

The dataset contains no labeled imaging cases for the primary model, so it cannot estimate an imaging-specific relationship at all.

There were 54 directly imaged planets in the full catalog, but none had the complete combination of:

- reported insolation flux;
- orbital period;
- stellar temperature;
- stellar radius;
- stellar mass

required for the reported-target model.

In data-science terms, this is not merely a small subgroup. It is **zero labeled support**.

A model cannot estimate an imaging-specific HZ probability any more reliably than a classifier can estimate performance for a class that never appears in its labeled training or test data.

The analysis can say that:

- imaging occupies a distinct region of the raw feature space;

- imaging contributes several extremely wide-orbit systems;
- a small number of imaging planets receive a physics-derived flux estimate.

It cannot say that:

- imaged planets are less likely to be HZ candidates;
- imaged planets have no HZ members;
- the period–temperature relationship is different for imaging;
- the transit/RV model generalizes to directly imaged systems.

Predictions in that regime would be unsupported extrapolation.

This is why the empty or nearly empty imaging panels in the figures are informative. They show where the dataset stops supporting the question.

4.7 The flux-regression result is powerful but leakage-adjacent

Continuous flux regression recovered the catalog’s irradiation calculation almost perfectly, but that is a consistency check rather than independent evidence of habitability prediction.

The log-flux ridge regression achieved:

$$\begin{aligned} & \backslash \\ & R^2=0.9970\quad[0.9954,0.9982], \\ & \backslash \end{aligned}$$

with:

$$\begin{aligned} & \backslash \\ & \text{MAE}=0.0300\ \mathrm{dex}\quad[0.0240,0.0369]. \\ & \backslash \end{aligned}$$

When its predictions were converted to too-cold, HZ, and too-hot categories, three-zone accuracy was:

\[
98.67\%\quad[97.35\%,99.81\%].
\]

Only four planets crossed one of the adopted zone boundaries.

These results are far more stable than the binary classifier. That stability arises for two reasons.

First, continuous regression preserves information that binary classification discards. It distinguishes a planet barely outside the HZ from a planet receiving hundreds of times Earth's stellar energy.

Second, and more importantly, catalog insolation is calculated from quantities closely related to the predictors:

- orbital period or orbital distance;
- stellar temperature;
- stellar radius;
- stellar mass.

This makes the regression **leakage-adjacent**.

It is not literal target leakage: the model was not given the reported flux column itself. But the target is mathematically constructed from nearly the same physical inputs. The model is therefore reproducing an existing calculation rather than discovering a new empirical law from unrelated predictors.

The even broader physics-derived HZ label is closer still to direct circularity because it was explicitly engineered from the features.

The proper use of the regression is to show that:

- the catalog fields are broadly physically consistent;
- the expected irradiation scaling is visible;
- the binary threshold loses information;
- the physical feature set is correctly chosen.

It should not be used to claim that the model can predict actual habitability with 99.7% explanatory power.

[Continuous-flux regression figure]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.png)

4.8 The clustering analysis describes irradiation regimes, not Earth likeness

The clearest cluster result was that short-period giant planets rarely receive moderate stellar flux; the analysis did not identify a uniquely Earth-like planet category.

The hot or warm giant cluster had:

- reported HZ overlap of 0.0% $\left([0.0\%, 3.0\%]\right)$;
- derived HZ overlap of 0.0% $\left([0.0\%, 0.45\%]\right)$.

That result is consistent with the broader period finding: very massive planets on extremely short orbits are usually highly irradiated.

The intermediate-mass, moderate-period cluster had a higher reported point estimate:

$\left[20.0\% \text{ quad } [4.2\%, 37.9\%],\right]$

while the long-period giant cluster had:

$\left[18.2\% \text{ quad } [0.0\%, 44.4\%].\right]$

Those intervals are too wide to rank the clusters reliably. Only 13 positive host systems entered the reported-label clustering comparison.

The broader derived-flux analysis suggested more HZ overlap among long-period giants, but this is partly mechanical because orbital period contributes both to the clustering and to the derived-flux calculation.

A giant planet can also occupy a radiatively moderate orbit without being remotely Earth-like. It may have no solid surface and an atmosphere radically different from Earth's.

Because planet radius was absent, the dataset could not reliably separate:

- rocky super-Earths;
- gas-rich sub-Neptunes;
- small Neptunes.

The cluster labels are therefore descriptive mass-period groups, not definitive planetary taxonomies.

The valid conclusion is:

> The clustering separates broad orbital and mass regimes and identifies overheated short-period giants, but it does not establish which cluster is most likely to contain genuinely habitable terrestrial worlds.

4.9 Physics-aware outlier handling preserved information about the observation process

The most extreme rows were scientifically meaningful cases, and automatically trimming them would have removed entire detection regimes.

The catalog included:

- 192 planets orbiting giant or evolved stars;
- 18 systems with compact remnants or hot subdwarfs;
- 114 planets with periods shorter than one day;
- 74 companions above 13 Jupiter masses;

- four systems with periods above 100,000 days.

These extremes were not randomly distributed:

- giant-host systems were predominantly radial-velocity discoveries;
- ultra-short-period planets were predominantly transit discoveries;
- the widest systems were imaging discoveries.

A generic outlier rule would therefore have altered the discovery-method composition of the data.

The analysis retained these observations, used log transformations, created physical annotations, and applied sensitivity analyses rather than winsorizing them into artificial values.

This choice supports a broader methodological principle:

> In scientific data, statistical rarity is not the same as measurement error. Domain-specific subpopulations should be identified before they are trimmed.

5. What is robust, suggestive, and unsupported?

The categories below apply to this dataset and its observed labeled cohort. “Robust” does not mean universal or causal.

5.1 Relatively robust findings

| Finding | Why it is classified as relatively robust |

|---|---|

| Orbital period is an important individual predictor | Strong held-out permutation effect and stable model contribution |

| The period relationship is curved rather than simply increasing | Negative quadratic term with host-bootstrap interval below zero |

| Stellar temperature, radius, and mass matter jointly | Large block-drop effect in logistic

regression and joint permutation effects in tree models |

| The full stellar ridge-logistic model outperforms the naive baseline | Large PR-AUC and log-loss improvement |

| The ridge-logistic model outperforms the constrained tree ensembles | Paired PR-AUC and log-loss intervals favor logistic regression |

| Discovery method produces strong covariate shift | Large differences in feature composition, missingness, and raw prevalence |

| A method indicator adds little after physical features are known | No improvement in PR-AUC or log-loss; adjusted odds ratio centered near one |

| Continuous flux can be reconstructed from the physical inputs | Very high out-of-host $\backslash(R^2\backslash)$, with the circularity caveat |

| Imaging is unsupported in the reported-label analysis | Zero eligible labeled imaging cases |

5.2 Suggestive findings

| Finding | Why it remains suggestive |

|---|---|

| The favorable period shifts outward for hotter stars | Correct directional response, but interaction-improvement intervals include zero |

| The period $\times$ temperature term independently improves classification | Small positive point improvement, insufficient precision |

| The interaction may operate differently in transit and RV samples | Method-specific positive signals but very small positive-host counts |

| Intermediate-mass, moderate-period planets may have increased reported HZ overlap | Higher point estimate but only five positive planets and overlapping intervals |

| Certain local regions in the tree response surfaces may contain additional structure | Flexible surfaces are unstable and do not improve held-out performance |

5.3 Unsupported claims

| Claim | Why it is unsupported |

|---|---|

| Any listed planet is actually habitable | Flux is only one condition; atmospheres, surfaces, water, and stellar activity are not modeled |

| The period $\times$ temperature interaction is definitively proven | Its incremental uncertainty

interval includes zero |
The model estimates Galactic HZ occurrence	No survey denominator or completeness correction is present
Radial-velocity surveys intrinsically find more habitable planets	Their raw rate is explained by different covariate distributions in the labeled subset
Imaging planets have a lower HZ rate	There are no reported-label imaging cases
One stellar variable has a uniquely causal effect	Stellar features are highly collinear and the analysis is observational
Long-period giant planets are the best Earth analogues	Moderate flux does not make a gas giant Earth-like
The regression's R^2 measures habitability prediction	It mainly measures reconstruction of a related catalog calculation

6. Limitations

6.1 The outcome was missing for more than 90% of planets

The primary target was available for only a small and selectively characterized subset of the catalog.

Reported insolation was missing for 4,205 of 4,575 planets. The labeled subset was not a random 8.1% sample.

Whether flux was available depended on:

- discovery method;
- quality of stellar characterization;
- follow-up observations;
- the history and priorities of the discovering survey;
- which literature values were incorporated into the catalog.

This is a missing-not-at-random problem. Host bootstrapping quantifies uncertainty within the labeled host sample; it cannot make the missing labels representative of all detected planets.

6.2 Only 26 independent positive host systems supported the classifier

The apparent sample of 366 labeled planets contained only 26 host systems with a positive example.

Several planets can share one host star. Treating them as independent would exaggerate the information available about stellar effects.

Host-grouped validation prevented direct leakage across sibling planets, but the small positive-host count still produced:

- broad performance intervals;
- unstable individual coefficients;
- low power for method-specific effects;
- sensitivity to a small number of unusual systems;
- limited ability to estimate nonlinear interactions.

Repeated folds provide a more stable estimate across possible partitions. They do not increase the number of independent positive systems.

The analysis also evaluated several related model variants. The consistency of effects and their uncertainty intervals are therefore more informative than isolated p-values.

6.3 The analysis is internally validated but not externally validated

Repeated host-grouped cross-validation tests new hosts from the same catalog, not new planets from a different survey era or measurement process.

A stronger external validation would use:

- planets confirmed after the 2021 snapshot;
- a later NASA Exoplanet Archive release;
- a survey or mission not dominating the training catalog;
- a prospectively defined target list;
- independently recomputed stellar and flux values.

Such a temporal or survey-level holdout would test whether the selected model survives changes in instrumentation, follow-up practices, and catalog construction.

6.4 Survey completeness was unavailable

The table records confirmed discoveries but not the systems that each survey could have detected and failed to find.

Estimating how common HZ planets are requires denominator information such as:

- which stars were monitored;
- survey duration and observing cadence;
- instrument sensitivity;
- detection thresholds;
- transit injection-and-recovery rates;
- radial-velocity noise and sampling;
- imaging contrast limits and inner working angles.

Without that information, this analysis cannot convert the observed catalog into an unbiased planetary occurrence estimate.

6.5 The HZ threshold is an operational label, not a natural binary truth

The boundaries at 0.35 and 1.75 Earth flux turn a continuous physical quantity into a sharp classification.

A planet just inside a boundary and one just outside it may be physically more similar than two planets on opposite ends of the same class.

This thresholding:

- discards information;
- creates sensitivity for boundary cases;
- combines too-hot and too-cold planets into one negative class;
- can change class balance under alternative HZ definitions.

The continuous regression showed that only four planets crossed the adopted boundaries, and several were close to an edge. Sensitivity analyses with alternative conservative or optimistic HZ limits would be appropriate in future work.

6.6 The catalog lacks many variables needed for actual habitability

Radiative flux is only one part of a planet's climate and biological suitability.

Important missing or incomplete information includes:

- planet radius;
- rocky versus gaseous composition;
- atmospheric pressure and composition;
- greenhouse warming;
- surface reflectivity or albedo;
- water inventory;
- stellar flare and ultraviolet activity;
- stellar age;
- tidal locking;
- magnetic environment;
- long-term orbital stability;
- obliquity and rotation;
- climate feedbacks.

Equilibrium temperature was available for only about 20% of planets and is itself an idealized thermal estimate rather than an observed surface temperature.

Planet radius was absent entirely, preventing a conventional Earth Similarity Index and weakening the physical interpretation of the clusters.

6.7 Planet mass was not measured uniformly

The single mass column combined values produced through different observational and modeling processes.

For radial-velocity planets, the value may be closer to a minimum mass when orbital inclination is unknown. For transiting planets, it may come from follow-up dynamical measurements. For imaged planets, it may depend on brightness and evolutionary models.

The snapshot did not include enough provenance information to standardize these meanings.

Planet mass was therefore excluded from the primary HZ classifier and used only in the explicitly selected mass-characterized clustering cohort.

6.8 Measurement uncertainty was not propagated

The table contained point estimates but not the complete uncertainty distributions needed for an errors-in-variables analysis.

Orbital periods are often precise, but stellar radius, mass, temperature, metallicity, and derived planetary properties can have meaningful uncertainty.

Ignoring predictor uncertainty can:

- attenuate coefficients;

- move planets across HZ boundaries;
- understate uncertainty in response surfaces;
- exaggerate the precision of cluster assignments.

A future analysis should propagate measurement posterior samples or reported upper and lower uncertainties through the full modeling pipeline.

6.9 Composite rows may combine values from different sources

A catalog row can be complete without every value coming from one internally consistent physical solution.

For multi-planet hosts, the audit found cases where different planet rows reported somewhat different values for the same host-star property. This can happen when catalog values are selected from different literature references.

The issue likely contributes to the small discrepancies between reported and physics-reconstructed flux.

A more rigorous physical study would harmonize stellar parameters at the host level before calculating planetary irradiation.

6.10 Giant stars and compact remnants require different evolutionary interpretations

Present-day stellar flux may not summarize the long-term climate history of planets around unusual stars.

A planet currently in a giant star's HZ may previously have experienced a very different energy environment. A planet around a white dwarf or hot subdwarf has also undergone a radically different stellar evolutionary history.

These systems were retained and flagged, but they should not be interpreted through exactly the same biological-habitability lens as stable planets around ordinary main-sequence stars.

6.11 The selected model is predictive, not causal

The logistic coefficients describe conditional associations in a selected catalog, not controlled causal effects.

The analysis has strong physical motivation, but no intervention occurred. We cannot manipulate:

- stellar temperature;
- stellar size;
- orbital period;
- discovery method.

The observed features are also linked through known physical relationships and survey selection.

Accordingly, statements such as “increasing the period causes habitability” are inappropriate. A different period represents a different orbiting system; it is not a treatment applied while everything else remains fixed.

7. Claims the Medium article must not make

The following wording should be treated as a publication guardrail.

| The article must **not** say | Defensible replacement |

|---|---|

| **“These are habitable planets.”** | “These are candidates receiving a broadly Earth-like amount of stellar energy.” |

| **“The model predicts life or Earth-like conditions.”** | “The model predicts a radiative screening label; atmosphere, surface conditions, water, and biology are not observed.” |

| **“We proved that cooler stars and moderate periods cause habitability.”** | “Within the labeled catalog, moderate periods and collective stellar structure predict radiative HZ membership.” |

| **“We proved the period × temperature interaction.”** | “The interaction had the expected direction and modest point improvement, but its 95% uncertainty interval included no improvement.” |

| **“A period of roughly 150, 200, or 600 days is optimal.”** | “The favorable period varies with the host star; exact model peaks are descriptive and poorly identified.” |

| **“Stellar temperature became unimportant after adding radius and mass.”** | “The stellar variables are highly collinear, so individual coefficients became unstable while the joint stellar block remained strongly informative.” |

| **“Radial-velocity surveys find more habitable worlds than transit surveys.”** | “The labeled RV subset had a higher raw HZ rate because it occupied a different region of orbital and stellar feature space.” |

| **“Discovery method does not matter.”** | “The method label added little after conditioning on measured physical features, but discovery method strongly affected which systems and measurements entered the catalog.” |

| **“Direct imaging found no HZ candidates.”** | “The primary dataset contained no imaging discoveries with the labeled support needed for that comparison.” |

| **“The model estimates how common habitable planets are.”** | “The model estimates HZ membership conditional on detection, catalog inclusion, and label availability; it is not an occurrence-rate analysis.” |

| **“The 99.7% regression $\backslash(R^2\backslash)$ proves we can predict habitability almost perfectly.”** | “The regression nearly reconstructed catalog insolation from the quantities used to calculate it; this is a leakage-adjacent physical consistency result.” |

| **“Random forests and gradient boosting are inferior for exoplanet research.”** | “In this small labeled cohort, constrained tree ensembles generalized less well than a physics-informed regularized logistic model.” |

| **“The long-period giant cluster is the most habitable planet type.”** | “Some long-period giants receive moderate flux, but radiative HZ membership does not make a gas giant Earth-like.” |

| **“Our confidence intervals correct detection bias.”** | “The intervals quantify sampling uncertainty among observed hosts; they do not correct the catalog’s selection mechanism.” |

One-sentence editorial test

Before publishing any sentence containing the word **habitable**, ask whether it would remain true after replacing that word with:

> **“receives between 0.35 and 1.75 times Earth’s stellar flux.”**

When the sentence becomes false or misleading after that replacement, it is probably overclaiming.

8. Implications for future research

8.1 More positive host systems matter more than more algorithms

The next major performance gain is more likely to come from broader labeled coverage than from another model family.

The analysis already compared:

- regularized logistic regression;
- random forest;
- gradient boosting;
- continuous ridge regression.

The flexible classifiers did not improve generalization. Continuing to search algorithms would increase researcher degrees of freedom without resolving the information bottleneck.

The priority should be obtaining more independent HZ-positive hosts across:

- stellar temperatures;
- stellar radii and masses;
- orbital periods;
- discovery methods.

8.2 Target reporting should be made more uniform

The main dataset limitation was not a missing predictor but a missing outcome.

Reported flux existed for only 8.1% of the catalog. A standardized, archive-wide flux field computed under documented assumptions would make descriptive comparison much broader.

For predictive science, the archive should retain both:

- published or reported flux;
- a uniformly recomputed flux with provenance and uncertainty.

That distinction would reduce ambiguity between observed catalog fields and deterministic derived targets.

8.3 Survey selection information is required for occurrence claims

Correcting detection bias requires the surveys' opportunities to detect planets, not just the planets that were detected.

Future work should combine planet records with:

- surveyed-star lists;
- observation duration;
- detection completeness;
- injection-and-recovery data;
- instrument sensitivity;
- follow-up selection criteria.

This would permit inverse-probability weighting, hierarchical occurrence modeling, or explicit joint models of detection and HZ membership.

8.4 Imaging needs labeled support before comparison

An imaging comparison requires enough directly imaged systems with consistent stellar, orbital, and flux measurements.

Until those data exist, the appropriate outputs are:

- descriptive imaging feature distributions;
- uncertainty-aware physical case studies;
- explicit “unsupported region” annotations.

A single pooled model should not silently extrapolate from transit and RV systems into imaging territory.

8.5 A stronger future model would combine physical structure with uncertainty

The next useful methodological step is not necessarily a deeper learner; it is a model that respects the physical relationships and measurement uncertainty simultaneously.

Possible extensions include:

- hierarchical models with host-level random effects;
- physically constrained generalized additive models;
- Bayesian errors-in-variables regression;
- monotonic or shape-constrained boosting;
- posterior propagation from stellar and orbital measurements;
- explicit latent-variable models for stellar structure.

Such models could preserve interpretability while allowing carefully controlled departures from the idealized irradiation equation.

8.6 External temporal validation should be pre-specified

A later catalog release can provide a genuine test rather than another cross-validation rearrangement.

The 2021 snapshot could serve as a locked training set. Planets confirmed after that snapshot could then form a temporal test set.

This would evaluate whether the model transfers across:

- later missions;
- changing follow-up practices;
- improved stellar measurements;
- different feature distributions.

The model, threshold, and preprocessing decisions should be fixed before inspecting that test set.

9. Discussion conclusion

The analysis supports a coherent but carefully bounded result.

Orbital period is the strongest individual predictor of radiative habitable-zone membership, but period must be interpreted relative to the host star. Stellar temperature, radius, and mass collectively determine where a moderate-energy orbit can occur. Their collinearity makes individual coefficients unstable, yet their block-level predictive value is substantial.

The data support an intermediate-period relationship and a cooler-star association. The more

specific period-by-temperature interaction appears in the expected direction but remains suggestive because only 26 positive host systems support the primary classifier.

Transit and radial-velocity discoveries have different raw HZ rates because they sample different planetary and stellar regimes. Once those physical differences are modeled, discovery method adds little residual information. This is evidence of covariate shift, not evidence that the catalog is free of detection bias. Imaging remains outside the supported labeled domain.

The full stellar-structure ridge-logistic model is the preferred classifier. It outperforms both constrained random forest and constrained gradient boosting on unseen host systems. The continuous flux regression provides a valuable physical consistency check, but its near-perfect performance is leakage-adjacent and should not be presented as independent habitability prediction.

The central scientific interpretation is:

> Potentially temperate stellar irradiation occurs in a star-dependent orbital band. The data reveal that band most reliably through a simple, regularized model combining orbital period with the host star's collective structure—not through a universal period cutoff, a discovery-method label, or a more flexible tree ensemble.

10. Research-paper synthesis mindmap

Exoplanet habitability drivers

	└─ CRISP-DM analysis	
	└─ Business understanding	COMPLETE
	└─ Data understanding	COMPLETE
	└─ Data preparation	COMPLETE
	└─ Exploratory and structural analysis	COMPLETE
	└─ Predictive modeling	COMPLETE
	└─ Evaluation and model selection	COMPLETE
	└─ Research-paper synthesis	
	└─ Part 1. Abstract, introduction, and methods	COMPLETE
	└─ Part 2. Results	COMPLETE
	└─ Part 3. Discussion and limitations	COMPLETE – THIS CHUNK
	└─ Conditional moderate-period interpretation	
	└─ Suggestive interaction finding	
	└─ Joint stellar-block interpretation	
	└─ Why simpler modeling generalized better	
	└─ Transit-RV covariate shift	
	└─ Imaging as missing support	
	└─ Flux regression as leakage-adjacent	
	└─ Clustering and outlier interpretation	
	└─ Robust / suggestive / unsupported claims	
	└─ Medium article overclaim guardrails	
	└─ Future research priorities	
	└─ Part 4. Conclusions and recommendations	NEXT
	└─ Formal conclusions by research question	
	└─ Modeling and data recommendations	
	└─ Future observing priorities	
	└─ General-audience executive summary	
	└─ Medium-ready ending	
	└─ GitHub README takeaways	

Next: Final Synthesis, Part 4 will provide the formal research conclusions, actionable recommendations, final plain-language summary, and publication-ready closing material for

the Medium article and GitHub README.

Standing requirement: determine which stellar and orbital features predict radiative habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the physical relationship from transit, radial-velocity, and imaging selection effects using a staged, limited-compute CRISP-DM workflow.

PROMPT · SEP 01, 2026 01:12 PM

continue with Final Synthesis, Part 4 — the final chunk.

Since this is the last synthesis piece and feeds directly into my actual remaining deliverables (a GitHub README.md organized as "Assignment 1," a public repo, and the Medium article), please make the outputs directly usable rather than just described:

1. For the "GitHub README takeaways" section, write an actual skeleton README.md structure (section headers plus one or two sentences of what goes under each) that organizes this whole project as Assignment 1 — something I can paste in and fill out with links to the notebook, transcript, and figures.
2. For the "Medium-ready ending," write an actual closing section of publishable prose (not a description of what it should contain) that a reader could scroll to and get the full payoff of the article, consistent with the plain-language voice from Parts 1-2 and the overclaim guardrails from Part 3.
3. In the general-audience executive summary, keep it short enough to work as the article's opening hook (a few tight paragraphs, not a restated abstract) and make sure it leads with the two most surprising findings for a data science reader: the joint-stellar-block-matters-but-individual-coefficients-don't result, and the simpler-model-beats-random-forest-and-gradient-boosting result.

Standing requirement, for the last time: determine which stellar and orbital features predict radiative habitable-zone membership, test whether moderate orbital periods around cooler stars are more favorable, and distinguish the physical relationship from transit, radial-velocity, and imaging selection effects using a staged, limited-compute CRISP-DM workflow.

CHATGPT · SEP 01, 2026 01:12 PM

I'll close with publication-ready conclusions, recommendations, a concise article hook, a finished Medium ending, and a paste-ready "Assignment 1" README skeleton.

Final Synthesis, Part 4

Conclusions, Recommendations, and Paste-Ready Deliverables

This final synthesis closes the CRISP-DM analysis of the uploaded `all_exoplanets_2021.csv` catalog. `filecite{turn2file0}{L1-L4}`

1. General-audience executive summary

Paste-ready opening hook

The first surprise was that no single stellar feature told the whole story. Stellar temperature, radius, and mass became statistically unstable when entered together because they describe overlapping aspects of the same star. Yet removing all three as a group caused a large loss of predictive performance. The individual coefficients were uncertain; the **joint stellar-structure signal was not**.

The second surprise was that more flexible machine-learning models performed worse. A physics-informed ridge-logistic model reached an out-of-host PR-AUC of **0.596** \([0.447, 0.789]\), compared with **0.475** for a constrained random forest and **0.366** for constrained gradient boosting. With only 26 host systems supplying positive examples, the simpler model used the available information more efficiently than the tree ensembles.

The scientific pattern was also more nuanced than “longer orbits are better.” Very short orbits were usually too hot, while extremely long orbits could be too cold. The favorable region was an intermediate orbital band whose location depended on the host star: cooler stars could support moderate irradiation at shorter periods, while hotter, larger stars generally required more distant orbits. The explicit period-by-temperature interaction appeared in the expected direction, but its uncertainty was too large to call it definitively proven.

Finally, the large raw difference between transit and radial-velocity discoveries was mainly a

covariate-shift story. The two methods detected and characterized different kinds of systems. Once orbital and stellar measurements were included, the method label itself added no predictive value. That does not eliminate detection bias—it shows where the bias entered the data-generating process.

2. Formal conclusions by research question

2.1 Which stellar and orbital features predict radiative habitable-zone membership?

Orbital period was the strongest individual predictor, while stellar temperature, radius, and mass formed the strongest correlated predictor block.

Held-out permutation analysis showed that destroying orbital-period information reduced random-forest PR-AUC by:

$$\begin{aligned} & \backslash[\\ & 0.235 \backslash \text{quad}[0.059, 0.446] \\ & \backslash] \end{aligned}$$

and increased log-loss by:

$$\begin{aligned} & \backslash[\\ & 0.091 \backslash \text{quad}[0.049, 0.140]. \\ & \backslash] \end{aligned}$$

Moving stellar temperature, radius, and mass together reduced PR-AUC by:

$$\begin{aligned} & \backslash[\\ & 0.197 \backslash \text{quad}[0.055, 0.376] \\ & \backslash] \end{aligned}$$

and increased log-loss by:

$$\begin{aligned} & \backslash[\\ & 0.036\backslashquad[0.012,0.062]. \\ & \backslash] \end{aligned}$$

The ridge-logistic block-drop analysis was even clearer. Removing the stellar-structure terms from the best classifier reduced out-of-host PR-AUC by:

$$\begin{aligned} & \backslash[\\ & 0.295\backslashquad[0.119,0.473] \\ & \backslash] \end{aligned}$$

and worsened log-loss by:

$$\begin{aligned} & \backslash[\\ & 0.058\backslashquad[0.022,0.100]. \\ & \backslash] \end{aligned}$$

The central feature conclusion is therefore:

> Orbital period supplies a distinct orbital signal, while the host star’s temperature, radius, and mass supply an important but strongly shared stellar signal.

It would be misleading to select one of the three stellar measurements as the uniquely dominant physical feature.

2.2 Are planets with moderate orbital periods around cooler stars more likely to be candidates?

Broadly yes, but “moderate period” is star-dependent rather than a universal range of days.

The negative period-curvature coefficient was:

$$\begin{aligned} & \backslash[\\ & -0.604\backslashquad[-0.651,-0.091]. \end{aligned}$$

\]

This supports a hill-shaped relationship:

very short periods intermediate periods very long periods
often too hot → more favorable → potentially too cold

HZ candidates also tended to orbit cooler stars. In the raw host-balanced comparison, stellar temperature had a Cliff's delta of:

\[

$-0.579 \text{ quad } [-0.735, -0.395]$.

\]

The explicit period-by-temperature interaction improved PR-AUC by:

\[

$0.0278 \text{ quad } [-0.0058, +0.0761]$

\]

and reduced log-loss by:

\[

$0.00337 \text{ quad } [-0.00098, +0.00802]$.

\]

Both point estimates favored the interaction, but both intervals included zero. After stellar radius and mass were included, the interaction coefficient shrank by approximately 80%.

The appropriately qualified conclusion is:

> **The evidence supports a moderate-period band whose location depends on the complete host-star system. The narrower claim that a separately identifiable period × temperature coefficient improves prediction is suggestive rather than conclusive.**

2.3 How did stellar collinearity affect interpretation?

Collinearity redistributed coefficient credit without removing the underlying stellar signal.

The correlations among the stellar variables were approximately:

- temperature–mass: 0.98;
- radius–mass: 0.98;
- temperature–radius: 0.95.

Their VIF values ranged from 5.4 to 11.2.

After radius and mass were added, the standardized temperature coefficient moved from:

$\backslash$
-1.772\quad[-2.265,-1.095]
 $\backslash$

to:

$\backslash$
-0.225\quad[-0.918,+0.522].
 $\backslash$

The interaction coefficient moved from $\backslash(+0.368\backslash)$ to $\backslash(+0.074\backslash)$.

Those shifts mean that the data could no longer assign unique credit to temperature once other measurements of stellar structure were available. They do not mean that the star stopped mattering.

The broader data-science lesson is:

> When several variables measure overlapping aspects of one physical system, interpret them jointly before drawing conclusions from individual coefficients or permutation

ranks.

2.4 Which classifier performed best?

The full stellar-structure ridge-logistic model was the preferred classifier.

Model	PR-AUC	Recall	Precision	F1	MCC	Log-loss						
Prevalence baseline	0.077	$\backslash([0.049,0.107])$	0	0	0	0.271	$\backslash([0.201,0.346])$					
Full stellar ridge logistic	0.596	$\backslash([0.447,0.789])$	0.802	$\backslash([0.631,0.957])$	0.542	$\backslash([0.379,0.698])$	0.647	$\backslash([0.496,0.775])$	0.626	$\backslash([0.476,0.761])$	0.140	$\backslash([0.102,0.181])$
Constrained random forest	0.475	$\backslash([0.332,0.678])$	0.691	$\backslash([0.497,0.870])$	0.535	$\backslash([0.358,0.709])$	0.603	$\backslash([0.432,0.742])$	0.571	$\backslash([0.394,0.721])$	0.162	$\backslash([0.119,0.207])$
Constrained gradient boosting	0.366	$\backslash([0.243,0.548])$	0.786	$\backslash([0.600,0.945])$	0.337	$\backslash([0.215,0.465])$	0.472	$\backslash([0.325,0.599])$	0.456	$\backslash([0.310,0.586])$	0.183	$\backslash([0.134,0.234])$

On the same held-out hosts:

- random forest lost **0.122 PR-AUC** $\backslash([0.019,0.235])$ and added **0.021 log-loss** $\backslash([0.004,0.040])$;
- gradient boosting lost **0.231 PR-AUC** $\backslash([0.121,0.364])$ and added **0.042 log-loss** $\backslash([0.020,0.067])$.

These are meaningful negative findings.

> With only 26 positive host systems, the scientifically structured model generalized more reliably than models able to construct many local nonlinear rules.

No further algorithm search or ensembling is justified by these results.

2.5 How did discovery method affect the observed relationship?

Discovery method strongly changed which systems entered the sample, but its label added little after the physical features were known.

The observed host-balanced HZ rates were:

- transit: 5.6% $\backslash([2.9\%,8.5\%])$;
- radial velocity: 20.3% $\backslash([9.5\%,32.9\%])$.

The physical model, without receiving discovery method, predicted:

- transit: 5.6% $\backslash([4.2\%,7.2\%])$;
- radial velocity: 21.3% $\backslash([15.5\%,27.7\%])$.

Adding a regularized method indicator:

- reduced PR-AUC by 0.017 $\backslash([0.002,0.042])$;
- worsened log-loss by 0.0023 $\backslash([0.0009,0.0043])$;
- produced an adjusted RV-versus-transit odds ratio of 0.94 $\backslash([0.30,2.38])$.

The correct interpretation is:

> Transit and radial velocity produced different observed HZ rates because they sampled different stellar and orbital populations. Once those physical differences were represented, the discovery-method label contained little additional information.

This is covariate shift, not evidence of an unbiased catalog.

Direct imaging remained unidentifiable because no imaging planet entered the reported-label modeling cohort.

2.6 What did clustering contribute?

Clustering clearly isolated an overheated short-period giant population, but it did not

identify one definitively “most habitable” planet type.

The hot/warm-giant cluster had:

- reported HZ overlap of 0.0% $\backslash[0.0\%,3.0\%]\backslash$;
- derived HZ overlap of 0.0% $\backslash[0.0\%,0.45\%]\backslash$.

The intermediate-mass, moderate-period cluster had a reported HZ rate of:

$\backslash$
20.0%\quad[4.2\%,37.9\%],
 $\backslash$

while the long-period giant cluster had:

$\backslash$
18.2%\quad[0.0\%,44.4\%].
 $\backslash$

The intervals were too wide to rank those clusters reliably.

Because planet radius was absent, the analysis could not cleanly separate rocky super-Earths from gas-rich sub-Neptunes. Cluster membership should therefore be interpreted as broad mass–period structure, not as a definitive habitability taxonomy.

2.7 What did continuous-flux regression add?

Continuous regression gave the clearest physical picture, but not the most independent scientific test.

The log-flux ridge regression achieved:

- MAE: 0.0300 dex $\backslash[0.0240,0.0369]\backslash$;
- RMSE: 0.0624 dex $\backslash[0.0478,0.0764]\backslash$;
- $\backslash(R^2\backslash)$: 0.9970 $\backslash[0.9954,0.9982]\backslash$.

After converting predicted flux back to hot/HZ/cold zones, accuracy was:

$\left[98.67\% \pm [97.35\%, 99.81\%] \right]$

This result confirms that the catalog's stellar and orbital quantities are highly consistent with the expected irradiation relationship.

It is also leakage-adjacent. Reported insolation is calculated from closely related quantities, so the regression primarily reconstructs the catalog's physical calculation.

It should be presented as:

> a physical consistency and continuous-target cross-check

rather than:

> a near-perfect model of actual habitability.

3. Final evidence classification

Relatively robust within this catalog

- Orbital period is an important individual predictor.
- The period relationship is curved rather than monotonically increasing.
- Stellar temperature, radius, and mass matter jointly.
- The full stellar ridge-logistic model substantially outperforms the naive baseline.
- The logistic model outperforms both constrained tree ensembles.
- Discovery method creates substantial covariate shift.
- A method label adds no useful residual prediction after physical adjustment.
- Flux can be reconstructed accurately from orbital and stellar quantities.

- The primary dataset cannot support an imaging-specific result.

Suggestive

- The favorable orbital period shifts outward for hotter stars.
- An explicit period $\times$ temperature term adds predictive value.
- The interaction may differ between transit and radial-velocity populations.
- Intermediate-mass, moderate-period planets may have elevated reported HZ overlap.

Unsupported

- Any candidate is actually habitable or inhabited.
- The period $\times$ temperature interaction has been definitively proven.
- The model estimates the prevalence of HZ planets in the Galaxy.
- Radial-velocity surveys intrinsically find more habitable worlds.
- Imaging planets are less likely to occupy habitable-zone orbits.
- One stellar variable has a uniquely causal effect.
- Long-period giant planets are Earth analogues.
- The flux-regression $\backslash(R^2\backslash)$ represents near-perfect habitability prediction.

4. Final recommendations

4.1 Modeling recommendation

Adopt the **full stellar-structure ridge-logistic model without discovery method** as the project's primary classifier.

Use it as a candidate-ranking or screening model rather than as an autonomous scientific decision rule. Its operating threshold should be selected according to the cost of missed candidates versus unnecessary follow-up, not treated as a universal constant.

Do not add an ensemble merely to increase apparent sophistication. Both evaluated tree ensembles lost performance on unseen host systems.

4.2 Interpretation recommendation

Report two levels of feature importance:

1. orbital period as the strongest individual predictor;
2. stellar temperature, radius, and mass as one correlated stellar-structure block.

Avoid claims such as “temperature is unimportant” or “radius is the dominant stellar variable.” The data do not support that level of attribution.

Describe the central physical finding as:

> **a moderate-period × stellar-structure relationship**

rather than reducing it to a definitive temperature-interaction coefficient.

4.3 Validation recommendation

Preserve host-grouped validation in every future version of the project.

A random row split would allow planets from the same stellar system to appear in both training and testing, creating optimistic leakage through shared host features.

For future model comparisons:

- reuse a locked set of host-level folds;
- make configuration decisions inside nested grouped validation;

- report paired whole-host uncertainty intervals;
- avoid choosing models from point metrics alone.

4.4 Data-preparation recommendation

Continue to avoid catalog-wide median imputation for highly incomplete physical fields.

In particular:

- do not impute missing insolation labels as non-HZ;
- do not impute missing eccentricity as zero;
- do not globally impute planet mass across discovery methods;
- do not use equilibrium temperature in the primary HZ classifier;
- do not treat a derived flux label as independent evidence.

High-missingness variables can be used in explicitly labeled subset or sensitivity analyses when their observational meaning is preserved.

4.5 Data-collection recommendation

The most valuable next step is more consistent data, not another algorithm.

Priorities are:

1. standardized reported and uniformly recomputed flux values with provenance;
2. planet radius, needed to distinguish terrestrial and gaseous worlds;
3. measurement uncertainties and posterior samples;
4. host-level harmonization of stellar parameters;
5. larger labeled imaging and microlensing samples;
6. survey completeness and detection-efficiency information;
7. stellar age, activity, flare environment, and atmospheric measurements.

4.6 External-validation recommendation

Use a later exoplanet catalog release as a pre-specified temporal test set.

The complete 2021 workflow—including preprocessing, selected features, regularization, and evaluation threshold—should be locked before viewing the later systems. This would test whether the result transfers across new missions, improved measurements, and changing catalog practices.

4.7 Communication recommendation

Use the phrase:

> **radiative habitable-zone candidate**

or:

> **planet receiving broadly Earth-like stellar energy**

throughout public communication.

Avoid shortening the result to “habitable planet” unless the sentence explicitly states that this is only a flux-based first-pass screen.

5. Medium-ready ending

Paste-ready closing section

What the model learned—and what it did not

The most important result of this project was not that one stellar variable won a feature-importance contest. In fact, the individual stellar variables refused to behave that neatly.

Stellar temperature, radius, and mass are different measurements, but they describe overlapping parts of the same physical system. Hotter stars tend to be larger and more massive, and all three measurements help determine how much energy reaches an orbiting planet. Once they entered the model together, their individual coefficients became unstable. Temperature's coefficient moved sharply toward zero, and the explicit period-by-temperature interaction became much smaller.

It would be easy to look at those coefficients and conclude that stellar temperature did not matter after all.

The group-level test showed the opposite.

When all of the stellar information was removed together, out-of-host PR-AUC fell by about 0.30 and log-loss became substantially worse. The model could not reliably divide the credit among temperature, radius, and mass, but it clearly needed the information they carried collectively.

That is one of the most transferable data-science lessons in the analysis: **an unstable individual coefficient is not the same thing as an unimportant feature family**. When predictors are strongly collinear, the scientifically meaningful question may be whether the group matters, not which member of the group finishes first in a ranking.

The second major lesson came from the model comparison.

Random forests and gradient boosting were given a fair but deliberately constrained opportunity to improve the result. Their depth, leaf sizes, and learning rates were restricted to reduce memorization, and every configuration decision was made inside nested host-grouped validation.

Neither ensemble won.

The full stellar-structure ridge-logistic model achieved a PR-AUC of 0.596 on unseen host systems. The random forest reached 0.475, and gradient boosting reached 0.366. Both flexible models also produced worse probability estimates.

This is not evidence that tree ensembles are bad algorithms. It is evidence that flexibility has to be paid for with data. Here, only 26 independent host systems contributed positive examples. The logistic model used a small amount of justified structure—a curved period relationship, a limited interaction hypothesis, and regularization—to extract more stable information from that small sample.

Sometimes the best model is not the one that can represent the most complicated boundary. It is the one whose assumptions most efficiently match the information the dataset can actually support.

The astrophysical story is equally important.

The analysis did not find that longer planetary years are always better. Very short orbits were usually too hot, but a planet can also orbit so far away that it receives too little energy. The favorable region was an intermediate orbital band.

And that band moved with the star.

A cooler star can provide moderate heating to a relatively close planet with a short year. A hotter or larger star generally requires the planet to orbit farther away. The explicit period-by-temperature interaction appeared in the expected direction, but the evidence was not precise enough to call that one coefficient proven. The stronger conclusion was broader: orbital period has to be interpreted together with the complete stellar environment.

Discovery method added another layer to the story.

Radial-velocity discoveries had a much higher raw habitable-zone rate than transit discoveries in the labeled sample. But the difference was almost completely reproduced by a model that did not know the discovery method. Transit and radial-velocity surveys had populated different regions of period and stellar-feature space.

That is covariate shift in a scientific setting. The method affected which systems appeared in the dataset before the classifier ever saw them. Adding the method label afterward did not

undo that selection process—and it did not improve prediction.

Direct imaging provided an even clearer warning. The reported-flux modeling sample contained no eligible imaging planets. That is not evidence that imaging finds no planets in habitable-zone orbits. It is a region with no labeled support, and responsible analysis has to say so rather than silently extrapolate.

Finally, the continuous regression reconstructed catalog insolation with an $\langle R^2 \rangle$ of 0.997. That number is visually impressive, but it is not the headline. Catalog insolation is calculated from the same kinds of stellar and orbital measurements used as predictors. The regression demonstrated physical and data consistency; it did not predict life with 99.7% accuracy.

This project never identified inhabited worlds. It identified which measured orbital and stellar conditions predict a broad range of Earth-like stellar heating among already detected and characterized planets.

The final result is narrower than a search for “Earth 2.0,” but it is also more defensible:

> Potentially temperate irradiation occurs in a star-dependent orbital band, and in this small labeled dataset, a simple physics-informed model described that band more reliably than more flexible machine-learning ensembles.

That is both an astrophysical finding and a data-science one. Good models do not merely fit the available rows. They respect how the target was constructed, how the observations were selected, where the dataset has no support, and how much complexity the evidence can sustain.

6. GitHub README.md skeleton

Paste-ready “Assignment 1” structure

The Markdown below can be pasted directly into `README.md`. Replace the uppercase placeholders and adjust relative paths to match the final repository.

`markdown

Assignment 1 – Exoplanet Habitability Drivers

Predicting Radiative Habitable–Zone Membership with CRISP–DM

This project investigates which stellar and orbital measurements predict whether a confirmed exoplanet receives a broadly Earth–like amount of stellar energy. It also tests whether the apparent relationship changes across transit, radial–velocity, and direct–imaging discovery methods.

> **Scope:** This is a radiative habitable–zone screening analysis. It does not determine whether any planet is actually habitable, Earth–like, or inhabited.

Project Links

- **Analysis notebook:** [Open the notebook]
(./notebooks/assignment_1_exoplanet_habitability.ipynb)
- **Rendered notebook:** [View the HTML report]
(./reports/assignment_1_exoplanet_habitability.html)
- **Analysis transcript:** [Read the staged CRISP–DM analysis]
(./docs/analysis_transcript.md)
- **Medium article:** [Read the public article](PASTE_MEDIUM_URL_HERE)
- **Source dataset:** [Kaggle / NASA Exoplanet Archive reference]
(PASTE_DATASET_URL_HERE)

Executive Summary

The strongest individual predictor of radiative habitable–zone membership was orbital period, but period was only meaningful in the context of the host star. Habitable–zone candidates tended to occupy an intermediate orbital range, with cooler stars allowing favorable irradiation at shorter periods.

Stellar temperature, radius, and mass had unstable individual coefficients because they were highly collinear. However, removing them as a group caused a large decline in out–of–host performance, showing that the joint stellar–structure signal was important even when individual attribution was uncertain.

A full stellar–structure ridge–logistic model outperformed constrained random

forest and gradient-boosting models. The result suggests that, with only 26 positive host systems, a physics-informed regularized model was more data-efficient than flexible tree ensembles.

Research Questions

1. Which stellar and orbital features predict radiative habitable-zone membership?
2. Are moderate-period planets around cooler stars more likely to be candidates?
3. Does a log-period $\times$ stellar-temperature interaction improve prediction?
4. How do transit, radial-velocity, and imaging selection effects influence the observed relationship?
5. Do nonlinear tree ensembles outperform a regularized, physics-informed logistic model?

What “Habitable Zone” Means Here

The habitable zone is used as a first-pass energy screen: the orbital region where a planet receives neither extremely high nor extremely low stellar energy.

The project labels a planet as a candidate when its reported insolation lies between 0.35 and 1.75 times the flux received by Earth. This label does not measure atmosphere, water, surface composition, climate stability, or biological activity.

Dataset

The source file contains 4,575 confirmed exoplanets and 23 columns, including:

- orbital period and orbital size;
- planet mass and eccentricity;
- reported insolation and equilibrium temperature;
- stellar temperature, radius, mass, metallicity, and surface gravity;
- discovery method, year, and facility.

Only 370 planets have reported insolation values. The final primary modeling cohort contains 366 planets from 264 host systems, including 32 positive planets

belonging to only 26 positive host systems.

Important Data Limitations

- Reported insolation is missing for 91.9% of planets.
- Planet radius is absent, preventing a conventional Earth Similarity Index.
- Planet mass and eccentricity are strongly missing and discovery-method dependent.
- No directly imaged planet enters the primary reported-flux modeling cohort.
- The catalog contains confirmed discoveries, not a random sample of all planetary systems.

CRISP-DM Workflow

1. Business Understanding

The project defines a narrow and measurable target: radiative habitable-zone candidacy rather than actual habitability. The primary scientific hypothesis is that favorable orbital periods depend on the host star rather than following one universal cutoff.

2. Data Understanding

The analysis audits column definitions, units, missingness, discovery-method composition, duplicate records, physical ranges, class imbalance, and the number of independent host systems.

3. Data Preparation

Missing outcome values remain unknown rather than being converted to negative labels. High-missingness physical fields are not globally median-imputed, and all preprocessing is fitted inside host-grouped validation.

4. Exploratory Analysis

Exploratory analysis examines the period-temperature pattern, missingness structure, discovery-method differences, physical outliers, feature collinearity, and mass-period clusters.

5. Modeling

The model sequence includes naive baselines, targeted and full ridge-logistic models, a method-adjusted logistic model, constrained random forest, constrained gradient boosting, and a continuous log-insolation regression cross-check.

6. Evaluation

All classifiers use repeated host-grouped cross-validation, whole-host bootstrap uncertainty intervals, and paired comparisons on identical held-out hosts. PR-AUC, recall, F1, Matthews correlation, and log-loss are emphasized instead of raw accuracy.

Leakage Controls

The primary HZ classifier does not use:

- reported insolation;
- derived insolation;
- equilibrium temperature;
- HZ labels or zone labels;
- Earth-equivalent orbital-period features;
- planet or host identifiers.

Derived flux is used only as a physics-based sensitivity and consistency check.

Validation Design

The analysis uses:

- 15 repeats;
- 6 outer host-grouped folds per repeat;
- 90 outer test folds in total;
- nested host-grouped configuration and threshold selection;
- whole-host bootstrap confidence intervals.

Grouping by `Planet Host` prevents planets from the same stellar system appearing in both training and testing.

Exploratory Findings

Period and Stellar Temperature

Habitable-zone candidates tend to occupy longer periods and cooler stellar environments than the dominant non-HZ population. The pattern is conditional: a favorable period around a cool star may be too short around a hotter or larger star.

![Orbital period versus stellar temperature]
(./figures/raw_period_vs_stellar_temperature_by_method.png)

Missingness and Selection

Missingness follows the discovery and follow-up process rather than appearing randomly. This motivates the decision not to apply broad global imputation.

![Missingness by column](./figures/raw_missingness_by_column.png)

Physics-Aware Outliers

Giant hosts, compact remnants, ultra-short-period planets, and wide imaging companions are retained and annotated rather than automatically removed. These cases are genuine astrophysical subpopulations and are strongly associated with discovery method.

Feature Selection and Collinearity

Orbital period is nearly independent of the stellar measurements, while stellar temperature, radius, and mass are strongly collinear.

Individual stellar coefficients become unstable when all stellar measurements enter the model. Joint block tests show that the stellar features remain collectively important.

![Joint stellar-block effect]
(./figures/coefficient_shift_and_joint_stellar_effect_final.png)

Clustering

A four-cluster mass-period solution identifies:

1. low-mass, short-period planets;
2. intermediate-mass, moderate-period planets;
3. hot or warm giants;
4. long-period giants.

The hot/warm-giant cluster has essentially no HZ overlap. The longer-period clusters cannot be ranked reliably using the sparse reported-flux labels.

![Mass-period clusters](./figures/mass_period_clusters_with_hz_overlap.png)

Model Results

Model	PR-AUC	Recall	F1	MCC	Log-loss
Prevalence baseline	0.077	0.000	0.000	0.000	0.271
Full stellar ridge logistic	0.596	0.802	0.647	0.626	0.140
Constrained random forest	0.475	0.691	0.603	0.571	0.162
Constrained gradient boosting	0.366	0.786	0.472	0.456	0.183

All reported metrics are based on repeated out-of-host predictions. Full uncertainty intervals are available in the results tables and notebook.

![Classifier comparison]

(./figures/chunk7_classifier_performance_and_selection.png)

Best Model

The selected classifier is the **full stellar-structure ridge-logistic model without discovery method**.

It combines:

- log orbital period;
- orbital-period curvature;
- stellar effective temperature;

- stellar radius;
- stellar mass;
- a targeted period $\times$ temperature term.

The model is retained because it provides the best out-of-host ranking, classification balance, probability quality, and interpretability.

Period $\times$ Temperature Result

The explicit interaction has the expected direction, and its point estimates improve PR-AUC and log-loss. However, the 95% uncertainty intervals include no improvement.

The interaction is therefore reported as ****suggestive, not conclusive****. The stronger result is that moderate orbital period depends on the complete stellar environment.

Joint Stellar-Structure Result

Adding radius and mass causes the individual temperature and interaction coefficients to shrink substantially because the stellar variables share information.

Removing the full stellar block reduces PR-AUC by approximately 0.295 and worsens log-loss by approximately 0.058. Stellar structure is therefore important jointly even though individual coefficient attribution is unstable.

Discovery-Method Bias

The raw HZ rate is higher for radial-velocity discoveries than for transit discoveries. A physical model without discovery method already reproduces that difference.

Adding a regularized discovery-method indicator does not improve prediction. This suggests that method bias is expressed mainly through covariate shift—which stars and orbital periods enter each sample—rather than through an additional method effect after conditioning on the measured physical features.

![Discovery-method adjustment](./figures/discovery_method_adjustment_final.png)

Direct imaging remains unsupported in the primary model because no imaging planet has the required reported label and complete core features.

Continuous Flux Cross-Check

A ridge regression predicts log reported insolation with:

- MAE ≈ 0.030 dex;
- RMSE ≈ 0.062 dex;
- $\backslash(R^2 \approx 0.997\backslash)$.

This confirms the physical consistency of the catalog fields. It is not an independent habitability result because reported flux is calculated from closely related stellar and orbital quantities.

![Continuous-flux regression](./figures/chunk7_log_insolation_regression_crosscheck.png)

Main Conclusions

- Orbital period is the strongest individual predictor.
- Very short and very long periods are less favorable than an intermediate orbital region.
- The favorable period depends on the host star.
- Stellar temperature, radius, and mass matter collectively.
- The period $\times$ temperature interaction is suggestive rather than proven.
- Discovery method creates covariate shift but adds little residual predictive information.
- The physics-informed ridge-logistic model outperforms random forest and gradient boosting.
- Continuous flux prediction is stable but leakage-adjacent.
- The results predict radiative HZ candidacy, not actual habitability.

Claims This Project Does Not Make

This project does not claim that:

- any listed planet is inhabited or genuinely habitable;
- the model estimates how common habitable planets are in the Galaxy;
- radial-velocity surveys intrinsically find more habitable planets;
- directly imaged planets have a lower HZ rate;
- one stellar feature has a uniquely causal effect;
- the period $\times$ temperature interaction is definitively proven;
- the log-flux regression predicts habitability with 99.7% accuracy.

See [Discussion and Limitations](./docs/discussion_and_limitations.md) for the complete claim guardrails.

Repository Structure

text

```
.
├── README.md
├── data
│   ├── raw
│   │   └── all_exoplanets_2021.csv
│   └── processed
│       └── exoplanets_prepared_no_statistical_imputation.csv
├── notebooks
│   └── assignment_1_exoplanet_habitability.ipynb
├── src
│   ├── prepare_exoplanets.py
│   ├── feature_outlier_analysis.py
│   ├── clustering_analysis.py
│   ├── logistic_models.py
│   ├── random_forest_model.py
│   └── gradient_boosting_and_flux_regression.py
├── figures
│   ├── raw_period_vs_stellar_temperature_by_method.png
│   ├── raw_missingness_by_column.png
│   └── coefficient_shift_and_joint_stellar_effect_final.png
```



```
| | └── discovery_method_adjustment_final.png
| | └── mass_period_clusters_with_hz_overlap.png
| | └── chunk7_classifier_performance_and_selection.png
| └── results
| | └── model_performance.csv
| | └── feature_importance.csv
| | └── interaction_tests.csv
| | └── method_bias_analysis.csv
| └── reports
| | └── assignment_1_exoplanet_habitability.html
| └── docs
| | └── analysis_transcript.md
| | └── discussion_and_limitations.md
| └── requirements.txt
└── LICENSE
```

```
---
```

```
## Reproducing the Analysis
```

```
Create an isolated Python environment and install the project dependencies:
```

```
bash
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

```
Run the notebook from the repository root:
```

```
bash
jupyter lab notebooks/assignment_1_exoplanet_habitability.ipynb
```

Record the exact Python and package versions used for the final run in
`requirements.txt` or an environment lock file.

Suggested Python Dependencies

- pandas
- numpy
- scipy
- scikit-learn
- matplotlib
- joblib
- jupyter

Add any additional packages actually imported by the final notebook.

Reproducibility Notes

All learned preprocessing, threshold selection, and configuration selection must occur inside the training portion of each host-grouped split.

Random seeds should be fixed and listed in the notebook. Generated tables should be written to `results/`, and publication figures should be written to `figures/`.

Future Work

The highest-priority extensions are:

- temporal validation on post-2021 discoveries;
- survey-completeness correction;
- standardized flux reporting;
- planet-radius and atmospheric data;
- measurement-uncertainty propagation;
- larger imaging and microlensing labeled samples;
- host-level harmonization of stellar parameters.

The next research gain is more likely to come from broader and better-characterized data than from additional algorithm search.

Data and Licensing

Document the Kaggle dataset source, NASA Exoplanet Archive attribution, and applicable data-distribution terms here.

Do not commit the raw dataset to the public repository when its distribution terms require users to download it from the original source. In that case, include a download script or clear setup instructions instead.

Citation

Use the citation format required by the course and acknowledge both the Kaggle dataset publisher and the NASA Exoplanet Archive.

A repository citation can be added after publication:

bibtex

```
@misc{YOUR_NAME_2026_exoplanet_habitability,  
  author    = {YOUR NAME},  
  title     = {Assignment 1: Exoplanet Habitability Drivers},  
  year      = {2026},  
  howpublished = {GitHub repository},  
  url       = {PASTE_REPOSITORY_URL_HERE}  
}
```

Author

YOUR NAME

Master's Program in Data Science

COURSE / UNIVERSITY

DATE

License

Add the selected code and documentation license here. Confirm that the chosen license is compatible with all included assets and data files.

,

7. Recommended public-repository figure set

Place these five figures in the repository's `figures/` directory and use the first four in the main Medium article:

1. [Raw period–temperature pattern]

(sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.png)

2. [Classifier performance comparison]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.png)

3. [Coefficient redistribution and joint stellar effect]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.png)

4. [Discovery-method covariate shift]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.png)

5. [Raw missingness structure]

(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.png)

Useful technical-appendix figures:

- [Mass–period clustering]

(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.png)

- [Physics-aware outlier landscape]

(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.png)

- [Continuous-flux regression]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.png)

- [Period–temperature response surface]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.png)

8. Final project conclusion

The project answers the standing research question with a qualified yes.

Stellar and orbital features can predict whether a detected, characterized exoplanet falls within the adopted radiative habitable-zone interval. Orbital period is the strongest individual predictor, while stellar temperature, radius, and mass matter collectively. The relevant orbital pattern is not “the longer the better”; it is a moderate-period band determined by the host star’s overall structure.

The evidence for a specific period-by-temperature interaction is directionally consistent with the expected physics but remains suggestive because the labeled sample contains only 26 positive host systems. Once radius and mass are included, much of the interaction signal is redistributed among the correlated stellar variables.

Transit and radial-velocity discoveries have different apparent HZ rates because they sample different regions of stellar and orbital feature space. Discovery method therefore creates substantial covariate shift. Its label adds no useful predictive information after the physical features are included, but this does not correct the upstream detection and characterization

bias. Imaging cannot be compared because the primary dataset provides no eligible labeled imaging cases.

The full stellar-structure ridge-logistic model is the selected classifier. It generalizes better than both constrained random forest and constrained gradient boosting under repeated host-grouped validation. The continuous flux regression confirms the expected irradiation relationship but remains a leakage-adjacent consistency check rather than independent evidence of planetary habitability.

The final plain-language conclusion is:

> A potentially temperate orbit is not defined by one universal year length. It is a moving target set by the star. In this small, selectively labeled catalog, the most reliable way to describe that target was not a more complicated machine-learning model, but a carefully validated, regularized model that respected the underlying physics and the limits of the data.

9. Completed project map

Assignment 1 – Exoplanet habitability drivers

└─ CRISP-DM

└─ Business understanding	COMPLETE
└─ Data understanding	COMPLETE
└─ Data preparation	COMPLETE
└─ Modeling	COMPLETE
└─ Evaluation	COMPLETE
└─ Communication	COMPLETE

└─ Scientific questions

└─ Predictive stellar/orbital features	ANSWERED
└─ Moderate-period × cooler-star pattern	SUPPORTED BROADLY
└─ Explicit period × temperature term	SUGGESTIVE
└─ Transit versus RV selection effects	CHARACTERIZED
└─ Imaging comparison	UNSUPPORTED BY LABELS

└─ Model decision

└─ Full stellar ridge logistic	SELECTED
└─ Method-adjusted logistic	NOT PROMOTED
└─ Random forest	NOT PROMOTED
└─ Gradient boosting	NOT PROMOTED
└─ Continuous flux regression	PHYSICS CROSS-CHECK
└─ Further model search	CLOSED

└─ Final deliverables

└─ Research-paper synthesis	COMPLETE
└─ Medium article opening hook	COMPLETE
└─ Medium-ready closing section	COMPLETE
└─ Assignment 1 README skeleton	COMPLETE
└─ Publication figure shortlist	COMPLETE
└─ Public-claim guardrails	COMPLETE

PROMPT · SEP 01, 2026 05:10 PM

Create a graphic for this to be used in the medium article:

The expected pattern is closer to an inverted U:

very short period → moderate period → very long period

usually too hot potentially temperate potentially too cold

PROMPT · SEP 01, 2026 05:19 PM

Make this table into an image that I can put in my medium article:

Group	Median orbital period	Median stellar temperature
-------	-----------------------	----------------------------

— — — —

Habitable-zone candidates	32.4 days	3,466 K
---------------------------	-----------	---------

Planets outside the adopted zone	6.7 days	5,055 K
----------------------------------	----------	---------

PROMPT · SEP 01, 2026 05:52 PM

Create an image of this table that I can use in my medium article.

Feature pair	Correlation	Plain-language interpretation
--------------	-------------	-------------------------------

--

-----	-----:	-----

Stellar temperature — stellar mass	0.98	Hotter stars in this sample are generally more massive.
------------------------------------	-------------	---

Stellar radius — stellar mass	0.98	Larger stars are generally more massive.
-------------------------------	-------------	--

--

Stellar temperature — stellar radius	0.95	Hotter stars also tend to be larger.
--------------------------------------	-------------	--------------------------------------

--

Orbital period — stellar temperature	0.00	A planet's year length has essentially no overall correlation with its star's temperature.
--------------------------------------	-------------	--

Orbital period — stellar mass	-0.02	A planet's year length is also almost independent of stellar mass.
-------------------------------	--------------	--

PROMPT · SEP 05, 2026 12:13 PM

[image / attachment]

This graphic has chart elements blocked and the legends don't fully explain all of the markers on the charts. Remake this chart using the same data that was previously made but move the legend off of the chart and make sure the legend includes all of the markers. Make sure that the markers on the chart appropriately represent the data and the values that were previously calculated.

CHATGPT · SEP 05, 2026 12:14 PM

I'm rebuilding it as a single, uncluttered forest plot so nothing is covered: all three model-stage markers plus the paired-shift marker will be explained in an external legend, and every point/interval will come directly from the previously calculated coefficient tables.

CHATGPT · SEP 05, 2026 12:14 PM

Remade with the same previously calculated coefficient and uncertainty values, a single uncluttered forest-plot layout, and an external legend that explains every marker type.

- [PNG for Medium](sandbox:/mnt/data/coefficient_shift_and_joint_stellar_effect_remade.png)
- [Vector PDF](sandbox:/mnt/data/coefficient_shift_and_joint_stellar_effect_remade.pdf)

PROMPT · SEP 05, 2026 12:17 PM

Create an image of a table that I can use in my medium article that represents this information:

Model	PR-AUC	Recall	F1	MCC	Log-loss
-------	--------	--------	----	-----	----------

-- --: --: --: --: --:

Prevalence baseline	0.077	0.000	0.000	0.000	0.271
---------------------	-------	-------	-------	-------	-------

Full stellar ridge logistic	0.596	0.802	0.647	0.626	0.140
------------------------------------	--------------	--------------	--------------	--------------	--------------

Constrained random forest	0.475	0.691	0.603	0.571	0.162
---------------------------	-------	-------	-------	-------	-------

Constrained gradient boosting	0.366	0.786	0.472	0.456	0.183
-------------------------------	-------	-------	-------	-------	-------

PROMPT · SEP 05, 2026 01:49 PM

Create an image of a table that I can use in my medium article that represents this information:

| Discovery method | Observed HZ prevalence |

| — | — |

| Transit | 5.6% [2.9%, 8.5%] |

| Radial velocity | 20.3% [9.5%, 32.9%] |

PROMPT · SEP 13, 2026 03:08 PM

regenerate this image: Discovery-method covariate shift

PROMPT · SEP 13, 2026 06:30 PM

list all of the artifacts from each chunk

Below is the complete artifact inventory I can reconstruct from the full project. I'm separating the **seven analytical chunks**, the **four final-synthesis chunks**, and the **Medium-specific graphics created afterward**.

I'm linking files that are still present in the current runtime. Some PNGs, scripts, JSON metadata files, and ZIP bundles were generated earlier but are no longer mounted at their original paths; I still list their filenames so you have the complete record.

Chunk 1 — Business Understanding & Initial Data Understanding

No new analytical output files were created in Chunk 1. The work used the original source dataset:

- [all_exoplanets_2021.csv](sandbox:/mnt/data/all_exoplanets_2021.csv)

Chunk 2 — Raw Visual Baseline, Data Preparation & Feature Engineering

Figures

- raw_period_vs_stellar_temperature_by_method.png — generated earlier
- [raw_period_vs_stellar_temperature_by_method.pdf](sandbox:/mnt/data/exoplanet_habitability/raw_period_vs_stellar_temperature_by_method.pdf)
- raw_missingness_by_column.png — generated earlier
- [raw_missingness_by_column.pdf](sandbox:/mnt/data/exoplanet_habitability/raw_missingness_by_column.pdf)

Prepared data / audit tables

- [preprocessing_decision_register.csv]

(sandbox:/mnt/data/exoplanet_habitability/preprocessing_decision_register.csv)
- [exoplanets_prepared_no_statistical_imputation.csv]
(sandbox:/mnt/data/exoplanet_habitability/exoplanets_prepared_no_statistical_imputation.csv)
)
- [modeling_cohort_sizes.csv]
(sandbox:/mnt/data/exoplanet_habitability/modeling_cohort_sizes.csv)
- [raw_missingness_summary.csv]
(sandbox:/mnt/data/exoplanet_habitability/raw_missingness_summary.csv)
- [raw_scatter_panel_summary.csv]
(sandbox:/mnt/data/exoplanet_habitability/raw_scatter_panel_summary.csv)

Reproducibility code

- `prepare_exoplanets.py` — generated earlier

Chunk 3 — Feature Selection, Outlier Analysis & Clustering

Figures

- `core_feature_collinearity_heatmap.png` — generated earlier
- [core_feature_collinearity_heatmap.pdf]
(sandbox:/mnt/data/exoplanet_habitability/core_feature_collinearity_heatmap.pdf)
- `physics_aware_outlier_landscape.png` — generated earlier
- [physics_aware_outlier_landscape.pdf]
(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_landscape.pdf)
- `mass_period_clusters_with_hz_overlap.png` — generated earlier
- [mass_period_clusters_with_hz_overlap.pdf]
(sandbox:/mnt/data/exoplanet_habitability/mass_period_clusters_with_hz_overlap.pdf)

Analysis datasets

- [exoplanets_chunk3_feature_outlier_cluster.csv]
(sandbox:/mnt/data/exoplanet_habitability/exoplanets_chunk3_feature_outlier_cluster.csv)

- [mass_period_cluster_assignments.csv]

(sandbox:/mnt/data/exoplanet_habitability/mass_period_cluster_assignments.csv)

Feature-selection outputs

- [core_host_balanced_spearman_correlation.csv]

(sandbox:/mnt/data/exoplanet_habitability/core_host_balanced_spearman_correlation.csv)

- [core_host_weighted_vif.csv]

(sandbox:/mnt/data/exoplanet_habitability/core_host_weighted_vif.csv)

- [core_feature_relevance_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/core_feature_relevance_host_bootstrap.csv)

Outlier outputs

- [physics_aware_outlier_summary.csv]

(sandbox:/mnt/data/exoplanet_habitability/physics_aware_outlier_summary.csv)

Clustering outputs

- [cluster_solution_diagnostics.csv]

(sandbox:/mnt/data/exoplanet_habitability/cluster_solution_diagnostics.csv)

- [cluster_profiles.csv](sandbox:/mnt/data/exoplanet_habitability/cluster_profiles.csv)

- [cluster_hz_overlap_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/cluster_hz_overlap_host_bootstrap.csv)

- [cluster_hz_overlap_by_discovery_method.csv]

(sandbox:/mnt/data/exoplanet_habitability/cluster_hz_overlap_by_discovery_method.csv)

- [cluster_outlier_sensitivity.csv]

(sandbox:/mnt/data/exoplanet_habitability/cluster_outlier_sensitivity.csv)

Reproducibility code

- analyze_feature_outliers_chunk3.py — generated earlier

- analyze_clusters_chunk3.py — generated earlier

Chunk 4 — Naive Baselines & Period × Temperature Logistic Model

Figures

- `chunk4_reviewed_logistic_performance_and_interaction_test.png` — generated earlier
- `[chunk4_reviewed_logistic_performance_and_interaction_test.pdf]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_logistic_performance_and_interaction_test.pdf`)
- `chunk4_reviewed_period_temperature_response_and_interaction_map.png` — generated earlier
- `[chunk4_reviewed_period_temperature_response_and_interaction_map.pdf]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_period_temperature_response_and_interaction_map.pdf`)

Model-performance outputs

- `[chunk4_reviewed_key_model_metrics.csv]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_key_model_metrics.csv`)
- `[chunk4_reviewed_interaction_increment_host_bootstrap.csv]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_interaction_increment_host_bootstrap.csv`)
- `[chunk4_reviewed_method_stratified_performance_host_bootstrap.csv]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_method_stratified_performance_host_bootstrap.csv`)

Statistical tests

- `[chunk4_reviewed_interaction_logloss_sign_flip_test.csv]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_interaction_logloss_sign_flip_test.csv`)
- `[chunk4_reviewed_method_sign_flip_tests.csv]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_method_sign_flip_tests.csv`)
- `[chunk4_reviewed_interaction_coefficient_summary.csv]`
(`sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_interaction_coefficient_summary.csv`)

csv)

- [chunk4_reviewed_peak_period_by_temperature_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_peak_period_by_temperature_host_bootstrap.csv)

- [chunk4_reviewed_always_hz_baseline_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_always_hz_baseline_host_bootstrap.csv)

Validation / predictions

- [chunk4_reviewed_repeated_group_cv_predictions_aggregated.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_repeated_group_cv_predictions_aggregated.csv)

- [chunk4_reviewed_repeated_group_cv_predictions_long.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_repeated_group_cv_predictions_long.csv)

- [chunk4_reviewed_outer_fold_audit.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_outer_fold_audit.csv)

- [chunk4_reviewed_inner_cv_thresholds.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk4_reviewed_inner_cv_thresholds.csv)

Metadata / code

- chunk4_reviewed_run_metadata.json — generated earlier

- analyze_interaction_logistic_chunk4_reviewed.py — generated earlier

- render_chunk4_reviewed_charts.py — generated earlier

Chunk 5 — Full Stellar-Structure Logistic Model & Discovery-Method Adjustment

Figures

- performance_and_incremental_value_final.png — generated earlier

- [performance_and_incremental_value_final.pdf]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/performance_and_incremental_value_final.pdf)

- coefficient_shift_and_joint_stellar_effect_final.png — generated earlier

- [coefficient_shift_and_joint_stellar_effect_final.pdf]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/coefficient_shift_and_joint_stellar_effect_final.pdf)

- discovery_method_adjustment_final.png — generated earlier

- [discovery_method_adjustment_final.pdf]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/discovery_method_adjustment_final.pdf)

Key model results

- [key_model_metrics.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_model_metrics.csv)

- [key_incremental_comparisons.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_incremental_comparisons.csv)

- [key_coefficient_estimates.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_coefficient_estimates.csv)

- [key_coefficient_shifts.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/key_coefficient_shifts.csv)

Discovery-method outputs

- [method_calibration_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/method_calibration_host_bootstrap.csv)

- [method_stratified_performance_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/method_stratified_performance_host_bootstrap.csv)

Statistical tests / validation

- [logloss_sign_flip_tests.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/logloss_sign_flip_tests.csv)

- [repeated_group_cv_predictions_aggregated.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/repeated_group_cv_predictions_aggregated.csv)

- [outer_fold_audit.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk5_final/outer_fold_audit.csv)

Metadata / code

- run_metadata.json — generated earlier
- analysis_core.py — generated earlier
- run_cv.py — generated earlier
- run_coefficients.py — generated earlier
- combine_results.py — generated earlier
- render_final_charts.py — generated earlier

Chunk 6 — Constrained Random Forest

Figures

- random_forest_performance_selection.png — generated earlier
- [random_forest_performance_selection.pdf]
- (sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_performance_selection.pdf)
- random_forest_grouped_permutation_importance.png — generated earlier
- [random_forest_grouped_permutation_importance.pdf]
- (sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_grouped_permutation_importance.pdf)
- random_forest_period_temperature_response_surface.png — generated earlier
- [random_forest_period_temperature_response_surface.pdf]
- (sandbox:/mnt/data/exoplanet_habitability/chunk6_final/random_forest_period_temperature_response_surface.pdf)

Model-performance outputs

- [model_performance_host_bootstrap.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/model_performance_host_bootstrap.csv)
- [rf_vs_logistic_paired_host_bootstrap.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_vs_logistic_paired_host_bootstrap.csv)
- [rf_vs_logistic_logloss_sign_flip_test.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_vs_logistic_logloss_sign_flip_test.csv)
- [method_stratified_performance_host_bootstrap.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/method_stratified_performance_host_bootstrap.csv)

Hyperparameter/configuration diagnostics

- [configuration_selection_frequency.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/configuration_selection_frequency.csv)
- [configuration_inner_cv_summary.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/configuration_inner_cv_summary.csv)

Feature importance / response surface

- [heldout_grouped_permutation_importance_host_bootstrap.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/heldout_grouped_permutation_importance_host_bootstrap.csv)
- [period_temperature_conditional_response_surface.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/period_temperature_conditional_response_surface.csv)
- [period_temperature_response_peak_periods.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/period_temperature_response_peak_periods.csv)

Validation / predictions

- [rf_repeated_group_cv_predictions_aggregated.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_repeated_group_cv_predictions_aggregated.csv)
- [rf_outer_fold_audit.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk6_final/rf_outer_fold_audit.csv)

Metadata / code / package

- run_metadata.json — generated earlier
- run_random_forest_chunk6.py — generated earlier
- analyze_random_forest_results.py — generated earlier
- render_chunk6_charts.py — generated earlier
- chunk6_random_forest_key_artifacts.zip — generated earlier

Chunk 7 — Gradient Boosting & Continuous Log-Insolation Regression

Figures

- chunk7_classifier_performance_and_selection.png — generated earlier
- [chunk7_classifier_performance_and_selection.pdf]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_performance_and_selection.pdf)
- chunk7_gradient_boosting_grouped_permutation_importance.png — generated earlier
- [chunk7_gradient_boosting_grouped_permutation_importance.pdf]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_grouped_permutation_importance.pdf)
- chunk7_gradient_boosting_period_temperature_surface.png — generated earlier
- [chunk7_gradient_boosting_period_temperature_surface.pdf]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_period_temperature_surface.pdf)
- chunk7_log_insolation_regression_crosscheck.png — generated earlier

- [chunk7_log_insolation_regression_crosscheck.pdf]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_insolation_regression_crosscheck.pdf)

Classifier-performance outputs

- [chunk7_classification_model_performance_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_model_performance_host_bootstrap.csv)

- [chunk7_classification_paired_model_contrasts_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_paired_model_contrasts_host_bootstrap.csv)

- [chunk7_classification_method_stratified_performance.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classification_method_stratified_performance.csv)

Gradient-boosting outputs

- [chunk7_gradient_boosting_configuration_summary.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_gradient_boosting_configuration_summary.csv)

- [gb_heldout_grouped_permutation_importance_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/gb_heldout_grouped_permutation_importance_host_bootstrap.csv)

Continuous-flux regression outputs

- [log_flux_regression_metrics_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_metrics_host_bootstrap.csv)

- [log_flux_regression_zone_metrics_host_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_zone_metrics_host_bootstrap.csv)

- [log_flux_regression_coefficient_bootstrap.csv]

(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/log_flux_regression_coefficient_bootstrap.csv)

- [chunk7_classifier_regression_hz_agreement_host_bootstrap.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_classifier_regression_hz_agreement_host_bootstrap.csv)
- [chunk7_log_flux_regression_zone_disagreements.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_log_flux_regression_zone_disagreements.csv)

QA / metadata / package

- [chunk7_qa_checks.csv]
(sandbox:/mnt/data/exoplanet_habitability/chunk7_release/chunk7_qa_checks.csv)
- chunk7_run_metadata_verified.json — generated earlier
- chunk7_analysis_summary.json — generated earlier
- chunk7_verified_key_artifacts.zip — generated earlier

Final Synthesis — Part 1

Abstract, introduction, scope statement, and methods.

No new analytical files were created. This chunk consolidated existing outputs into publication-ready prose.

Final Synthesis — Part 2

Results section and figure shortlist.

No new analytical files were created. It reused the main figures from Chunks 2–7 and selected the strongest public-article figures.

Final Synthesis — Part 3

Discussion, limitations, claim guardrails, robust/suggestive/unsupported evidence map.

No new analytical files were created.

Final Synthesis — Part 4

Conclusions, recommendations, Medium ending, and paste-ready GitHub README skeleton.

No new analytical files were created.

Post-synthesis / Medium article graphics

These were created after the analytical chunks specifically for your Medium article.

Inverted-U orbital-period explainer

- [Radiative HZ likelihood vs orbital period infographic]
(sandbox:/mnt/data/a_clean_infographic_chart_diagram_on_a_white_bac.png)

Candidate vs non-candidate median table

- [Typical orbits and host stars infographic]

(sandbox:/mnt/data/a_clean_infographic_table_image_on_a_white_light_b.png)

Correlation table infographic

- [Stellar and orbital feature correlations infographic]

(sandbox:/mnt/data/a_clean_infographic_table_slide_style_image_white.png)

Model-performance table infographic

A model-performance table image was generated, but its generic generated filename was subsequently reused by another infographic, so I would not treat the current file at that old generic path as a reliable copy of the original model-performance image.

Observed HZ prevalence by discovery method infographic

Likewise, this was generated under a generic image filename that was reused later.

Regenerated discovery-method covariate-shift infographic

- [Discovery-method covariate shift infographic]

(sandbox:/mnt/data/a_clean_infographic_figure_style_image_with_a_whit.png)

Original / remade stellar-collinearity figure

- [Original currently mounted PNG]

(sandbox:/mnt/data/coefficient_shift_and_joint_stellar_effect_final.png)

- [Remade uncluttered PDF]

(sandbox:/mnt/data/coefficient_shift_and_joint_stellar_effect_remade.pdf)

Total analytical artifact groups

Across Chunks 2–7, the project produced:

- **18 major analytical figures** in PNG/PDF pairs
- **50+ CSV result/audit tables**
- **multiple prepared modeling datasets**
- **repeated grouped-CV prediction files**
- **model-selection and bootstrap diagnostics**
- **reproducibility scripts for every major modeling stage**
- **metadata/QA JSON files and ZIP bundles**
- **6+ Medium-specific explanatory graphics** after the formal analysis

Chunk 1 and the four Final Synthesis chunks were primarily narrative/methodological and did not add new analytical result files.